
BuzzCalculus

完整題庫題本 • Complete Problem Set

總題數 **1118** 題

極限 159 題

微分 307 題

積分 488 題

級數 164 題

編譯 XeLaTeX + XeCJK

姓名 _____

日期 _____

分數 _____

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1 lim-001

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

2 lim-002

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

3 lim-003

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

4 lim-004

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x + 7}{2x^2 + 5}$$

5 lim-005

R2 Basic • 30s • rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

6 lim-006

R2 Basic • 35s • rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{x}$$

7 lim-007

R2 Basic • 35s • rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\tan(4x)}{\sin(2x)}$$

8 lim-008

R2 Basic • 45s • rank-2 / beginner-friendly

$$\lim_{x \rightarrow \infty} x \left(\sqrt{x^2 + 1} - x \right)$$

9 lim-009

R3 Standard • 45s • rank-3

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$$

10 lim-010

R3 Standard • 50s • rank-3

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

11 lim-011

R3 Standard • 45s • rank-3

$$\lim_{x \rightarrow 0^+} x \ln x$$

12 lim-012

R3 Standard • 65s • rank-3

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x) - x + \frac{x^2}{2}}{x^3}$$

13 lim-013

R3 Standard • 75s • rank-3

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

14 lim-014

R4 Advanced • 75s • rank-4

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x) - \sin x}{x^2}$$

15 lim-015

R3 Standard • 70s • rank-3

$$\lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2}$$

16 lim-016

R4 Advanced • 90s • rank-4

$$\lim_{x \rightarrow \infty} x \left(\sqrt{x^2 + x} - x - \frac{1}{2} \right)$$

17 lim-017

R3 Standard • 75s • rank-3

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3}$$

18 lim-018

R3 Standard • 70s • rank-3

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3}$$

19 lim-019

R4 Advanced • 95s • rank-4

$$\lim_{x \rightarrow 0} \frac{(1 + x)^{1/x} - e}{x}$$

20 lim-020

R3 Standard • 70s • rank-3

$$\lim_{x \rightarrow 0} \frac{\arctan x - x}{x^3}$$

21 td-lim-001

R6 Boss+ • 90s • rank-6 / boss-rank

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{k}{n^2}\right)$$

22 td-lim-002

R6 Boss+ • 120s • rank-6 / boss-rank

$$\lim_{n \rightarrow \infty} n \left(\prod_{k=1}^n \left(1 + \frac{k}{n^2}\right) - \sqrt{e} \right)$$

23 td-lim-004

R4 Advanced • 80s • rank-4

$$\lim_{x \rightarrow 0} \frac{(1+x)^x - 1 - x^2}{x^3}$$

24 lim-021

R2 Basic • 55s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$$

25 lim-022

R2 Basic • 55s • rationalize / infinity

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$$

26 lim-024

R2 Basic • 50s • taylor / log

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x}{x^2}$$

27 lim-025

R4 Advanced • 55s • multivariable / standard-limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{xy}$$

28 lim-026

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$$

29 lim-027

R4 Advanced • 70s • multivariable / path-test

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

30 lim-028

R4 Advanced • 75s • multivariable / rationalize

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{1+x^2+y^2} - 1}{x^2 + y^2}$$

31 lim-029

R4 Advanced • 60s • multivariable / squeeze

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 + y^4}{x^2 + y^2}$$

32 lim-030

R2 Basic • 50s • standard-limit / trig-limit

$$\lim_{x \rightarrow 0} \frac{1 - \cos(3x)}{x^2}$$

33 lim-031

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}$$

34 lim-032

R4 Advanced • 70s • multivariable / path-test

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

35 lim-033

R4 Advanced • 60s • multivariable / standard-limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{e^{x+y} - 1}{x + y}$$

36 lim-034

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2}$$

37 lim-035

R2 Basic • 50s • standard-limit / trig-limit

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(2x)}$$

38 lim-036

R3 Standard • 70s • taylor / log

$$\lim_{x \rightarrow 0} \frac{\log(1+x+x^2) - x}{x^2}$$

39 lim-037

R2 Basic • 50s • rationalize / rank-2

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+4x} - 1}{x}$$

40 lim-039

R4 Advanced • 80s • multivariable / path-test

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

41 lim-040

R2 Basic • 55s • standard-limit / log

$$\lim_{x \rightarrow \infty} x \log \left(1 + \frac{2}{x} \right)$$

42 gap-tech-001

R2 Basic • 45s • technique-recognition / rationalize

$$\text{Technique for } \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$$

43 gap-tech-002

R2 Basic • 45s • technique-recognition / taylor

$$\text{Technique for } \lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

44 gap-tech-008

R2 Basic • 55s • technique-recognition / multivariable-limit

$$\text{Technique for } \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

45 mob-tech-001

R2 Basic • 45s • technique-sprint / technique-recognition

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x) \quad \text{Use which technique?}$$

46 mob-tech-002

R2 Basic • 45s • technique-sprint / technique-recognition

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2} \quad \text{Use which technique?}$$

47 mob-tech-009

R2 Basic • 45s • technique-sprint / technique-recognition

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \quad \text{Use which technique?}$$

48 mob-trap-009

R2 Basic • 45s • trap-drill / rank-2

$$\frac{0}{0} \text{ limit form} \quad \text{Main trap?}$$

49 mob-limtrap-003

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x \rightarrow 0} \frac{1 - \cos x - x^2/2}{x^4}$$

50 mob-limtrap-004

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2}{x^3}$$

51 mob-limtrap-005

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2}{x^3}$$

52 mob-limtrap-006

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - x/2}{x^2}$$

53 mob-limtrap-008

R3 Standard • 70s • limit-trap / taylor

$$\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} \sin \frac{1}{\sqrt{x^2 + y^2}}$$

54 rel-basic-001

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$$

55 rel-basic-003

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x \rightarrow 0} \frac{e^{4x} - 1}{x}$$

56 rel-basic-004

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{x}$$

57 rel-basic-005

R1 Warm-up • 45s • rationalize / beginner-foundation

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

58 rel-adv-002

R3 Standard • 75s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2 - x^3/6}{x^4}$$

59 rel-boss-019

R5 Boss • 110s • asymptotic / limit-trap

$$\lim_{n \rightarrow \infty} n (\log(n+1) - \log n)$$

60 rel-hard-lim-001

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6 - x^5/120}{x^7}$$

61 rel-hard-lim-002

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{\tan x - x - x^3/3 - 2x^5/15}{x^7}$$

62 rel-hard-lim-003

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2 - x^3/3 + x^4/4}{x^5}$$

63 rel-hard-lim-004

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2 - x^2 - x^4/12}{x^6}$$

64 rel-hard-lim-007

R5 Boss • 95s • taylor / limit-trap

$$\lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(xy)}{x^2 y^2}$$

65 rel-hard-lim-008

R5 Boss • 95s • taylor / limit-trap

$$\lim_{(x,y) \rightarrow (0,0)} \frac{e^{x^2+y^2} - 1}{x^2 + y^2}$$

66 rel-hard-lim-009

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x \rightarrow 0} \frac{\arctan x - x + x^3/3}{x^5}$$

67 rel-hard-lim-010

R5 Boss • 95s • taylor / limit-trap

$$\lim_{n \rightarrow \infty} n^2 \left(\left(1 + \frac{1}{n} \right)^n - e + \frac{e}{2n} \right)$$

68 rel-hard-lim-006

R5 Boss • 80s • path-test / multivariable

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{(x^2 + y^2)^2}$$

69 exam-lim-001

R5 Boss • 180s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x + \frac{9}{2}x^3}{x^5}$$

70 exam-lim-002

R5 Boss • 180s • taylor / log

$$\lim_{x \rightarrow 0} \frac{\log(1+2x) - 2x + 2x^2}{x^3}$$

71 exam-lim-003

R5 Boss • 180s • taylor / exponential

$$\lim_{x \rightarrow 0} \frac{e^x - \cos x - \sin x}{x^2}$$

72 exam-lim-005

R5 Boss • 180s • trig-limit / exam-style

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$$

73 exam-lim-006

R5 Boss • 180s • taylor / radical

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+4x} - 1 - 2x}{x^2}$$

74 exam-lim-007

R5 Boss • 180s • taylor / infinity-limit

$$\lim_{x \rightarrow \infty} x^2 \left(\sqrt{1 + \frac{3}{x}} - 1 - \frac{3}{2x} \right)$$

75 exam-lim-008

R5 Boss • 180s • rational / exam-style

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{3x^2 - 2x}$$

76 exam-lim-009

R5 Boss • 180s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x(1 - \cos x)}$$

77 exam-lim-010

R5 Boss • 180s • taylor / inverse-trig

$$\lim_{x \rightarrow 0} \frac{\arctan(2x) - 2x}{x^3}$$

78 exam-lim-011

R5 Boss • 180s • log-limit / exam-style

$$\lim_{x \rightarrow 0^+} x^x$$

79 exam-lim-012

R5 Boss • 180s • exponential-limit / exam-style

$$\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x}$$

80 exam-lim-015

R5 Boss • 180s • chain-rule / exponential-limit

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x}$$

81 exam-lim-017

R5 Boss • 180s • taylor / exam-style

$$\lim_{x \rightarrow 0} \frac{\cos x - e^{-x^2/2}}{x^4}$$

82 exam-lim-019

R5 Boss • 180s • log-limit / exam-style

$$\lim_{x \rightarrow \infty} x(\log(x+1) - \log x)$$

83 exam-lim-020

R5 Boss • 180s • taylor / inverse-trig

$$\lim_{x \rightarrow 0} \frac{\arcsin x - x}{x^3}$$

84 uni-lim-001

R6 Boss+ • 210s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$$

85 uni-lim-002

R6 Boss+ • 210s • taylor / log

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + \frac{x^2}{2} - \frac{x^3}{3}}{x^4}$$

86 uni-lim-003

R6 Boss+ • 210s • taylor / exponential

$$\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2}$$

87 uni-lim-005

R6 Boss+ • 210s • taylor / radical

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x} - x}{x^3}$$

88 uni-lim-006

R6 Boss+ • 210s • log-limit / infinity-limit

$$\lim_{x \rightarrow \infty} x \left((x+1) \log \left(1 + \frac{1}{x} \right) - 1 \right)$$

89 uni-lim-007

R6 Boss+ • 210s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{\sin(2x) - 2 \sin x}{x^3}$$

90 uni-lim-008

R6 Boss+ • 210s • log-limit / exponential-limit

$$\lim_{x \rightarrow 0} \frac{(1+3x)^{1/x} - e^3}{x}$$

91 uni-lim-009

R6 Boss+ • 210s • taylor / hyperbolic

$$\lim_{x \rightarrow 0} \frac{\cosh x - \cos x}{x^2}$$

92 uni-lim-010

R6 Boss+ • 210s • taylor / inverse-trig

$$\lim_{x \rightarrow 0} \frac{\arctan x - \arcsin x}{x^3}$$

93 uni-lim-011

R6 Boss+ • 210s • taylor / log

$$\lim_{x \rightarrow 0} \frac{\log(1 + \sin x) - x}{x^2}$$

94 uni-lim-012

R6 Boss+ • 210s • taylor / exponential-limit

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{1 - \cos x}$$

95 uni-lim-014

R6 Boss+ • 210s • taylor / radical

$$\lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1+x^2}} - \cos x}{x^4}$$

96 depth-lim-001

R6 Boss+ • 270s • nested-taylor / taylor

$$\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6} - \frac{x^5}{120}}{x^7}$$

97 depth-lim-002

R6 Boss+ • 270s • nested-taylor / taylor

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4}}{x^5}$$

98 depth-lim-003

R5 Boss • 220s • composite-taylor / taylor

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 - x}{x^2}$$

99 depth-lim-004

R6 Boss+ • 270s • asymptotic-balance / taylor

$$\lim_{x \rightarrow 0} \frac{\cos x - \sqrt{1-x^2}}{x^4}$$

100 depth-lim-005

R5 Boss • 220s • asymptotic-balance / taylor

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x(1 - \cos x)}$$

101 depth-lim-006

R6 Boss+ • 270s • infinity-limit / log-limit

$$\lim_{x \rightarrow \infty} x^2 \left(\log \left(1 + \frac{1}{x} \right) - \frac{1}{x+1} \right)$$

102 depth-lim-008

R5 Boss • 220s • inverse-trig / taylor

$$\lim_{x \rightarrow 0} \frac{\arcsin x - \arctan x}{x^3}$$

103 depth-lim-009

R5 Boss • 220s • hyperbolic / taylor

$$\lim_{x \rightarrow 0} \frac{\sinh x - \sin x}{x^3}$$

104 depth-lim-010

R5 Boss • 220s • trig-limit / asymptotic-balance

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$$

105 depth-lim-011

R5 Boss • 220s • log / taylor

$$\lim_{x \rightarrow 0} \frac{\log(\cos x)}{x^2}$$

106 depth-lim-012

R6 Boss+ • 270s • exponential-limit / log-limit

$$\lim_{x \rightarrow 0} \frac{(1+x+x^2)^{1/x} - e}{x}$$

107 burst-boss-lim-001

R5 Boss • 240s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\tan x - x - x^3/3}{x^5}$$

108 burst-boss-lim-002

R5 Boss • 240s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^5}$$

109 burst-boss-lim-003

R5 Boss • 240s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2 - x^3/3}{x^4}$$

110 burst-boss-lim-004

R6 Boss+ • 300s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 - x - x^2/2}{x^4}$$

111 burst-boss-lim-005

R6 Boss+ • 300s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4}$$

112 burst-boss-lim-006

R6 Boss+ • 300s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\arctan x - x + x^3/3 - x^5/5}{x^7}$$

113 burst-boss-lim-008

R6 Boss+ • 300s • true-boss / boss-rank

$$\lim_{x \rightarrow \infty} x \left(\left(1 + \frac{1}{x} \right)^x - e \right)$$

114 burst-boss2-lim-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$$

115 burst-boss2-lim-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{x/\tan x - 1 + x^2/3}{x^4}$$

116 burst-boss2-lim-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{1/\sin x - 1/x - x/6}{x^3}$$

117 burst-boss2-lim-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\log(\cos x) + x^2/2}{x^4}$$

118 burst-boss2-lim-005

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\int_0^x e^{-t^2} dt - x + x^3/3}{x^5}$$

119 burst-boss2-lim-006

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\int_0^x \log(1+t^2) dt - x^3/3 + x^5/10}{x^7}$$

120 burst-boss2-lim-007

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \frac{1}{n+k} - \log 2 \right)$$

121 burst-boss2-lim-008

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n^2 \left(\sum_{k=1}^n \frac{1}{n^2 + k^2} - \frac{\pi}{4n} \right)$$

122 burst-boss2-lim-009

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \frac{1}{\sqrt{n^2 + k^2}} - \log(1 + \sqrt{2}) \right)$$

123 burst-boss2-lim-010

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n \left(\int_0^1 \left(1 + \frac{x}{n} \right)^n dx - (e - 1) \right)$$

124 burst-boss3-lim-001

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n \sup_{0 \leq x \leq 1} x^n (1 - x)$$

125 burst-boss3-lim-002

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n^2 \sup_{0 \leq x \leq 1} x^n (1 - x)^2$$

126 burst-boss3-lim-003

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n \int_0^1 \frac{x^n}{1 + x} dx$$

127 burst-boss3-lim-004

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n^2 \left(\frac{1}{n} \sum_{k=1}^n e^{k/n} - (e - 1) - \frac{e - 1}{2n} \right)$$

128 burst-boss3-lim-005

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n^2 \left(\frac{1}{n} \sum_{k=1}^n \log \left(1 + \frac{k}{n} \right) - (2 \log 2 - 1) - \frac{\log 2}{2n} \right)$$

129 burst-boss3-lim-006

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n \rightarrow \infty} n^2 \left(\frac{1}{n} \sum_{k=1}^n \frac{1}{1 + (k/n)^2} - \frac{\pi}{4} + \frac{1}{4n} \right)$$

130 burst-boss3-lim-007

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{e^{t^2} - \cos t}{t^2} dt - \frac{3}{2}x}{x^3}$$

131 burst-boss3-lim-008

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\frac{1}{x^3} \int_0^x \sin(t^2) dt - \frac{1}{3}}{x^4}$$

132 burst-boss3-lim-009

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\log(\sin x / x) + x^2 / 6}{x^4}$$

133 burst-boss3-lim-010

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\log(\cosh x) - x^2 / 2}{x^4}$$

134 burst-boss3-lim-011

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - x/2 + x^2/8}{x^3}$$

135 burst-boss3-lim-012

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^{1/x^2}$$

136 world-001

R1 Warm-up • 35s • trig-limit / world-universities

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(3x)}$$

137 world-002

R2 Basic • 55s • exponential-limit / world-universities

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x}$$

138 world-003

R2 Basic • 55s • exponential-limit / world-universities

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$$

139 world-007

R2 Basic • 55s • taylor / hyperbolic

$$\lim_{x \rightarrow 0} \frac{\sinh x - x}{x^3}$$

140 world-008

R2 Basic • 55s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x}$$

141 world-011

R3 Standard • 90s • exponential-limit / world-universities

$$\lim_{x \rightarrow 0} \left(\frac{2^x + 8^x}{2}\right)^{1/x}$$

142 world-012

R4 Advanced • 150s • taylor / trig-limit

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \cot^2 x\right)$$

143 world-013

R4 Advanced • 150s • exponential-limit / taylor

$$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x}\right)^{1/x}$$

144 world-026

R4 Advanced • 150s • taylor / world-universities

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$$

145 world-030

R4 Advanced • 150s • exponential-limit / taylor

$$\lim_{x \rightarrow 0} (\cos x)^{1/\sin^2 x}$$

146 world-035

R5 Boss • 240s • stirling / world-universities

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! n^n}\right)^{1/n}$$

147 world-040

R5 Boss • 240s • exponential-limit / taylor

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{x^2}$$

148 world-046

R5 Boss • 240s • asymptotics / world-universities

$$\lim_{x \rightarrow \infty} \frac{x^2 \left(x^{1/x} - 1 - \frac{\log x}{x}\right)}{(\log x)^2}$$

149 world-049

R5 Boss • 240s • asymptotics / world-universities

$$\lim_{n \rightarrow \infty} n^2 \left(\left(1 + \frac{1}{n}\right)^n - e + \frac{e}{2n}\right)$$

150 world-066

R6 Boss+ • 340s • product-limit / world-universities

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{k^2}{n^3}\right)$$

151 world-067

R6 Boss+ • 340s • stirling / world-universities

$$\lim_{n \rightarrow \infty} \left(\frac{(3n)!}{n! (2n)!}\right)^{1/n}$$

152 world-071

R6 Boss+ • 340s • riemann-sum / world-universities

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2}$$

153 world-074

R6 Boss+ • 340s • stirling / central-binomial

$$\lim_{n \rightarrow \infty} \frac{4^n}{\sqrt{n} \binom{2n}{n}}$$

154 world-091

R6 Boss+ • 340s • asymptotics / world-universities

$$\lim_{n \rightarrow \infty} n \left(\left(1 + \frac{1}{n} \right)^n - e \right)$$

155 world-096

R6 Boss+ • 340s • exponential-limit / taylor

$$\lim_{x \rightarrow 0} \left(\frac{\sinh x}{x} \right)^{1/x^2}$$

156 world-099

R6 Boss+ • 340s • taylor / world-universities

$$\lim_{x \rightarrow 0} \frac{\tan(\sin x) - \sin(\tan x)}{x^7}$$

157 putnam-003

R5 Boss • 240s • putnam / product-limit

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{k}{n} \right)^{1/n}$$

158 putnam-008

R5 Boss • 240s • putnam / limit

$$\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x (1 + \sin 2t)^{1/t} dt$$

159 putnam-010

R6 Boss+ • 340s • putnam / riemann-sum

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + (k/n)^2}$$

微分 Derivatives

307 題

160 der-001

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx} (x^5 - 3x^2 + 7)$$

161 der-002

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx} (\sin x + \cos x)$$

162 der-003

R1 Warm-up • 40s • rank-1 / beginner-friendly

$$\frac{d}{dx} (e^x \ln x)$$

163 der-004

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx} (\sqrt{x})$$

164 der-005

R2 Basic • 40s • rank-2 / beginner-friendly

$$\frac{d}{dx} (\ln(1 + x^2))$$

165 der-006

R2 Basic • 45s • rank-2 / beginner-friendly

$$\frac{d}{dx} (x^2 \sin x)$$

166 der-007

R2 Basic • 35s • rank-2 / beginner-friendly

$$\frac{d}{dx} (\arctan x)$$

167 der-008

R2 Basic • 45s • rank-2 / beginner-friendly

$$\frac{d}{dx} \left(\frac{x}{x+1} \right)$$

168 der-009

R3 Standard • 55s • rank-3

$$\frac{d}{dx} (x^x)$$

169 der-010

R3 Standard • 60s • rank-3

$$\frac{d}{dx} (\sin(x^2)e^{-x})$$

170 der-011

R3 Standard • 45s • rank-3

$$\frac{d}{dx} (\ln(\sin x))$$

171 der-012

R4 Advanced • 70s • rank-4

$$\frac{d}{dx} \left(\frac{\ln x}{x^2 + 1} \right)$$

172 der-013

R4 Advanced • 90s • rank-4

$$\frac{d}{dx} (x^{\sin x})$$

173 der-014

R3 Standard • 85s • rank-3

$$\frac{d}{dx} \left(\arctan \frac{1+x}{1-x} \right)$$

174 der-015

R3 Standard • 75s • rank-3

$$\frac{d}{dx} \left(\ln \left(x + \sqrt{1+x^2} \right) \right)$$

175 der-016

R3 Standard • 80s • rank-3

$$\frac{d}{dx} \left(\frac{\sin x}{1 + \cos x} \right)$$

176 der-017

R3 Standard • 80s • rank-3

$$\frac{d}{dx} \left(e^{x^2} \cos(3x) \right)$$

177 der-018

R3 Standard • 80s • rank-3

$$\frac{d}{dx} \left(\ln \left(x + \sqrt{x^2 - 1} \right) \right)$$

178 der-019

R2 Basic • 55s • rank-2 / beginner-friendly

$$\frac{d}{dx} ((x^2 + 1)^5)$$

179 der-020

R3 Standard • 75s • rank-3

$$\frac{d}{dx} (\arctan \sqrt{x})$$

180 td-der-001

R4 Advanced • 100s • rank-4

$$\frac{d^3}{dx^3} \arctan \frac{1}{x}$$

181 td-der-002

R2 Basic • 60s • rank-2 / beginner-friendly

$$\frac{d}{dx} \left[\frac{1}{8} (35x^4 - 30x^2 + 3) \right]$$

182 td-der-003

R3 Standard • 55s • rank-3

$$\frac{d}{dx} (x \log(1+x))$$

183 td-der-004

R3 Standard • 50s • rank-3

$$\frac{d}{dx} (\log(5 + 3 \cos x))$$

184 der-021

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial x} (x^2 y + \sin(xy))$$

185 der-022

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial y} (xe^{xy} + y^3)$$

186 der-023

R4 Advanced • 70s • multivariable / mixed-partial

$$\frac{\partial^2}{\partial x \partial y} e^{xy}$$

187 der-024

R4 Advanced • 75s • multivariable / directional-derivative

Directional derivative of $f = x^2 + y^2$ at $(1, 2)$ in direction $(3, 4)/5$ **188 der-025**

R4 Advanced • 60s • multivariable / partial-derivative

$$\frac{\partial}{\partial x} \log(x^2 + y^2)$$

189 der-026

R2 Basic • 55s • product-rule / chain-rule

$$\frac{d}{dx} (x^2 \sin(3x))$$

190 der-027

R2 Basic • 55s • log / chain-rule

$$\frac{d}{dx} \log \sqrt{1+x^2}$$

191 der-028

R2 Basic • 60s • inverse-trig / chain-rule

$$\frac{d}{dx} \arctan(e^x)$$

192 der-029

R2 Basic • 60s • second-derivative / product-rule

$$\frac{d^2}{dx^2} (xe^x)$$

193 der-030

R3 Standard • 80s • implicit-differentiation / rank-3

Find y' from $x^2 + xy + y^2 = 7$ **194 der-031**

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial x} (e^{x^2+y^2})$$

195 der-032

R4 Advanced • 75s • multivariable / partial-derivative

$$\frac{\partial}{\partial y} \arctan\left(\frac{y}{x}\right)$$

196 der-033

R4 Advanced • 65s • multivariable / second-derivative

$$\frac{\partial^2}{\partial x^2} (x^2y + \sin x)$$

197 der-034

R4 Advanced • 75s • multivariable / jacobian

Jacobian determinant of $u = x + y$, $v = xy$ **198 der-035**

R4 Advanced • 80s • multivariable / chain-rule

If $z = x^2y$, $x = t^2$, $y = e^t$, find dz/dt **199 der-036**

R4 Advanced • 55s • multivariable / second-derivative

Laplacian of $f = x^2 + y^2$ **200 der-037**

R4 Advanced • 90s • multivariable / hessian

Hessian determinant of $f = x^3 + y^3 - 3xy$ at $(1, 1)$ **201 der-038**

R2 Basic • 65s • quotient-rule / rank-2

$$\frac{d}{dx} \left(\frac{x^2 + 1}{x - 1} \right)$$

202 der-039

R2 Basic • 45s • chain-rule / trig

$$\frac{d}{dx} (\sin^2 x)$$

203 der-040

R3 Standard • 80s • logarithmic-differentiation / rank-3

$$\frac{d}{dx} (x^{\log x})$$

204 der-041

R4 Advanced • 80s • total-differential / linearization

Use total differential to estimate Δf for $f = x^2y$ at $(2, 1)$, $dx = 0.01$, $dy = -0.02$ **205 der-042**

R4 Advanced • 85s • total-differential / linearization

Use total differential for $f = \sqrt{x^2 + y^2}$ at $(3, 4)$, $dx = 0.06$, $dy = -0.08$ **206 der-043**

R4 Advanced • 75s • total-differential / linearization

Estimate Δz for $z = \log(xy)$ at $(1, 2)$, $dx = 0.02$, $dy = -0.04$ **207 der-044**

R4 Advanced • 90s • total-differential / linearization

Linear part df of $f = e^{xy}$ **208 der-045**

R4 Advanced • 100s • total-differential / linearization

Linear part df of $f = \arctan(y/x)$ **209 der-046**

R4 Advanced • 85s • total-differential / linearization

Linear estimate of $x^2 + y^2$ at $(1.02, 1.97)$ using base $(1, 2)$ **210 der-047**

R5 Boss • 100s • total-differential-min / optimization

$$\min_{x,y} (x^2 + 2xy + 3y^2 - 4x - 6y)$$

211 der-048

R5 Boss • 100s • total-differential-min / optimization

$$\min_{x,y} (3x^2 + 2xy + 2y^2 - 8x - 6y)$$

212 der-049

R5 Boss • 70s • total-differential-min / optimization

x-coordinate of the minimum of $x^2 + y^2 + 2x - 4y + 7$

213 der-050

R5 Boss • 95s • total-differential-min / optimization

 $\min (x^2 + y^2)$ subject to $x + 2y = 5$

214 der-051

R4 Advanced • 75s • hessian / second-derivative

Hessian determinant of $f = x^2 + xy + y^2$

215 der-052

R4 Advanced • 75s • hessian / second-derivative

Hessian determinant of $f = x^4 + y^4$ at $(1, 1)$

216 der-053

R4 Advanced • 95s • hessian / second-derivative

Hessian determinant of $f = e^{xy}$ at $(0, 0)$

217 der-054

R4 Advanced • 90s • hessian / second-derivative

Hessian determinant of $f = x^4 + y^4 - 4xy$ at $(1, 1)$

218 der-055

R4 Advanced • 70s • hessian / critical-point

Classify the critical point $(0, 0)$ of $f = x^2 + y^2$

219 der-056

R4 Advanced • 70s • hessian / critical-point

Classify the critical point $(0, 0)$ of $f = x^2 - y^2$

220 der-057

R4 Advanced • 70s • hessian / degenerate

Hessian determinant of $f = x^4 + y^2$ at $(0, 0)$

221 der-058

R4 Advanced • 100s • hessian / log

Hessian determinant of $f = \log(x^2 + y^2)$ at $(1, 0)$

222 der-059

R4 Advanced • 65s • wronskian / linear-independence

Wronskian $W(\sin x, \cos x)$

223 der-060

R4 Advanced • 65s • wronskian / linear-independence

Wronskian $W(\cos x, \sin x)$

224 der-061

R4 Advanced • 70s • wronskian / linear-independence

Wronskian $W(e^x, e^{2x})$

225 der-062

R4 Advanced • 60s • wronskian / polynomial

Wronskian $W(x, x^2)$

226 der-063

R4 Advanced • 85s • wronskian / polynomial

Wronskian $W(1, x, x^2)$

227 der-064

R4 Advanced • 75s • wronskian / exponential

Wronskian $W(e^x, xe^x)$

228 der-065

R4 Advanced • 65s • wronskian / log

Wronskian $W(1, \log x)$

229 der-066

R4 Advanced • 65s • wronskian / polynomial

Wronskian $W(x^2, x^3)$

230 der-067

R5 Boss • 80s • jacobian-chain / chain-rule

If $u = x + y$, $v = x - y$, $F = u^2 + v^2$, find dF/dx

231 der-068

R5 Boss • 95s • jacobian-chain / chain-rule

If $u = x^2$, $v = \sin x$, $F = ue^v$, find dF/dx

232 der-069

R6 Boss+ • 80s • jacobian-chain / jacobian

Jacobian determinant of $u = x^2 - y^2$, $v = 2xy$ **233 der-070**

R5 Boss • 100s • jacobian-chain / jacobian

Let $r = u + v$, $s = uv$, $u = x + y$, $v = x - y$. Find $\frac{\partial(r, s)}{\partial(x, y)}$ **234 der-071**

R5 Boss • 75s • jacobian-chain / jacobian

Jacobian $\frac{\partial(x, y)}{\partial(r, t)}$ for $x = r \cos t$, $y = r \sin t$ **235 der-072**

R5 Boss • 90s • jacobian-chain / jacobian

Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ for $u = xy$, $v = x/y$ **236 der-073**

R6 Boss+ • 115s • jacobian-chain / jacobian

Let $u = 2x + y$, $v = x + 3y$, $p = u^2 - v^2$, $q = 2uv$. Find $\frac{\partial(p, q)}{\partial(x, y)}$ **237 der-074**

R5 Boss • 95s • jacobian-chain / jacobian

If $f(x, y) = (x, xy)$, $g(u, v) = (u + v, u - v)$, find $\det D(g \circ f)$ at $(2, 3)$ **238 der-075**

R6 Boss+ • 75s • complex / real-part

Real part of $(x + iy)^3$ **239 der-076**

R6 Boss+ • 75s • complex / imaginary-part

Imaginary part of $(x + iy)^3$ **240 der-077**

R6 Boss+ • 45s • complex / modulus

 $|1 + i|$ **241 der-078**

R6 Boss+ • 55s • complex / power

Real part of $(1 + i)^4$ **242 der-079**

R6 Boss+ • 70s • complex / cauchy-riemann

Is $u = x^2 - y^2$, $v = 2xy$ holomorphic by Cauchy-Riemann?**243 der-080**

R6 Boss+ • 70s • complex / cauchy-riemann

Is $u = x^2 + y^2$, $v = 0$ holomorphic on an open set?**244 der-081**

R6 Boss+ • 80s • complex / residue

 $\left| \text{Res}_{z=i} \frac{1}{z^2 + 1} \right|$ **245 der-082**

R6 Boss+ • 60s • complex / residue

 $\text{Res}_{z=2} \frac{1}{(z-2)^3}$ **246 der-083**

R6 Boss+ • 75s • complex / laurent

Coefficient of z^{-1} in e^z/z^3 **247 der-084**

R6 Boss+ • 70s • complex / contour-integral

 $\frac{1}{2\pi i} \int_{|z|=2} \frac{1}{z-1} dz$

248 der-085

R6 Boss+ • 75s • complex / complex-derivative

Real part of $f'(1+i)$ for $f(z) = z^4$ **249 der-086**

R6 Boss+ • 80s • complex / jacobian

Jacobian determinant of the complex map $z \mapsto z^2$ at $z = x + iy$ **250 der-087**

R6 Boss+ • 70s • complex / harmonic

 $\Delta \operatorname{Re}(z^3)$ ($z = x + iy$)**251 der-088**

R6 Boss+ • 65s • complex / harmonic

Is $u = x^3 - 3xy^2$ harmonic?**252 gap-der-param-001**

R2 Basic • 65s • parametric / parametric-differentiation

 $x = t^2, y = t^3. \frac{dy}{dx} = ?$ **253 gap-der-param-002**

R3 Standard • 70s • parametric / parametric-differentiation

 $x = \cos t, y = \sin t. \frac{dy}{dx} = ?$ **254 gap-der-param-003**

R2 Basic • 60s • parametric / second-derivative

 $x = t, y = t^2. \frac{d^2y}{dx^2} = ?$ **255 gap-der-param-004**

R2 Basic • 65s • parametric / parametric-differentiation

 $x = t^2 + 1, y = t^3. \frac{dy}{dx} \Big|_{t=2}$ **256 gap-der-polar-001**

R4 Advanced • 95s • polar-curve / parametric

 $r = 1 + \cos \theta. \frac{dy}{dx} \Big|_{\theta=\pi/2}$ **257 gap-der-param-005**

R2 Basic • 55s • parametric / speed

 $x = 3t, y = 4t.$ Speed $\sqrt{(dx/dt)^2 + (dy/dt)^2}$ **258 gap-der-app-001**

R3 Standard • 80s • related-rates / volume

A sphere has $r = 3, dr/dt = 2.$ $\frac{dV}{dt} = ?$ **259 gap-der-app-002**

R2 Basic • 70s • related-rates / area

A circle has $r = 5, dr/dt = 0.2.$ $\frac{dA}{dt} = ?$ **260 gap-der-app-003**

R3 Standard • 85s • related-rates / implicit-differentiation

Ladder length 10, $x = 6, dx/dt = 2.$ $dy/dt = ?$ **261 gap-der-app-004**

R2 Basic • 60s • tangent-normal / derivative

Normal slope to $y = x^2$ at $x = 2$ **262 gap-der-app-005**

R2 Basic • 65s • linear-approximation / application

Linear approximation of $\sqrt{4.04}$ **263 gap-der-app-006**

R2 Basic • 70s • newton-method / application

One Newton step for $f(x) = x^2 - 2$ from $x_0 = 1$ **264 gap-der-app-007**

R3 Standard • 80s • curvature / application

Curvature of $y = x^2$ at $x = 0$ **265 gap-der-ode-001**

R2 Basic • 75s • ode-intro / separable

 $y' + y = 0, y(0) = 2. y = ?$ **266 gap-der-ode-002**

R2 Basic • 70s • ode-intro / separable

 $y' = 2y, y(0) = 3. y = ?$ **267 gap-der-ode-003**

$$y' + y = e^x, y(0) = 0. y = ?$$

268 gap-der-ode-004

R3 Standard • 90s • ode-intro / second-order

$$y'' + y = 0, y(0) = 0, y'(0) = 1. y = ?$$

269 gap-der-ode-005

R4 Advanced • 105s • ode-intro / logistic

$$y' = y(1 - y), y(0) = 1/2. y = ?$$

270 gap-tech-007

R2 Basic • 45s • technique-recognition / parametric

$$x = t^2, y = t^3. \text{ Which differentiation technique?}$$

271 mob-tech-012

R2 Basic • 45s • technique-sprint / technique-recognition

Optimize $f(x, y)$ subject to $g(x, y) = 0$ Use which technique?

272 mob-tech-014

R2 Basic • 45s • technique-sprint / technique-recognition

$\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ Use which technique?

273 mob-tech-018

R2 Basic • 45s • technique-sprint / technique-recognition

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0 \text{ Use which technique?}$$

274 mob-tech-020

R2 Basic • 45s • technique-sprint / technique-recognition

$$y = x^x \text{ Use which technique?}$$

275 mob-trap-001

R2 Basic • 45s • trap-drill / rank-2

$$\frac{d}{dx} \sin(3x) \text{ Main trap?}$$

276 mob-trap-007

R2 Basic • 45s • trap-drill / rank-2

$$\frac{\partial}{\partial x} y^2 \text{ Main trap?}$$

277 mob-trap-011

R2 Basic • 45s • trap-drill / rank-2

$$\nabla \times (\nabla f) \text{ Main trap?}$$

278 mob-lm-001

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max xy \text{ subject to } x + y = 10 (x, y > 0)$$

279 mob-lm-002

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\min x^2 + y^2 \text{ subject to } x + y = 6$$

280 mob-lm-003

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max xyz \text{ subject to } x + y + z = 12 (x, y, z > 0)$$

281 mob-lm-004

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\min x^2 + y^2 \text{ subject to } xy = 8 (x, y > 0)$$

282 mob-lm-005

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max x^2 y \text{ subject to } x + 2y = 6 (x, y > 0)$$

283 mob-lm-006

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\min x^2 + y^2 + z^2 \text{ subject to } x + y + z = 3$$

284 mob-lm-007

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max A \text{ rectangle with perimeter } 20$$

285 mob-lm-008

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\min x^2 + y^2 + z^2 \text{ subject to } x + y + z = 1$$

286 mob-lm-009

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max xy \text{ subject to } x^2 + y^2 = 50$$

287 mob-lm-010

R4 Advanced • 85s • lagrange-multiplier / optimization

$$\max(x + y) \text{ subject to } x^2 + y^2 = 2$$

288 mob-bessel-001

R6 Boss+ • 70s • bessel / special-functions

$$J_0(0)$$

289 mob-bessel-002R6 Boss+ • 70s • [bessel](#) / [special-functions](#)

$J_1(0)$

290 mob-bessel-003R6 Boss+ • 70s • [bessel](#) / [special-functions](#)

coefficient of x^2 in $J_0(x) = 1 - x^2/4 + \dots$

291 mob-bessel-004R6 Boss+ • 70s • [bessel](#) / [special-functions](#)

coefficient of x in $J_1(x) = x/2 + \dots$

292 mob-bessel-005R6 Boss+ • 45s • [bessel](#) / [special-functions](#)

$J'_0(x)$

293 mob-bessel-006R6 Boss+ • 45s • [bessel](#) / [special-functions](#)

$x^2 y'' + xy' + (x^2 - \nu^2)y = 0$

294 mob-bessel-007R6 Boss+ • 45s • [bessel](#) / [special-functions](#)Singularity of Bessel equation at $x = 0$ **295 mob-bessel-008**R6 Boss+ • 45s • [bessel](#) / [special-functions](#)

$J_{-2}(x)$ equals

296 mob-bessel-009R6 Boss+ • 45s • [bessel](#) / [special-functions](#)

$J_{-1}(x)$ equals

297 mob-bessel-010R6 Boss+ • 45s • [bessel](#) / [special-functions](#)

Bessel functions most often appear after which ODE type?

298 mob-nabla-001R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$\nabla \cdot \langle x, y, z \rangle$

299 mob-nabla-002R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

z-component of $\nabla \times \langle y, -x, 0 \rangle$

300 mob-nabla-003R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$\Delta(x^2 + y^2 + z^2)$

301 mob-nabla-004R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$D_{(1,1)/\sqrt{2}}(x^2 + y^2)$ at $(1, 1)$

302 mob-nabla-005R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$\nabla \times (\nabla f)$

303 mob-nabla-006R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$\nabla \cdot (\nabla \times \mathbf{F})$

304 mob-nabla-007R4 Advanced • 75s • [nabla](#) / [vector-calculus](#)

$\int_C \nabla(x^2 + y^2) \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$

305 mob-nabla-008R4 Advanced • 45s • [nabla](#) / [vector-calculus](#)

$\nabla \cdot \nabla f$ is called

306 mob-nabla-009R4 Advanced • 45s • [nabla](#) / [vector-calculus](#)

$\nabla \times \mathbf{F} = 0$ usually suggests

307 mob-nabla-010

Green theorem circulation uses which scalar derivative?

308 rel-basic-007

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}x^3\right|_{x=2}$$

309 rel-basic-008

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\sin x\right|_{x=0}$$

310 rel-basic-009

R1 Warm-up • 45s • chain-rule / beginner-foundation

$$\left.\frac{d}{dx}e^{2x}\right|_{x=0}$$

311 rel-basic-010

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\sqrt{x}\right|_{x=4}$$

312 rel-basic-011

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\log x\right|_{x=e}$$

313 rel-basic-012

R1 Warm-up • 45s • product-rule / beginner-foundation

$$\left.\frac{d}{dx}(x\log x)\right|_{x=1}$$

314 rel-adv-005

R4 Advanced • 75s • jacobian / multivariable

$$\frac{\partial(u,v)}{\partial(x,y)}\text{ at } (1,2), \quad u = x^2 - y^2, \quad v = 2xy$$

315 rel-adv-006

R4 Advanced • 75s • directional-derivative / multivariable

$$D_{(2,-1,2)/3}(x^2y + yz)\text{ at } (1,2,3)$$

316 rel-adv-007

R4 Advanced • 75s • hessian / optimization

$$\det H_f(1,1), \quad f = x^4 + y^4 - 4xy$$

317 rel-adv-008

R4 Advanced • 75s • lagrange-multiplier / optimization

$$\min(x^2 + y^2 + z^2) \quad \text{subject to } x + 2y + 2z = 9$$

318 rel-boss-007

R6 Boss+ • 110s • bessel / special-functions

$$\text{coefficient of } x^4 \text{ in } J_0(x)$$

319 rel-hard-mv-005

R5 Boss • 120s • jacobian / multivariable

$$\frac{\partial(u,v)}{\partial(x,y)}\text{ at } (2,1), \quad u = \frac{x}{y}, \quad v = xy$$

320 rel-hard-mv-006

R5 Boss • 120s • lagrange-multiplier / optimization

$$\max xyz \quad \text{subject to } x + y + z = 1, \quad x, y, z > 0$$

321 rel-hard-mv-007

R5 Boss • 120s • lagrange-multiplier / optimization

$$\min(x^2 + y^2) \quad \text{subject to } xy = 1, \quad x, y > 0$$

322 rel-hard-der-001

R5 Boss • 95s • wronskian / boss-rank

$$W(e^x, xe^x)\text{ at } x = 0$$

323 rel-hard-der-002

R5 Boss • 95s • wronskian / boss-rank

$$W(\sin x, \cos x)\text{ at } x = 0$$

324 rel-hard-der-003

R5 Boss • 95s • hessian / boss-rank

$$\det H_f(1,0), \quad f = \log(x^2 + y^2)$$

325 rel-hard-der-004

R5 Boss • 95s • curvature / boss-rank

$$\kappa \text{ for } y = \log x \text{ at } x = 1$$

326 rel-hard-der-005

R5 Boss • 95s • parametric-differentiation / boss-rank

$$\frac{dy}{dx} \text{ at } t = 2, \quad x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}$$

327 rel-hard-der-006

R5 Boss • 95s • parametric-differentiation / boss-rank

$$\frac{d^2y}{dx^2} \text{ at } t = 2, \quad x = t^2, \quad y = t^3$$

328 rel-hard-der-007

R5 Boss • 95s • directional-derivative / nabla

$$D_{(1,2,2)/3}(xyz) \text{ at } (1, 1, 1)$$

329 rel-hard-der-008

R5 Boss • 95s • nabla / vector-calculus

$$\nabla \cdot \langle xy, yz, zx \rangle \text{ at } (1, 2, 3)$$

330 rel-hard-der-009

R5 Boss • 95s • nabla / vector-calculus

$$z\text{-component of } \nabla \times \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, 0 \right\rangle \text{ at } (1, 0, 0)$$

331 rel-hard-der-010

R5 Boss • 95s • nabla / laplacian

$$\Delta e^{x+y} \text{ at } (0, 0)$$

332 hc-der-003

R6 Boss+ • 135s • exponential / chain-rule

$$\frac{d}{dx} \left(e^{e^{2x} + x^2} \right)$$

333 hc-der-004

R6 Boss+ • 150s • higher-derivative / exponential

$$\frac{d^5}{dx^5} \left(x^4 e^{3x} \right) \Big|_{x=0}$$

334 hc-der-005

R6 Boss+ • 150s • higher-derivative / taylor

$$\frac{d^7}{dx^7} (x^2 \sin x) \Big|_{x=0}$$

335 hc-der-006

R6 Boss+ • 150s • higher-derivative / taylor

$$\frac{d^8}{dx^8} e^{x^2} \Big|_{x=0}$$

336 hc-der-007

R6 Boss+ • 150s • higher-derivative / taylor

$$\frac{d^{10}}{dx^{10}} \left(\frac{\sin x}{x} \right) \Big|_{x=0}$$

337 hc-der-008

R6 Boss+ • 150s • implicit-differentiation / higher-derivative

$$y = e^{xy}, \quad y(0) = 1. \text{ Find } y'''(0)$$

338 hc-der-009

R6 Boss+ • 150s • higher-derivative / trig

$$\frac{d^{12}}{dx^{12}} \cos(2x) \Big|_{x=0}$$

339 hc-der-010

R6 Boss+ • 150s • higher-derivative / exponential

$$\frac{d^9}{dx^9} (e^x \sin x) \Big|_{x=0}$$

340 exam-der-003

R5 Boss • 180s • quotient-rule / log

$$\frac{d}{dx} \left(\frac{\log x}{x} \right)$$

341 exam-der-004

R5 Boss • 180s • chain-rule / inverse-trig

$$\frac{d}{dx} \arctan\left(\frac{1+x}{1-x}\right)$$

342 exam-der-005

R5 Boss • 180s • product-rule / exponential

$$\frac{d}{dx} (x^2 e^{-x} \sin x)$$

343 exam-der-006

R5 Boss • 180s • log / radical

$$\frac{d}{dx} \log(\sqrt{1+x^2} + x)$$

344 exam-der-007

R5 Boss • 180s • higher-derivative / chain-rule

$$\frac{d^2}{dx^2} (e^{x^2})$$

345 exam-der-009

R5 Boss • 180s • quotient-rule / trig

$$\frac{d}{dx} \left(\frac{\cos x}{1 + \sin x} \right)$$

346 exam-der-010

R5 Boss • 180s • product-rule / log

$$\frac{d}{dx} (x \log(1+x^2))$$

347 exam-der-011

R5 Boss • 180s • quotient-rule / chain-rule

$$\frac{d}{dx} \left(\frac{(x^2+1)^3}{x-1} \right)$$

348 exam-der-012

R5 Boss • 180s • log-differentiation / exam-style

$$\text{若 } y = x^2 e^x, \text{ 求 } \frac{y'}{y}$$

349 exam-der-013

R5 Boss • 180s • parametric / exam-style

$$x = t^2 + 1, y = t^3 - t. \text{ 求 } \frac{dy}{dx} \Big|_{t=2}$$

350 exam-der-014

R5 Boss • 180s • fundamental-theorem / chain-rule

$$\frac{d}{dx} \left(\int_0^{x^2} e^{t^2} dt \right)$$

351 exam-der-015

R5 Boss • 180s • fundamental-theorem / chain-rule

$$\frac{d}{dx} \left(\int_x^{x^2} \log(1+t^2) dt \right)$$

352 exam-der-016

R5 Boss • 180s • inverse-function / exam-style

$$f(x) = x + e^x. \text{ 若 } f(0) = 1, \text{ 求 } (f^{-1})'(1)$$

353 exam-der-017

R5 Boss • 180s • chain-rule / inverse-trig

$$\frac{d}{dx} \arctan(\sqrt{x})$$

354 exam-der-018

R5 Boss • 180s • trig / log

$$\frac{d}{dx} \log(\sec x + \tan x)$$

355 exam-der-019

R5 Boss • 180s • quotient-rule / radical

$$\frac{d}{dx} \left(\frac{x}{\sqrt{1+x^2}} \right)$$

356 exam-der-020

R5 Boss • 180s • quotient-rule / exponential

$$\frac{d}{dx} \left(\frac{e^x}{1+e^x} \right)$$

357 exam-multi-002

R5 Boss • 180s • multivariable / higher-derivative

$$\frac{\partial^2}{\partial x \partial y} (x^2 y^3 + e^{xy})$$

358 exam-multi-003

R5 Boss • 180s • multivariable / gradient

若 $f(x, y) = x^2 + y^2$, 求 $|\nabla f(1, 2)|^2$

359 exam-multi-004

R5 Boss • 180s • multivariable / hessian

求 $f(x, y) = x^3 + y^3 - 3xy$ 在 $(1, 1)$ 的 Hessian determinant

360 exam-multi-005

R5 Boss • 180s • multivariable / directional-derivative

求 $f(x, y) = x^2 y$ 在 $(1, 2)$ 沿 $(3/5, 4/5)$ 的方向導數

361 exam-multi-008

R5 Boss • 180s • multivariable / jacobian

若 $u = x^2 - y^2$, $v = 2xy$, 求 $\frac{\partial(u, v)}{\partial(x, y)}$

362 exam-multi-009

R5 Boss • 180s • multivariable / nabra

$$\nabla \cdot (x^2 y, y^2 z, z^2 x)$$

363 exam-multi-010

R5 Boss • 180s • multivariable / nabra

求 $\nabla \times (y^2, z^2, x^2)$ 的 z -component

364 uni-der-001

R6 Boss+ • 210s • log-differentiation / trig

$$\frac{d}{dx} (x^{\cos x})$$

365 uni-der-002

R6 Boss+ • 210s • log-differentiation / trig

$$\frac{d}{dx} ((\sin x)^x)$$

366 uni-der-003

R6 Boss+ • 210s • higher-derivative / exponential

$$\frac{d^3}{dx^3} (e^{2x} \sin x) \Big|_{x=0}$$

367 uni-der-004

R6 Boss+ • 210s • higher-derivative / taylor

$$\frac{d^4}{dx^4} (x^2 e^x) \Big|_{x=0}$$

368 uni-der-005

R6 Boss+ • 210s • higher-derivative / log

$$\frac{d^2}{dx^2} \log(x + \sqrt{1+x^2})$$

369 uni-der-006

R6 Boss+ • 210s • fundamental-theorem / chain-rule

$$\frac{d}{dx} \left(\int_0^{\sin x} e^{t^2} dt \right)$$

370 uni-der-007

R6 Boss+ • 210s • fundamental-theorem / chain-rule

$$\frac{d}{dx} \left(\int_x^{2x} \log(1+t^2) dt \right)$$

371 uni-der-008

R6 Boss+ • 210s • inverse-trig / chain-rule

$$\frac{d}{dx} \arctan \left(\frac{2x}{1-x^2} \right)$$

求 $\nabla \times (xy, yz, zx)$ 的 z -component

372 uni-der-009

R6 Boss+ • 210s • parametric / exam-style

$$x = t + \frac{1}{t}, y = t - \frac{1}{t}. \text{ 求 } \left. \frac{dy}{dx} \right|_{t=2}$$

373 uni-der-010

R6 Boss+ • 210s • polar-curve / parametric

$$r = 1 + \cos \theta. \text{ 求切線斜率於 } \theta = \frac{\pi}{2}$$

374 uni-der-011

R6 Boss+ • 210s • implicit-differentiation / exam-style

$$x^2 + xy + y^2 = 3. \text{ 求 } \left. \frac{dy}{dx} \right|_{(1,1)}$$

375 uni-der-012

R6 Boss+ • 210s • multivariable / hessian

求 $f(x, y) = x^4 + y^4 - 4xy$ 在 $(1, 1)$ 的 Hessian determinant

376 uni-der-013

R6 Boss+ • 210s • lagrange-multiplier / exam-style

在 $x^2 + 4y^2 = 8$ 上求 xy 的最大值

377 uni-der-014

R6 Boss+ • 210s • multivariable / directional-derivative

求 $f(x, y) = e^{xy}$ 在 $(0, 2)$ 沿 $(3/5, 4/5)$ 的方向導數

378 uni-der-015

R6 Boss+ • 210s • multivariable / nabla

$$\nabla \cdot (xe^y, ye^z, ze^x)$$

379 uni-der-016

R6 Boss+ • 210s • multivariable / nabla

380 uni-der-017

R6 Boss+ • 210s • chain-rule / log

$$\frac{d}{dx} \log(\log(\log x))$$

381 uni-der-018

R6 Boss+ • 210s • chain-rule / log

$$\frac{d}{dx} \exp((\log x)^3)$$

382 uni-der-019

R6 Boss+ • 210s • higher-derivative / trig

$$\left. \frac{d^5}{dx^5} \sin(2x) \right|_{x=0}$$

383 uni-der-020

R6 Boss+ • 210s • log-differentiation / exam-style

$$\frac{d}{dx} ((1 + x^2)^x)$$

384 depth-der-001

R5 Boss • 220s • log-differentiation / power-exponential

$$\frac{d}{dx} (x^{x^2})$$

385 depth-der-002

R6 Boss+ • 270s • log-differentiation / trig

$$\frac{d}{dx} ((\sin x)^{\cos x})$$

386 depth-der-004

R5 Boss • 220s • higher-derivative / log

$$\frac{d^3}{dx^3} (x^2 \log x)$$

387 depth-der-005

R6 Boss+ • 270s • fundamental-theorem / chain-rule

$$\frac{d}{dx} \left(\int_{x^2}^{x^3} \cos(t^2) dt \right)$$

388 depth-der-006

R5 Boss • 220s • chain-rule / log

$$\frac{d}{dx} \log \left(x + \sqrt{x^2 + 4} \right)$$

389 depth-der-007

R5 Boss • 220s • implicit-differentiation / exam-style

$$x^3 + y^3 = 6xy, \quad \frac{dy}{dx} \Big|_{(3,3)}$$

390 depth-der-008

R5 Boss • 220s • parametric / exam-style

$$x = e^t, \quad y = te^t. \quad \frac{dy}{dx} \Big|_{t=0}$$

391 depth-der-009

R6 Boss+ • 270s • parametric / higher-derivative

$$x = t^2, \quad y = t^3. \quad \frac{d^2y}{dx^2} \Big|_{t=1}$$

392 depth-der-010

R6 Boss+ • 270s • inverse-function / higher-derivative

$$f(x) = e^x + x^2, \quad f(0) = 1. \text{ find } (f^{-1})''(1)$$

393 depth-der-011

R5 Boss • 220s • multivariable / hessian

$$\text{Hessian determinant of } f(x, y) = x^2y + y^3 \text{ at } (1, 1)$$

394 depth-der-012

R6 Boss+ • 270s • multivariable / nabla

$$\Delta \log(x^2 + y^2)$$

395 depth-der-013

R5 Boss • 220s • multivariable / nabla

$$\nabla \cdot (xyz, x^2z, xy^2)$$

396 depth-der-014

R5 Boss • 220s • multivariable / nabla

$$\text{z-component of } \nabla \times (x^2y, xy^2, z)$$

397 depth-der-015

R6 Boss+ • 270s • inverse-trig / chain-rule

$$\frac{d}{dx} \arccos \left(\frac{1 - x^2}{1 + x^2} \right)$$

398 depth-der-016

R5 Boss • 220s • quotient-rule / log

$$\frac{d}{dx} \left(\frac{(\log x)^2}{x} \right)$$

399 depth-der-017

R5 Boss • 220s • higher-derivative / trig

$$\frac{d^4}{dx^4} \cos(3x) \Big|_{x=0}$$

400 depth-der-018

R5 Boss • 220s • chain-rule / radical

$$\frac{d}{dx} \sqrt{x + \sqrt{x}}$$

401 burst-der-001

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{20}}{dx^{20}} e^{2x} \Big|_{x=0}$$

402 burst-der-002

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{15}}{dx^{15}} \sin(3x) \Big|_{x=0}$$

403 burst-der-003

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{18}}{dx^{18}} \cosh(2x) \Big|_{x=0}$$

404 burst-der-004

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{10}}{dx^{10}} (x^3 e^x) \Big|_{x=0}$$

405 burst-der-005

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{11}}{dx^{11}} (x^2 \sin(2x)) \Big|_{x=0}$$

406 burst-der-006

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^8}{dx^8} \log(1+x) \Big|_{x=0}$$

407 burst-der-007

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{12}}{dx^{12}} \frac{1}{1-x} \Big|_{x=0}$$

408 burst-der-008

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{12}}{dx^{12}} (e^x \cos x) \Big|_{x=0}$$

409 burst-der-009

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^{14}}{dx^{14}} (x^4 e^{-x}) \Big|_{x=0}$$

410 burst-der-010

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^9}{dx^9} (e^{2x} \sin x) \Big|_{x=0}$$

411 burst-boss-der-001

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{12}}{dx^{12}} (\log(1+x))^2 \Big|_{x=0}$$

412 burst-boss-der-002

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{11}}{dx^{11}} (\log(1+x))^2 \Big|_{x=0}$$

413 burst-boss-der-003

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}} (e^{x^2} \cos x) \Big|_{x=0}$$

414 burst-boss-der-004

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{14}}{dx^{14}} (e^x \sin(2x)) \Big|_{x=0}$$

415 burst-boss-der-005

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}} (e^x \cos(2x)) \Big|_{x=0}$$

416 burst-boss-der-006

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}} (x^4 e^{3x}) \Big|_{x=0}$$

417 burst-boss-der-007

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}}(x^3 \sin(2x)) \Big|_{x=0}$$

418 burst-boss-der-008

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}(x^2 \cos(3x)) \Big|_{x=0}$$

419 burst-boss-der-009

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}} \frac{x^2}{(1-x)^4} \Big|_{x=0}$$

420 burst-boss-der-010

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{15}}{dx^{15}} \frac{x^3}{1+x^2} \Big|_{x=0}$$

421 burst-boss2-der-001

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{21}}{dx^{21}}(e^x \sin x) \Big|_{x=0}$$

422 burst-boss2-der-002

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}}(e^x \cos x) \Big|_{x=0}$$

423 burst-boss2-der-003

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{17}}{dx^{17}}(x^5 e^{2x}) \Big|_{x=0}$$

424 burst-boss2-der-004

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}(x^6 \cos(2x)) \Big|_{x=0}$$

425 burst-boss2-der-005

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{19}}{dx^{19}}(x^4 \sin(3x)) \Big|_{x=0}$$

426 burst-boss2-der-006

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}} \frac{1}{(1-x)^3} \Big|_{x=0}$$

427 burst-boss2-der-007

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}} \frac{x^2}{(1-x)^5} \Big|_{x=0}$$

428 burst-boss2-der-008

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}} \frac{x^4}{(1+x)^2} \Big|_{x=0}$$

429 burst-boss3-der-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}}(\log(1+x))^2 \Big|_{x=0}$$

430 burst-boss3-der-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}} \frac{x^4}{(1-x)^7} \Big|_{x=0}$$

431 burst-boss3-der-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}} \frac{x^5}{(1-x)^6} \Big|_{x=0}$$

432 burst-boss3-der-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}} \frac{x^3}{(1-x)^8} \Big|_{x=0}$$

433 burst-boss3-der-005

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{31}}{dx^{31}} \left(x^7 e^{-x^2} \right) \Big|_{x=0}$$

434 burst-boss3-der-006

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{30}}{dx^{30}} \left(x^6 \cosh(x^2) \right) \Big|_{x=0}$$

435 burst-boss3-der-007

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{27}}{dx^{27}} \left(x^5 \sin(x^2) \right) \Big|_{x=0}$$

436 burst-boss3-der-008

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{29}}{dx^{29}} \left(x^5 \cos(x^2) \right) \Big|_{x=0}$$

437 world-004

R1 Warm-up • 35s • inverse-trig / world-universities

$$\frac{d}{dx} \arctan x$$

438 world-024

R4 Advanced • 150s • higher-derivative / complex

$$\frac{d^4}{dx^4} \left(e^x \cos x \right) \Big|_{x=0}$$

439 world-025

R4 Advanced • 150s • taylor / coefficient

Coefficient of x^5 in $\tan x$

440 world-036

R5 Boss • 240s • taylor / coefficient

Coefficient of x^6 in $\sec x$

441 world-068

R6 Boss+ • 340s • taylor / coefficient

Coefficient of x^4 in $(1+x)^x$

442 world-093

R6 Boss+ • 340s • taylor / coefficient

Coefficient of x^4 in $e^{\sin x}$

443 hd-001

R5 Boss • 240s • higher-derivative / Cambridge

$$\frac{d^7}{dx^7} \frac{1}{1-x} \Big|_{x=0}$$

444 hd-002

R4 Advanced • 150s • higher-derivative / taylor

$$\frac{d^4}{dx^4} e^{x^2} \Big|_{x=0}$$

445 hd-003

R5 Boss • 240s • higher-derivative / taylor

$$\frac{d^6}{dx^6} e^{x^2} \Big|_{x=0}$$

446 hd-004

R5 Boss • 240s • higher-derivative / taylor

$$\frac{d^{10}}{dx^{10}} (x \sin x) \Big|_{x=0}$$

447 hd-005

R5 Boss • 240s • higher-derivative / taylor

$$\frac{d^4}{dx^4} \frac{1}{1+x^2} \Big|_{x=0}$$

448 hd-006

R6 Boss+ • 340s • higher-derivative / complex

$$\frac{d^5}{dx^5} \left(e^x \sin x \right) \Big|_{x=0}$$

449 hd-007

R6 Boss+ • 340s • higher-derivative / taylor

$$\left. \frac{d^7}{dx^7} \arctan x \right|_{x=0}$$

450 hd-008

R5 Boss • 240s • taylor / coefficient

Coefficient of x^7 in $\tan x$ **451 hd-009**

R5 Boss • 240s • inverse-trig / chain-rule

$$\frac{d}{dx} (\arcsin x)^2$$

452 hd-010

R6 Boss+ • 340s • log-differentiation / Caltech

$$\frac{d}{dx} x^{x^2}$$

453 hd-011

R5 Boss • 240s • trig / chain-rule

$$\frac{d}{dx} \log \tan \frac{x}{2}$$

454 hd-012

R5 Boss • 240s • taylor / coefficient

Coefficient of x^4 in $\tan^2 x$ **455 app-001**

R4 Advanced • 150s • applications / related-rates

球形氣球以 $100 \frac{cm^3}{s}$ 充氣， $r = 5$ 時半徑變化率 $\frac{dr}{dt}$ **456 app-002**

R5 Boss • 240s • applications / optimization

橢圓 $\frac{x^2}{9} + \frac{y^2}{4} = 1$ 內接矩形的最大面積**457 app-003**

R4 Advanced • 150s • applications / curvature

曲線 $y = x^2$ 在原點的曲率半徑**458 app-004**

R5 Boss • 240s • applications / curvature

曲線 $y = e^x$ 在 $x = 0$ 的曲率半徑**459 app-005**

R4 Advanced • 150s • applications / newton-method

對 $\sqrt{2}$ 用牛頓法， $x_0 = 1$ 的一次迭代 x_1 **460 app-006**

R4 Advanced • 150s • applications / linear-approximation

用線性近似估 $\sqrt{4.1}$ **461 app-007**

R4 Advanced • 150s • applications / optimization

$$\min_{x>0} \left(x + \frac{4}{x} \right)$$

462 app-008

R5 Boss • 240s • applications / optimization

原點到直線 $3x + 4y = 10$ 的最短距離**463 app-009**

R5 Boss • 240s • applications / optimization

半徑1 球內接圓柱的最大體積

464 app-010

R6 Boss+ • 340s • applications / optimization

半徑1 球內接圓錐的最大體積

465 app-011

R5 Boss • 240s • applications / related-rates

13 尺梯子下端以 $5 \frac{ft}{s}$ 滑離牆，距牆12 尺時頂端下滑速率

466 app-012

R6 Boss+ • 340s • applications / curvature

曲線 $y = \log x$ 的最大曲率**積分** Integrals

488 題

467 int-001

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\int 6x^2 dx$$

468 int-002

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\int \cos x dx$$

469 int-003

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\int \frac{1}{x} dx$$

470 int-004

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\int e^{3x} dx$$

471 int-005

R2 Basic • 45s • rank-2 / beginner-friendly

$$\int xe^{x^2} dx$$

472 int-006

R2 Basic • 40s • rank-2 / beginner-friendly

$$\int \frac{2x}{1+x^2} dx$$

473 int-007

R2 Basic • 45s • rank-2 / beginner-friendly

$$\int x \cos(x^2) dx$$

474 int-008

R2 Basic • 35s • rank-2 / beginner-friendly

$$\int_0^1 3x^2 dx$$

475 int-009

R3 Standard • 65s • rank-3

$$\int x \ln x dx$$

476 int-010

R3 Standard • 70s • rank-3

$$\int \sin^3 x dx$$

477 int-011

R3 Standard • 60s • rank-3

$$\int \frac{1}{x^2+4} dx$$

478 int-012

R2 Basic • 70s • rank-2 / beginner-friendly

$$\int_0^{\pi/2} \sin^2 x dx$$

479 int-013

R3 Standard • 100s • rank-3

$$\int (\ln x)^2 dx$$

480 int-014

R3 Standard • 85s • rank-3

$$\int \frac{x^3}{x^2+1} dx$$

481 int-015

R4 Advanced • 95s • rank-4

$$\int e^x \sin x dx$$

482 int-016

R3 Standard • 85s • rank-3

$$\int_0^1 \ln(1+x) dx$$

483 int-017

R3 Standard • 80s • rank-3

$$\int_0^{\pi/2} x \sin x dx$$

484 int-018

R3 Standard • 70s • rank-3

$$\int \frac{\sin x}{1 + \cos x} dx$$

485 int-019

R4 Advanced • 100s • rank-4

$$\int \frac{1}{1 + \sqrt{x}} dx$$

486 int-020

R3 Standard • 70s • rank-3

$$\int_0^1 \frac{x^2}{1 + x^3} dx$$

487 td-int-001

R6 Boss+ • 95s • rank-6 / boss-rank

$$\int_0^\infty \frac{\cos(3x)}{1 + x^2} dx$$

488 td-int-002

R6 Boss+ • 100s • rank-6 / boss-rank

$$\int_0^\pi \log(5 + 3 \cos x) dx$$

489 td-int-003

R4 Advanced • 85s • rank-4

$$\int_0^\infty \frac{e^{-5x} - e^{-2x}}{x} dx$$

490 td-int-004

R4 Advanced • 100s • rank-4

$$\int_0^\infty \frac{\sin(2x) \sin(5x)}{x} dx$$

491 td-int-005

R6 Boss+ • 80s • rank-6 / boss-rank

$$\text{Area of } 3x^2 + 2xy + 2y^2 \leq 1$$

492 td-int-006

R6 Boss+ • 90s • rank-6 / boss-rank

$$\int_{-\infty}^\infty x^4 e^{-3x^2} dx$$

493 int-021

R4 Advanced • 60s • multivariable / double-integral

$$\int_0^1 \int_0^1 (x + y) dy dx$$

494 int-022

R4 Advanced • 55s • multivariable / area

$$\iint_D 1 dA, \quad D = \{x \geq 0, y \geq 0, x + y \leq 2\}$$

495 int-023

R4 Advanced • 65s • multivariable / double-integral

$$\int_0^1 \int_0^x xy dy dx$$

496 int-024

R4 Advanced • 70s • multivariable / double-integral

$$\int_0^1 \int_0^{1-x} (x + y) dy dx$$

497 int-025

R4 Advanced • 75s • multivariable / polar-coordinates

$$\iint_{x^2 + y^2 \leq 1} (x^2 + y^2) dA$$

498 int-026

R4 Advanced • 55s • multivariable / double-integral

$$\int_0^2 \int_0^3 xy dy dx$$

499 int-027

R2 Basic • 55s • substitution / log

$$\int \frac{x}{1 + x^2} dx$$

500 int-028

R3 Standard • 60s • integration-by-parts / log

$$\int \log x dx$$

501 int-029

R2 Basic • 55s • inverse-trig / standard-integral

$$\int \frac{1}{x^2 + 9} dx$$

502 int-030

R2 Basic • 55s • substitution / definite-integral

$$\int_0^{\pi/2} \sin^3 x \cos x dx$$

503 int-031

R4 Advanced • 70s • multivariable / double-integral

$$\int_0^1 \int_y^1 x^2 dx dy$$

504 int-032

R4 Advanced • 60s • multivariable / polar-coordinates

$$\iint_{1 \leq x^2 + y^2 \leq 4} 1 \, dA$$

505 int-033

R4 Advanced • 85s • multivariable / polar-coordinates

$$\iint_{x^2 + y^2 \leq 4} x^2 \, dA$$

506 int-034

R4 Advanced • 60s • multivariable / area

$$\int_0^1 \int_0^{\sqrt{1-x^2}} 1 \, dy \, dx$$

507 int-035

R4 Advanced • 60s • multivariable / polar-coordinates

$$\int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta$$

508 int-036

R4 Advanced • 65s • multivariable / double-integral

$$\int_0^1 \int_0^{1-y} x \, dx \, dy$$

509 int-038

R3 Standard • 65s • integration-by-parts / rank-3

$$\int x \cos x \, dx$$

510 int-039

R3 Standard • 50s • inverse-trig / definite-integral

$$\int_0^1 \frac{1}{1+x^2} \, dx$$

511 int-040

R5 Boss • 70s • improper-integral / gamma

$$\int_0^\infty x e^{-x} \, dx$$

512 int-041

R3 Standard • 70s • substitution / u-sub

$$\int \frac{x^3}{1+x^4} \, dx$$

513 int-042

R3 Standard • 75s • substitution / u-sub

$$\int x^2 \sqrt{1+x^3} \, dx$$

514 int-043

R4 Advanced • 95s • substitution / u-sub

$$\int \frac{x}{\sqrt{1+x^2} (1+\sqrt{1+x^2})} \, dx$$

515 int-044

R3 Standard • 85s • substitution / u-sub

$$\int \frac{x e^{x^2}}{1+e^{x^2}} \, dx$$

516 int-045

R3 Standard • 70s • substitution / u-sub

$$\int \frac{(\log x)^2}{x} \, dx$$

517 int-046

R4 Advanced • 80s • substitution / u-sub

$$\int_0^1 \frac{2x}{(1+x^2)^3} \, dx$$

518 int-047

R3 Standard • 75s • trig-substitution / definite-integral

$$\int_0^2 \sqrt{4-x^2} \, dx$$

519 int-048

R4 Advanced • 90s • trig-substitution / definite-integral

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}} \, dx$$

520 int-049

R4 Advanced • 95s • trig-substitution / definite-integral

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} \, dx$$

521 int-050

R4 Advanced • 95s • trig-substitution / inverse-trig

$$\int_1^2 \frac{1}{x\sqrt{x^2-1}} \, dx$$

522 int-051

R4 Advanced • 95s • trig-substitution / radical

$$\int \frac{1}{(x^2+4)^{3/2}} \, dx$$

523 int-052

R3 Standard • 90s • integration-by-parts / ibp

$$\int x^2 e^x \, dx$$

524 int-053

R4 Advanced • 100s • integration-by-parts / ibp

$$\int x^2 \sin x \, dx$$

525 int-055

R4 Advanced • 100s • integration-by-parts / ibp

$$\int (\log x)^2 \, dx$$

526 int-056

R4 Advanced • 100s • integration-by-parts / ibp

$$\int_0^1 x \arctan x \, dx$$

527 int-057

R5 Boss • 110s • ode-style / convolution

Find $F(x) = \int_0^x e^{x-t} \sin t \, dt$

528 int-058

R5 Boss • 105s • ode-style / convolution

Find $F(x) = \int_0^x e^{2(x-t)} t \, dt$

529 int-059

R5 Boss • 110s • ode-style / convolution

Find $F(x) = \int_0^x e^{-(x-t)} t^2 \, dt$

530 int-060

R5 Boss • 110s • ode-style / convolution

Find $F(x) = \int_0^x \sin(x-t) \cos t \, dt$

531 int-061

R5 Boss • 85s • kings-property / symmetry

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} \, dx$$

532 int-062

R5 Boss • 80s • kings-property / symmetry

$$\int_0^1 \frac{e^x}{e^x + e^{1-x}} \, dx$$

533 int-063

R5 Boss • 90s • kings-property / symmetry

$$\int_0^{\pi/2} \frac{1 + \sin x}{2 + \sin x + \cos x} \, dx$$

534 int-064

R5 Boss • 110s • kings-property / symmetry

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$$

535 int-065

R5 Boss • 95s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-3x} - e^{-7x}}{x} \, dx$$

536 int-066

R5 Boss • 90s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-x} - e^{-4x}}{x} \, dx$$

537 int-067

R5 Boss • 90s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-2x} - e^{-8x}}{x} \, dx$$

538 int-068

R5 Boss • 100s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos(2x) - \cos(5x)}{x} \, dx$$

539 int-069

R5 Boss • 95s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos x - \cos(4x)}{x} \, dx$$

540 int-070

R5 Boss • 95s • frullani / cosine-integral

$$\int_0^\infty \frac{\cos(3x) - \cos(9x)}{x} dx$$

541 int-071

R5 Boss • 110s • frullani / trig

$$\int_0^\infty \frac{\sin(3x) \sin(8x)}{x} dx$$

542 int-072

R5 Boss • 105s • frullani / trig

$$\int_0^\infty \frac{\sin(2x) \sin(6x)}{x} dx$$

543 int-073

R5 Boss • 105s • frullani / trig

$$\int_0^\infty \frac{\sin x \sin(4x)}{x} dx$$

544 int-074

R5 Boss • 115s • frullani / inverse-trig

$$\int_0^\infty \frac{\arctan(5x) - \arctan(2x)}{x} dx$$

545 int-075

R5 Boss • 110s • frullani / rational

$$\int_0^\infty \frac{1}{x} \left(\frac{1}{1+2x} - \frac{1}{1+5x} \right) dx$$

546 int-076

R5 Boss • 105s • frullani / rational

$$\int_0^\infty \frac{1}{x} \left(\frac{1}{1+x} - \frac{1}{1+6x} \right) dx$$

547 gap-int-pf-001

R3 Standard • 85s • partial-fraction / rational-integral

$$\int \frac{1}{x^2 - 1} dx$$

548 gap-int-pf-002

R3 Standard • 60s • partial-fraction / rational-integral

$$\int \frac{x+1}{x^2+x} dx$$

549 gap-int-pf-003

R3 Standard • 75s • partial-fraction / rational-integral

$$\int \frac{1}{x(x+2)} dx$$

550 gap-int-pf-004

R3 Standard • 75s • partial-fraction / rational-integral

$$\int \frac{2x+3}{x^2+3x+2} dx$$

551 gap-int-pf-005

R3 Standard • 55s • partial-fraction / repeated-factor

$$\int \frac{1}{(x+1)^2} dx$$

552 gap-int-pf-006

R4 Advanced • 95s • partial-fraction / rational-integral

$$\int \frac{3x+5}{x^2+x-2} dx$$

553 gap-int-pf-007

R4 Advanced • 90s • partial-fraction / rational-integral

$$\int \frac{x+4}{x^2+5x+6} dx$$

554 gap-int-polar-001

R2 Basic • 60s • polar-curve / polar-area

Area enclosed by $r = 2$ from $0 \leq \theta \leq \pi/2$ **555 gap-int-polar-002**

R3 Standard • 80s • polar-curve / polar-area

Area enclosed by $r = \sin \theta$ for $0 \leq \theta \leq \pi$ **556 gap-int-triple-001**

R6 Boss+ • 75s • triple-integral / iterated-integral

$$\int_0^1 \int_0^1 \int_0^1 xyz \, dz \, dy \, dx$$

557 gap-int-triple-002

R6 Boss+ • 75s • triple-integral / iterated-integral

$$\int_0^1 \int_0^1 \int_0^1 (x+y+z) dz dy dx$$

558 gap-int-triple-003

R6 Boss+ • 80s • triple-integral / volume

$$\iiint_{x+y+z \leq 1, x, y, z \geq 0} 1 dV$$

559 gap-int-triple-004

R6 Boss+ • 80s • triple-integral / iterated-integral

$$\int_0^1 \int_0^x \int_0^y 1 dz dy dx$$

560 gap-int-triple-005

R6 Boss+ • 70s • triple-integral / cylindrical-coordinates

Volume of the cylinder $x^2 + y^2 \leq 4$, $0 \leq z \leq 3$

561 gap-int-triple-006

R6 Boss+ • 70s • triple-integral / spherical-coordinates

Volume of the ball $x^2 + y^2 + z^2 \leq 4$

562 gap-int-cov-001

R6 Boss+ • 75s • change-of-variables / jacobian

$$u = x + y, v = x - y. \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = ?$$

563 gap-int-cov-002

R6 Boss+ • 80s • change-of-variables / jacobian

Area of $0 \leq x + y \leq 1$, $0 \leq x - y \leq 2$

564 gap-tech-003

R2 Basic • 45s • technique-recognition / substitution

Technique for $\int x e^{x^2} dx$

565 gap-tech-004

R3 Standard • 45s • technique-recognition / integration-by-parts

Technique for $\int x \log x dx$

566 gap-tech-006

R3 Standard • 45s • technique-recognition / partial-fraction

Technique for $\int \frac{1}{x^2 - 1} dx$

567 mob-tech-003

R2 Basic • 45s • technique-sprint / technique-recognition

$\int x e^{x^2} dx$ Use which technique?

568 mob-tech-004

R2 Basic • 45s • technique-sprint / technique-recognition

$\int x \log x dx$ Use which technique?

569 mob-tech-005

R2 Basic • 45s • technique-sprint / technique-recognition

$\int \frac{dx}{x^2 - 1}$ Use which technique?

570 mob-tech-006

R2 Basic • 45s • technique-sprint / technique-recognition

$\int \frac{dx}{\sqrt{a^2 - x^2}}$ Use which technique?

571 mob-tech-010

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^1 f(x) dx$ with $f(x) + f(1-x)$ Use which technique?

572 mob-tech-011

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx$ Use which technique?

573 mob-tech-013

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_C \nabla \phi \cdot d\mathbf{r}$ Use which technique?

574 mob-tech-015

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^1 x^{a-1} (1-x)^{b-1} dx$ Use which technique?

575 mob-tech-016

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^\infty x^{s-1} e^{-x} dx$ Use which technique?

576 mob-tech-017

R2 Basic • 45s • technique-sprint / technique-recognition

$$\int_0^{\pi/2} \sin^n x \, dx \quad \text{Use which technique?}$$

577 mob-trap-002

R2 Basic • 45s • trap-drill / rank-2

$$\int \cos(3x) \, dx \quad \text{Main trap?}$$

578 mob-trap-003

R2 Basic • 45s • trap-drill / rank-2

$$\int \frac{1}{x^2 + 1} \, dx \quad \text{Main trap?}$$

579 mob-trap-006

R2 Basic • 45s • trap-drill / rank-2

$$\int_0^\infty e^{-ax} \, dx \quad \text{Main trap?}$$

580 mob-trap-008

R2 Basic • 45s • trap-drill / rank-2

$$\iint_R f(x, y) \, dA \quad \text{under change of variables} \quad \text{Main trap?}$$

581 mob-trap-010

R2 Basic • 45s • trap-drill / rank-2

$$\int \sec x \tan x \, dx \quad \text{Main trap?}$$

582 mob-trap-012

R2 Basic • 45s • trap-drill / rank-2

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \text{Main trap?}$$

583 mob-beta-001

R5 Boss • 70s • beta-function / special-functions

$$B(1, 1)$$

584 mob-beta-002

R5 Boss • 70s • beta-function / special-functions

$$B(2, 1)$$

585 mob-beta-003

R5 Boss • 70s • beta-function / special-functions

$$B(2, 2)$$

586 mob-beta-004

R5 Boss • 70s • beta-function / special-functions

$$B(3, 2)$$

587 mob-beta-005

R5 Boss • 70s • beta-function / special-functions

$$B(1/2, 1/2)$$

588 mob-beta-006

R5 Boss • 70s • beta-function / special-functions

$$B(1/2, 1)$$

589 mob-beta-007

R5 Boss • 70s • beta-function / special-functions

$$B(3/2, 1/2)$$

590 mob-beta-009

R5 Boss • 70s • beta-function / special-functions

$$B(3, 3)$$

591 mob-beta-010

R5 Boss • 70s • beta-function / special-functions

$$B(4, 1)$$

592 mob-gamma-001

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(1)$$

593 mob-gamma-002

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(1/2)$$

594 mob-gamma-003

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(3/2)$$

595 mob-gamma-004

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(5/2)$$

596 mob-gamma-005

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(6)$$

597 mob-gamma-006

R5 Boss • 70s • gamma-function / special-functions

$$\int_0^{\infty} x^2 e^{-x} dx$$

598 mob-gamma-007

R5 Boss • 70s • gamma-function / special-functions

$$\int_0^{\infty} x^{1/2} e^{-x} dx$$

599 mob-gamma-008

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(4)$$

600 mob-gamma-009

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(7/2)$$

601 mob-gamma-010

R5 Boss • 70s • gamma-function / special-functions

$$\Gamma(0+1)$$

602 mob-wallis-002

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^3 x dx$$

603 mob-wallis-003

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^4 x dx$$

604 mob-wallis-004

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^5 x dx$$

605 mob-wallis-005

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos^2 x dx$$

606 mob-wallis-006

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos^4 x dx$$

607 mob-wallis-007

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^2 x \cos^2 x dx$$

608 mob-wallis-008

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin x dx$$

609 mob-wallis-009

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos x \, dx$$

610 mob-wallis-010

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin x \cos x \, dx$$

611 rel-basic-013

R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^1 x^2 \, dx$$

612 rel-basic-014

R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^{\pi} \sin x \, dx$$

613 rel-basic-015

R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^1 e^{2x} \, dx$$

614 rel-basic-016

R1 Warm-up • 45s • basic-integral / log

$$\int_0^1 \frac{1}{1+x} \, dx$$

615 rel-basic-018

R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^2 3x \, dx$$

616 rel-adv-011

R5 Boss • 75s • improper-integral / gamma-function

$$\int_0^{\infty} x e^{-3x} \, dx$$

617 rel-adv-014

R4 Advanced • 75s • double-integral / iterated-integral

$$\int_0^1 \int_0^1 (x+y)^2 \, dy \, dx$$

618 rel-adv-009

R3 Standard • 80s • substitution / u-sub

$$\int x^3 e^{x^2} \, dx$$

619 rel-adv-010

R3 Standard • 80s • substitution / log

$$\int x \log(1+x^2) \, dx$$

620 rel-boss-001

R5 Boss • 110s • improper-integral / dirichlet-integral

$$\int_0^{\infty} \frac{\sin(3x)}{x} \, dx$$

621 rel-boss-002

R5 Boss • 110s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-2x} - e^{-5x}}{x} \, dx$$

622 rel-boss-003

R5 Boss • 110s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos(2x) - \cos(7x)}{x} \, dx$$

623 rel-boss-004

R5 Boss • 110s • gamma-function / special-functions

$$\int_0^{\infty} x^{5/2} e^{-x} \, dx$$

624 rel-boss-005

R5 Boss • 110s • beta-function / special-functions

$$\int_0^1 x^{-1/2} (1-x)^{3/2} \, dx$$

625 rel-boss-008

R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\sin z}{z^5}$$

626 rel-boss-009

R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^z}{z^4}$$

627 rel-boss-010

R6 Boss+ • 110s • change-of-variables / jacobian

$$\iint_T \frac{xy}{x+y} dA, \quad T = \{x, y > 0, x + y < 1\}$$

628 rel-boss-011

R6 Boss+ • 110s • triple-integral / simplex

$$\iiint_{x,y,z \geq 0, x+y+z \leq 1} xyz \, dV$$

629 rel-boss-012

R6 Boss+ • 110s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2 \leq 4} (x^2 + y^2 + z^2) \, dV$$

630 rel-boss-017

R5 Boss • 110s • improper-integral / gamma-function

$$\int_0^1 (\log x)^2 \, dx$$

631 rel-boss-020

R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1 - \cos z}{z^3}$$

632 rel-hard-int-001

R5 Boss • 125s • gamma-function / improper-integral

$$\int_0^\infty x^5 e^{-2x} \, dx$$

633 rel-hard-int-002

R5 Boss • 125s • gamma-function / improper-integral

$$\int_0^\infty x^{3/2} e^{-4x} \, dx$$

634 rel-hard-int-004

R5 Boss • 125s • parameter-integral / gamma-function

$$\int_0^1 x^3 (\log x)^2 \, dx$$

635 rel-hard-int-005

R5 Boss • 125s • wallis / special-functions

$$\int_0^{\pi/2} \sin^8 x \, dx$$

636 rel-hard-int-007

R5 Boss • 125s • beta-function / wallis

$$\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$$

637 rel-hard-int-008

R5 Boss • 125s • frullani / improper-integral

$$\int_0^\infty \frac{e^{-3x} - e^{-11x}}{x} \, dx$$

638 rel-hard-int-009

R5 Boss • 125s • frullani / cosine-integral

$$\int_0^\infty \frac{\cos(4x) - \cos(10x)}{x} \, dx$$

639 rel-hard-int-011

R5 Boss • 125s • improper-integral / substitution

$$\int_0^\infty \frac{x}{x^4 + 1} \, dx$$

640 rel-hard-int-012

R5 Boss • 125s • improper-integral / residue

$$\int_0^\infty \frac{dx}{x^4 + 1}$$

641 rel-hard-int-015

R5 Boss • 125s • parameter-integral / improper-integral

$$\int_0^1 x^{-1/2} \log x \, dx$$

642 rel-hard-int-017

R5 Boss • 125s • kings-property / improper-integral

$$\int_0^{\pi/2} (\log(\sin x) + \log(\cos x)) \, dx$$

643 rel-hard-int-018

R5 Boss • 125s • improper-integral / trig-substitution

$$\int_0^\infty \frac{dx}{(x^2 + 4)^2}$$

644 rel-hard-cpx-001

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\sin z}{z^4}$$

645 rel-hard-cpx-002

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\cos z}{z^3}$$

646 rel-hard-cpx-003

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^z - 1 - z}{z^4}$$

647 rel-hard-cpx-004

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\log(1+z)}{z^3}$$

648 rel-hard-cpx-005

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1}{z^2(1-z)}$$

649 rel-hard-cpx-006

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=1} \frac{e^z}{(z-1)^2}$$

650 rel-hard-cpx-007

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{z}{\sin^2 z}$$

651 rel-hard-cpx-008

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \cot z$$

652 rel-hard-cpx-009

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^{2z} - 1}{z^3}$$

653 rel-hard-cpx-010

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1 - \cos z - z^2/2}{z^5}$$

654 rel-hard-mv-001

R5 Boss • 120s • double-integral / simplex

$$\iint_{x,y \geq 0, x+y \leq 1} x^2 y \, dA$$

655 rel-hard-mv-002

R5 Boss • 120s • double-integral / polar

$$\iint_{x^2+y^2 \leq 1} x^2 y^2 \, dA$$

656 rel-hard-mv-003

R6 Boss+ • 120s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2 \leq 1} (x^2 + y^2 + z^2) dV$$

657 rel-hard-mv-004

R6 Boss+ • 120s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2 \leq 9} 1 dV$$

658 rel-hard-mv-008

R6 Boss+ • 120s • change-of-variables / jacobian

$$\iint_R (x+y) dA, \quad 0 \leq x+y \leq 1, \quad 0 \leq x-y \leq 2$$

659 rel-hard-mv-009

R5 Boss • 120s • double-integral / polar

$$\iint_{x^2+y^2 \leq 4} (x^2 + y^2)^2 dA$$

660 rel-hard-mv-010

R6 Boss+ • 120s • triple-integral / simplex

$$\iiint_{x,y,z \geq 0, x+y+z \leq 1} x^2 y z dV$$

661 hc-usub-001

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^3}{\sqrt{1+x^2}} dx$$

662 hc-usub-002

R6 Boss+ • 160s • substitution / u-sub

$$\int x^5 e^{x^2} dx$$

663 hc-usub-003

R6 Boss+ • 160s • substitution / u-sub

$$\int x (\log(1+x^2))^2 dx$$

664 hc-usub-004

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^2}{1+x^6} dx$$

665 hc-usub-005

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^3}{\sqrt{1+x^4} (1+\sqrt{1+x^4})} dx$$

666 hc-usub-006

R6 Boss+ • 160s • substitution / u-sub

$$\int e^x \sqrt{1+e^x} dx$$

667 hc-usub-007

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^2}{\sqrt{1-x^6}} dx$$

668 hc-usub-008

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^3}{(1+x^2)^3} dx$$

669 hc-usub-009

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^{\pi/2} \frac{\sin x \cos x}{(1+\sin^2 x)^2} dx$$

670 hc-usub-010

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^2 e^{x^3}}{1+e^{x^3}} dx$$

671 hc-ibp-002

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^2 \cos(3x) dx$$

672 hc-ibp-003

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^2 e^x \sin x dx$$

673 hc-ibp-005

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x(\arctan x)^2 dx$$

674 hc-ibp-006

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int e^{2x} \sin(3x) dx$$

675 hc-ibp-007

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^3 \log x dx$$

676 hc-ibp-008

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^1 x^4 \log(1+x) dx$$

677 hc-ibp-009

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^{\pi/2} x^2 \cos x dx$$

678 hc-ibp-010

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^1 x^3 \arctan x dx$$

679 hc-rad-001

R6 Boss+ • 160s • trig-substitution / radical

$$\int \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

680 hc-rad-002

R6 Boss+ • 160s • complete-square / inverse-trig

$$\int \frac{dx}{x^2 + 2x + 5}$$

681 hc-rad-003

R6 Boss+ • 160s • partial-fraction / rational

$$\int \frac{x^2 + 1}{x^3 - x} dx$$

682 hc-rad-004

R6 Boss+ • 160s • partial-fraction / rational

$$\int \frac{dx}{x^4 - 1}$$

683 hc-rad-005

R6 Boss+ • 150s • trig-substitution / radical

$$\int_2^4 \frac{\sqrt{x^2 - 4}}{x} dx$$

684 hc-rad-006

R6 Boss+ • 150s • trig / rationalize

$$\int_0^{\pi/2} \frac{dx}{1 + \sin x}$$

685 hc-rad-007

R6 Boss+ • 150s • trig / beta-function

$$\int_0^{\pi/2} \sin^5 x \cos^4 x dx$$

686 hc-rad-008

R6 Boss+ • 150s • improper-integral / rational

$$\int_0^{\infty} \frac{dx}{1+x^6}$$

687 hc-rad-010

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x}{\sqrt{1-x^4}} dx$$

688 hc-spec-002

R6 Boss+ • 160s • beta-function / improper-integral

$$\int_0^{\infty} \frac{\sqrt{x}}{(1+x)^3} dx$$

689 hc-spec-005

R6 Boss+ • 160s • gamma-function / zeta

$$\int_0^{\infty} \frac{x^5}{e^x - 1} dx$$

690 hc-spec-006

R6 Boss+ • 160s • gamma-function / gaussian-integral

$$\int_0^{\infty} x^4 e^{-2x^2} dx$$

691 exam-int-001

R5 Boss • 210s • substitution / rational

$$\int \frac{x}{(1+x^2)^2} dx$$

692 exam-int-002

R5 Boss • 210s • algebra / rational

$$\int \frac{x^3}{1+x^2} dx$$

693 exam-int-004

R5 Boss • 210s • integration-by-parts / log

$$\int x \log x dx$$

694 exam-int-005

R5 Boss • 210s • substitution / log

$$\int \frac{\log x}{x} dx$$

695 exam-int-007

R5 Boss • 210s • trig-integral / substitution

$$\int \sec^2 x \tan x dx$$

696 exam-int-010

R5 Boss • 210s • substitution / radical

$$\int \frac{\sqrt{x}}{1+x^{3/2}} dx$$

697 exam-int-011

R5 Boss • 210s • substitution / radical

$$\int \frac{x}{\sqrt{4-x^2}} dx$$

698 exam-int-012

R5 Boss • 210s • integration-by-parts / exponential

$$\int e^{2x} \sin x dx$$

699 exam-int-013

R5 Boss • 210s • integration-by-parts / inverse-trig

$$\int \arctan x dx$$

700 exam-int-014

R5 Boss • 210s • algebra / inverse-trig

$$\int \frac{x^2}{x^2+1} dx$$

701 exam-int-015

R5 Boss • 210s • substitution / log

$$\int \frac{1}{x \log x} dx$$

702 exam-int-016

R5 Boss • 210s • substitution / trig

$$\int \frac{\cos(\log x)}{x} dx$$

703 exam-int-017

R5 Boss • 210s • substitution / rational

$$\int \frac{2x+3}{x^2+3x+5} dx$$

704 exam-int-018

R5 Boss • 210s • substitution / inverse-trig

$$\int \frac{x}{x^4+1} dx$$

705 exam-int-019

R5 Boss • 210s • substitution / radical

$$\int \frac{1}{\sqrt{x}(1+x)} dx$$

706 exam-int-020

R5 Boss • 210s • substitution / radical

$$\int \frac{x^2}{\sqrt{1+x^3}} dx$$

707 exam-int-021

R5 Boss • 180s • definite-integral / substitution

$$\int_0^1 \frac{x}{1+x^2} dx$$

708 exam-int-024

R5 Boss • 180s • definite-integral / substitution

$$\int_0^{\pi/2} \frac{\sin x}{1+\cos x} dx$$

709 exam-int-025

R5 Boss • 180s • definite-integral / improper-integral

$$\int_0^1 \log x dx$$

710 exam-int-026

R5 Boss • 180s • definite-integral / improper-integral

$$\int_0^\infty e^{-2x} dx$$

711 exam-int-028

R5 Boss • 180s • definite-integral / substitution

$$\int_0^1 x \log(1+x^2) dx$$

712 exam-int-029

R5 Boss • 180s • definite-integral / integration-by-parts

$$\int_0^\pi x \sin x dx$$

713 exam-int-030

R5 Boss • 180s • definite-integral / exam-style

$$\int_0^1 (1-x)^5 dx$$

714 exam-multi-006

R5 Boss • 180s • multivariable / double-integral

$$\int_0^1 \int_0^x (x+y) dy dx$$

715 exam-multi-007

R5 Boss • 180s • multivariable / polar-coordinates

求 $r = 2 \cos \theta$ 所圍面積

716 uni-int-003

R6 Boss+ • 240s • integration-by-parts / log

$$\int \frac{\log x}{x^2} dx$$

717 uni-int-004

R6 Boss+ • 240s • integration-by-parts / inverse-trig

$$\int \frac{\arctan x}{x^2} dx$$

718 uni-int-005

R6 Boss+ • 240s • substitution / radical

$$\int x \sqrt{1+x^2} dx$$

719 uni-int-006

R6 Boss+ • 240s • substitution / trig-integral

$$\int \sin^5 x \cos x dx$$

720 uni-int-007

R6 Boss+ • 240s • trig-integral / exam-style

$$\int \sec^4 x dx$$

721 uni-int-008

R6 Boss+ • 240s • inverse-trig / rational

$$\int \frac{1}{x^2 + 2x + 5} dx$$

722 uni-int-009

R6 Boss+ • 240s • substitution / rational

$$\int \frac{2x+1}{x^2+x+1} dx$$

723 uni-int-010

R6 Boss+ • 240s • integration-by-parts / exponential

$$\int e^x \cos x dx$$

724 uni-int-011

R6 Boss+ • 240s • integration-by-parts / log

$$\int x^2 \log x dx$$

725 uni-int-012

R6 Boss+ • 240s • integration-by-parts / inverse-trig

$$\int x \arctan x dx$$

726 uni-int-013

R6 Boss+ • 240s • definite-integral / algebra

$$\int_0^1 \frac{x^3}{1+x^2} dx$$

727 uni-int-014

R6 Boss+ • 240s • improper-integral / inverse-trig

$$\int_0^\infty \frac{1}{(1+x^2)^2} dx$$

728 uni-int-015

R6 Boss+ • 240s • definite-integral / integration-by-parts

$$\int_0^1 x \log(1+x) dx$$

729 uni-int-016

R6 Boss+ • 240s • definite-integral / beta-function

$$\int_0^{\pi/2} \sin^3 x \cos^2 x dx$$

730 uni-int-018

R6 Boss+ • 240s • improper-integral / substitution

$$\int_0^\infty x e^{-x^2} dx$$

731 uni-int-020

R6 Boss+ • 240s • definite-integral / inverse-trig

$$\int_0^1 \frac{1}{1+x+x^2} dx$$

732 uni-int-023

R6 Boss+ • 240s • parametric / area

求參數曲線 $x = 2 \cos t$, $y = 3 \sin t$ 所圍面積

733 uni-int-025

R6 Boss+ • 240s • improper-integral / inverse-trig

$$\int_0^\infty \frac{1}{x^2+4} dx$$

734 depth-int-002

R6 Boss+ • 270s • algebra / rational

$$\int \frac{x^5}{1+x^3} dx$$

735 depth-int-003

R5 Boss • 220s • substitution / radical

$$\int \frac{x}{(1+x^2)^{3/2}} dx$$

736 depth-int-005

R6 Boss+ • 270s • integration-by-parts / exponential

$$\int x e^x \sin x dx$$

737 depth-int-006

R5 Boss • 220s • integration-by-parts / exponential

$$\int e^{2x} \cos(3x) dx$$

738 depth-int-007

R6 Boss+ • 270s • substitution / integration-by-parts

$$\int \frac{\arctan \sqrt{x}}{\sqrt{x}} dx$$

739 depth-int-008

R5 Boss • 220s • substitution / log

$$\int \frac{1}{x \sqrt{\log x}} dx$$

740 depth-int-009

R6 Boss+ • 270s • substitution / exponential

$$\int \sin(\log x) dx$$

741 depth-int-010

R5 Boss • 220s • substitution / radical

$$\int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$$

742 depth-int-011

R6 Boss+ • 270s • rational / inverse-trig

$$\int \frac{x^2}{(1+x^2)^2} dx$$

743 depth-int-013

R5 Boss • 220s • substitution / trig-integral

$$\int \frac{\sec^2 x}{1+\tan x} dx$$

744 depth-int-014

R6 Boss+ • 270s • integration-by-parts / log

$$\int x \log(1+x) dx$$

745 depth-int-015

R6 Boss+ • 260s • definite-integral / rational

$$\int_0^1 \frac{x}{1+x+x^2} dx$$

746 depth-int-016

R5 Boss • 260s • definite-integral / integration-by-parts

$$\int_0^{\pi/2} x \cos x \, dx$$

747 depth-int-017

R6 Boss+ • 260s • definite-integral / taylor

$$\int_0^1 \frac{\log(1+x)}{x} \, dx$$

748 depth-int-018

R6 Boss+ • 260s • definite-integral / substitution

$$\int_0^1 \frac{x^3}{(1+x^2)^2} \, dx$$

749 depth-int-019

R5 Boss • 260s • improper-integral / substitution

$$\int_0^\infty \frac{x}{(1+x^2)^2} \, dx$$

750 depth-int-020

R6 Boss+ • 260s • improper-integral / partial-fraction

$$\int_0^\infty \frac{1}{(x^2+1)(x^2+4)} \, dx$$

751 depth-int-021

R5 Boss • 260s • definite-integral / trig-integral

$$\int_0^\pi \sin^4 x \, dx$$

752 depth-int-022

R5 Boss • 260s • double-integral / region

$$\int_0^1 \int_y^1 x \, dx \, dy$$

753 depth-int-023

R6 Boss+ • 260s • polar-coordinates / polar-curve

area enclosed by $r = 1 + \cos \theta$

754 depth-int-026

R5 Boss • 260s • improper-integral / gamma-function

$$\int_0^\infty x^3 e^{-x^2} \, dx$$

755 depth-int-027

R5 Boss • 260s • definite-integral / trig-substitution

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} \, dx$$

756 depth-int-028

R5 Boss • 260s • improper-integral / integration-by-parts

$$\int_0^1 x^2 \log x \, dx$$

757 burst-int-001

R6 Boss+ • 330s • multi-ibp / gamma-function

$$\int_0^\infty x^4 e^{-3x} \, dx$$

758 burst-int-002

R6 Boss+ • 330s • multi-ibp / exponential

$$\int_0^\infty x^2 e^{-x} \sin x \, dx$$

759 burst-int-003

R6 Boss+ • 330s • multi-ibp / exponential

$$\int_0^\infty x^2 e^{-2x} \cos x \, dx$$

760 burst-int-004

R5 Boss • 330s • integration-by-parts / multi-ibp

$$\int_0^1 x^5 \log x \, dx$$

761 burst-int-005

R6 Boss+ • 330s • multi-ibp / parameter-integral

$$\int_0^1 x^4 (\log x)^2 dx$$

762 burst-int-006

R6 Boss+ • 330s • multi-ibp / parameter-integral

$$\int_0^1 x^2 (\log x)^3 dx$$

763 burst-int-007

R5 Boss • 330s • beta-function / special-function

$$\int_0^1 x^3 (1-x)^4 dx$$

764 burst-int-008

R6 Boss+ • 330s • beta-function / wallis

$$\int_0^1 \sqrt{x(1-x)} dx$$

765 burst-int-009

R6 Boss+ • 330s • gamma-function / special-function

$$\int_0^\infty x^4 e^{-x^2} dx$$

766 burst-int-010

R5 Boss • 330s • gamma-function / special-function

$$\int_0^\infty x^5 e^{-x^2} dx$$

767 burst-int-011

R6 Boss+ • 330s • wallis / special-function

$$\int_0^{\pi/2} \log(\sin x) dx$$

768 burst-int-012

R6 Boss+ • 330s • wallis / special-function

$$\int_0^{\pi/2} (\log(\sin x))^2 dx$$

769 burst-int-013

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^\pi x \sin^8 x dx$$

770 burst-int-014

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^{10} x dx$$

771 burst-int-015

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^{11} x dx$$

772 burst-int-016

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^6 x \cos^4 x dx$$

773 burst-int-017

R5 Boss • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^4 x \cos^4 x dx$$

774 burst-int-018

R5 Boss • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^5 x \cos^3 x dx$$

775 burst-int-019

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^8 x \cos^2 x dx$$

776 burst-int-020

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi} \sin^8 x \, dx$$

777 burst-int-021

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^7 x \, dx$$

778 burst-int-022

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^8 x \, dx$$

779 burst-int-023

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^3 x \cos^6 x \, dx$$

780 burst-int-024

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^2 x \cos^6 x \, dx$$

781 burst-int-025

R6 Boss+ • 330s • special-function / improper-integral

$$\int_0^{\infty} \frac{x^3}{e^x - 1} \, dx$$

782 burst-int-026

R6 Boss+ • 330s • special-function / improper-integral

$$\int_0^{\infty} \frac{x}{e^x - 1} \, dx$$

783 burst-int-027

R6 Boss+ • 330s • improper-integral / special-function

$$\int_0^{\infty} \frac{1}{1 + x^4} \, dx$$

784 burst-int-028

R6 Boss+ • 330s • improper-integral / kings-property

$$\int_0^{\infty} \frac{\log x}{1 + x^2} \, dx$$

785 burst-int-029

R6 Boss+ • 330s • beta-function / special-function

$$\int_0^{\infty} \frac{\sqrt{x}}{1 + x^2} \, dx$$

786 burst-int-030

R6 Boss+ • 330s • series-integral / special-function

$$\int_0^1 \frac{\log x}{1 + x} \, dx$$

787 burst-int-031

R5 Boss • 330s • trig-power / wallis

$$\int_0^{\pi/2} \sin^4(2x) \, dx$$

788 burst-int-032

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^6 x \cos^6 x \, dx$$

789 burst-int-033

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^{12} x \, dx$$

790 burst-int-034

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^{13} x \, dx$$

791 burst-int-035

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^9 x \cos^5 x \, dx$$

792 burst-int-036

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^3 e^{2x} \, dx$$

793 burst-int-037

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^2 \sin(3x) \, dx$$

794 burst-int-038

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int (\log x)^3 \, dx$$

795 burst-int-039

R5 Boss • 330s • integration-by-parts / log

$$\int x^4 \log x \, dx$$

796 burst-int-040

R6 Boss+ • 330s • integration-by-parts / inverse-trig

$$\int x^3 \arctan x \, dx$$

797 burst-int-041

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^3 e^{-x} \, dx$$

798 burst-int-042

R6 Boss+ • 330s • integration-by-parts / log

$$\int x^2 \log(1+x) \, dx$$

799 burst-rec-001

R6 Boss+ • 270s • recurrence-formula / parameter-integral

$$I_n = \int_0^1 x^n \log x \, dx. \text{ Find } I_n$$

800 burst-rec-002

R6 Boss+ • 270s • recurrence-formula / parameter-integral

$$I_n = \int_0^1 x^n (\log x)^2 \, dx. \text{ Find } I_n$$

801 burst-rec-003

R6 Boss+ • 270s • recurrence-formula / beta-function

$$I_n = \int_0^1 x^n (1-x)^2 \, dx. \text{ Find } I_n$$

802 burst-rec-004

R6 Boss+ • 270s • recurrence-formula / wallis

$$W_n = \int_0^{\pi/2} \sin^n x \, dx. \text{ Find } W_n/W_{n-2}$$

803 burst-rec-005

R6 Boss+ • 270s • recurrence-formula / wallis

$$A_n = \int_0^{\pi/2} \sin^{2n} x \, dx. \text{ Find } A_n/A_{n-1}$$

804 burst-rec-006

R6 Boss+ • 270s • recurrence-formula / wallis

$$B_n = \int_0^{\pi/2} \sin^{2n+1} x \, dx. \text{ Find } B_n/B_{n-1}$$

805 burst-rec-007

R6 Boss+ • 270s • recurrence-formula / gamma-function

$$G_n = \int_0^\infty x^n e^{-x} \, dx. \text{ Find } G_n/G_{n-1}$$

806 burst-rec-008

R6 Boss+ • 270s • recurrence-formula / integration-by-parts

$J_n = \int_0^1 x^n e^x dx$, $J_n = e - n J_{n-1}$. Find the coefficient of J_{n-1}

807 burst-rec-009

R5 Boss • 270s • recurrence-formula / definite-integral

$K_n = \int_0^1 x^n (1+x) dx$. Find K_n

808 burst-rec-010

R5 Boss • 270s • recurrence-formula / definite-integral

$L_n = \int_0^1 x^n (1-x) dx$. Find L_n

809 burst-boss-int-001

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^\infty \frac{(\log x)^2}{1+x^2} dx$

810 burst-boss-int-002

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^\infty \frac{\log x}{1+x^4} dx$

811 burst-boss-int-003

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^\infty \frac{\sqrt{x} \log x}{1+x^2} dx$

812 burst-boss-int-004

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^\infty \frac{x^4}{1+x^6} dx$

813 burst-boss-int-005

R5 Boss • 330s • true-boss / boss-rank

$\int_0^\infty \frac{x^2}{1+x^6} dx$

814 burst-boss-int-006

R5 Boss • 330s • true-boss / boss-rank

$\int_0^\infty \frac{1}{1+x^6} dx$

815 burst-boss-int-007

R5 Boss • 330s • true-boss / boss-rank

$\int_0^\infty \frac{x}{1+x^4} dx$

816 burst-boss-int-008

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^\infty \frac{x^2}{1+x^4} dx$

817 burst-boss-int-009

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^1 \log x \log(1-x) dx$

818 burst-boss-int-010

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^1 \frac{\log x}{1-x} dx$

819 burst-boss-int-011

R5 Boss • 330s • true-boss / boss-rank

$\int_0^1 (\log x)^4 dx$

820 burst-boss-int-012

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^1 x^3 (\log x)^4 dx$

821 burst-boss-int-013

R6 Boss+ • 420s • true-boss / boss-rank

$\int_0^1 x^2 (\log x)^5 dx$

822 burst-boss-int-014

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \log(\sin x) \log(\cos x) dx$$

823 burst-boss-int-015

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} (\log(\tan x))^2 dx$$

824 burst-boss-int-016

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi} x \sin^{10} x dx$$

825 burst-boss-int-017

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^8 x \cos^8 x dx$$

826 burst-boss-int-018

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^{14} x dx$$

827 burst-boss-int-019

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^{15} x dx$$

828 burst-boss-int-020

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\infty} x^5 e^{-2x} \sin x dx$$

829 burst-boss-anti-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 e^x dx$$

830 burst-boss-anti-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 e^{-2x} dx$$

831 burst-boss-anti-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \sin(2x) dx$$

832 burst-boss-anti-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \cos(2x) dx$$

833 burst-boss-anti-005

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 (\log x)^2 dx$$

834 burst-boss-anti-006

R6 Boss+ • 420s • true-boss / boss-rank

$$\int (\log x)^4 dx$$

835 burst-boss-anti-007

R5 Boss • 330s • true-boss / boss-rank

$$\int x^2 \arctan x dx$$

836 burst-boss-anti-008

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 \arctan x dx$$

837 burst-boss-anti-009

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^2 \log(1+x^2) dx$$

838 burst-boss-anti-010

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \log(1+x^2) dx$$

839 burst-boss-param-001

R6 Boss+ • 360s • true-boss / boss-rank

$$I(a) = \int_0^\infty e^{-ax} \sin x dx. \text{ Find } I(a)$$

840 burst-boss-param-002

R5 Boss • 300s • true-boss / boss-rank

$$I(a) = \int_0^\infty x e^{-ax} dx. \text{ Find } I(a)$$

841 burst-boss-param-003

R5 Boss • 300s • true-boss / boss-rank

$$I(a) = \int_0^\infty \frac{1}{1+ax^2} dx. \text{ Find } I(a)$$

842 burst-boss-param-004

R6 Boss+ • 360s • true-boss / boss-rank

$$I(a) = \int_0^{\pi/2} \log(1+a \sin^2 x) dx. \text{ Find } I(a)$$

843 burst-boss-param-005

R6 Boss+ • 360s • true-boss / boss-rank

$$J_n = \int_0^\infty \frac{x^n}{1+x^2} dx \text{ for } -1 < n < 1. \text{ Find } J_n$$

844 burst-boss-param-006

R6 Boss+ • 360s • true-boss / boss-rank

$$K_n = \int_0^1 x^n (\log x)^3 dx. \text{ Find } K_n$$

845 burst-boss2-int-001

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\log x}{1+x^3} dx$$

846 burst-boss2-int-002

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{(\log x)^2}{1+x^3} dx$$

847 burst-boss2-int-003

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\sqrt{x}}{1+x^3} dx$$

848 burst-boss2-int-004

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{x^{1/3}}{1+x^2} dx$$

849 burst-boss2-int-005

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\arctan x}{x(1+x^2)} dx$$

850 burst-boss2-int-006

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \left(\frac{\sin x}{x} \right)^2 dx$$

851 burst-boss2-int-007

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx$$

852 burst-boss2-int-008

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^3}{1-x} dx$$

853 burst-boss2-int-009

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^5}{1-x} dx$$

854 burst-boss2-int-010

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^3}{1+x} dx$$

855 burst-boss2-int-011

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^5}{1+x} dx$$

856 burst-boss2-int-012

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi/2} x \sin^2 x dx$$

857 burst-boss2-int-013

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi/2} x \sin x \cos x dx$$

858 burst-boss2-int-014

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi} x^2 \sin x dx$$

859 burst-boss2-param-001

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a, b) = \int_0^{\infty} e^{-ax} \sin(bx) dx. \text{ Find } F(a, b)$$

860 burst-boss2-param-002

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a, b) = \int_0^{\infty} e^{-ax} \cos(bx) dx. \text{ Find } F(a, b)$$

861 burst-boss2-param-003

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a) = \int_0^{\infty} x e^{-ax} \sin x dx. \text{ Find } F(a)$$

862 burst-boss2-param-004

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a) = \int_0^{\infty} x e^{-ax} \cos x dx. \text{ Find } F(a)$$

863 burst-boss2-param-005

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a) = \int_0^1 \frac{x^a - 1}{\log x} dx. \text{ Find } F(a)$$

864 burst-boss2-param-006

R6 Boss+ • 420s • true-boss / boss-rank

$$F(a) = \int_0^1 x^a (\log x)^5 dx. \text{ Find } F(a)$$

865 burst-boss2-anti-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^5 e^{2x} dx$$

866 burst-boss2-anti-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 \sin x dx$$

867 burst-boss2-anti-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 \cos x dx$$

868 burst-boss2-anti-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\int (\log x)^5 dx$$

869 burst-boss2-anti-005

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^5 \arctan x dx$$

870 burst-boss2-anti-006

R6 Boss+ • 420s • true-boss / boss-rank

$$\int \frac{x^5}{1+x^2} dx$$

871 burst-boss3-int-001

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty \int_0^\infty x^3 y^4 e^{-2x-3y} dx dy$$

872 burst-boss3-int-002

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^1 \int_0^1 xy^2 (\log x)^2 (\log y)^3 dx dy$$

873 burst-boss3-int-003

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^1 (\log(x(1-x)))^2 dx$$

874 burst-boss3-int-004

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^1 \left(\log \frac{x}{1-x} \right)^2 dx$$

875 burst-boss3-int-005

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\pi x \log(\sin x) dx$$

876 burst-boss3-int-006

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\pi x^3 \sin x dx$$

877 burst-boss3-int-007

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\pi x^4 \sin x dx$$

878 burst-boss3-int-008

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty x^3 e^{-2x} \cos(3x) dx$$

879 burst-boss3-int-009

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty x^3 e^{-2x} \sin(3x) dx$$

880 burst-boss3-int-010

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty x^4 e^{-x} \cos(2x) dx$$

881 burst-boss3-int-011

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty x^4 e^{-x} \sin(2x) dx$$

882 burst-boss3-int-012

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^\infty \frac{x^5}{e^x + 1} dx$$

883 burst-boss3-int-013

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x^3}{e^x + 1} dx$$

884 burst-boss3-param-001

R6 Boss+ • 480s • true-boss / boss-rank

$$F(a) = \int_0^{\infty} e^{-ax} \sin(3x) dx. \text{ Find } F''(a)$$

885 burst-boss3-param-002

R6 Boss+ • 480s • true-boss / boss-rank

$$G(a) = \int_0^{\infty} x^2 e^{-ax} \cos x dx. \text{ Find } G(a)$$

886 burst-boss3-param-003

R6 Boss+ • 480s • true-boss / boss-rank

$$H(a) = \int_0^{\infty} x^3 e^{-ax} \sin x dx. \text{ Find } H(a)$$

887 burst-boss3-param-004

R6 Boss+ • 480s • true-boss / boss-rank

$$B(a) = \int_0^1 x^a (\log x)^4 dx. \text{ Find } B(a)$$

888 burst-boss3-param-005

R6 Boss+ • 480s • true-boss / boss-rank

$$C(a) = \int_0^1 x^a (\log x)^6 dx. \text{ Find } C(a)$$

889 burst-boss3-param-006

R6 Boss+ • 480s • true-boss / boss-rank

$$Q(a) = \frac{d^3}{da^3} \int_0^1 \frac{x^a - 1}{\log x} dx. \text{ Find } Q(a)$$

890 burst-boss3-anti-001

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 e^{-3x} dx$$

891 burst-boss3-anti-002

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^5 \sin(2x) dx$$

892 burst-boss3-anti-003

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^5 \cos(2x) dx$$

893 burst-boss3-anti-004

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 \arctan x dx$$

894 burst-boss3-anti-005

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 \log(1+x) dx$$

895 burst-boss3-anti-006

R6 Boss+ • 540s • true-boss / boss-rank

$$\int \frac{x^7}{1+x^2} dx$$

896 world-005

R2 Basic • 55s • wallis / world-universities

$$\int_0^{\pi/2} \cos^5 x dx$$

897 world-010

R2 Basic • 55s • wallis / world-universities

$$\int_0^{\pi} \cos^4 x dx$$

898 world-014

R4 Advanced • 150s • kings-property / log-integral

$$\int_0^{\pi/4} \log(1 + \tan x) dx$$

899 world-015

R4 Advanced • 150s • feynman / world-universities

$$\int_0^1 \frac{\log(1+x)}{1+x^2} dx$$

900 world-016

R4 Advanced • 150s • frullani / world-universities

$$\int_0^\infty \frac{\cos(2x) - \cos(3x)}{x} dx$$

901 world-017

R4 Advanced • 150s • beta-function / world-universities

$$\int_0^\infty \frac{dx}{1+x^3}$$

902 world-018

R4 Advanced • 150s • integration-by-parts / world-universities

$$\int_0^1 \frac{\log x}{(1+x)^2} dx$$

903 world-019

R4 Advanced • 150s • frullani / feynman

$$\int_0^1 \frac{x^2 - 1}{\log x} dx$$

904 world-027

R4 Advanced • 150s • integration-by-parts / complex

$$\int_0^\infty x e^{-x} \sin x dx$$

905 world-029

R4 Advanced • 150s • feynman / world-universities

$$\int_0^\infty \frac{\log x}{(1+x^2)^2} dx$$

906 world-031

R5 Boss • 240s • hyperbolic / series

$$\int_0^\infty \frac{x}{\sinh x} dx$$

907 world-032

R5 Boss • 240s • dirichlet-beta / world-universities

$$\int_0^1 \frac{(\log x)^2}{1+x^2} dx$$

908 world-033

R5 Boss • 240s • dirichlet / world-universities

$$\int_0^\infty \frac{\cos x - \cos 3x}{x^2} dx$$

909 world-037

R5 Boss • 240s • gaussian / feynman

$$\int_0^\infty e^{-x^2} \cos(2x) dx$$

910 world-038

R5 Boss • 240s • dirichlet / world-universities

$$\int_0^\infty \frac{\sin^3 x}{x^3} dx$$

911 world-041

R5 Boss • 240s • frullani / world-universities

$$\int_0^1 \frac{x-1}{\log x} dx$$

912 world-043

R5 Boss • 240s • contour / log-integral

$$\int_0^\infty \frac{\log x}{1+x^6} dx$$

913 world-044

R5 Boss • 240s • fourier / log-integral

$$\int_0^{\pi/2} x \cot x dx$$

914 world-045

R5 Boss • 240s • contour / fourier

$$\int_{-\infty}^\infty \frac{\cos x}{1+x^2} dx$$

915 world-047

R5 Boss • 240s • dirichlet / parseval

$$\int_0^\infty \frac{\sin^4 x}{x^4} dx$$

916 world-050

$$\int_0^1 x^3 \log x \, dx$$

917 world-051

R6 Boss+ • 340s • ahmed-integral / world-universities

$$\int_0^1 \frac{\arctan(\sqrt{x^2+2})}{(x^2+1)\sqrt{x^2+2}} dx$$

918 world-052

R6 Boss+ • 340s • fourier / log-integral

$$\int_0^{\pi/2} \log^2(\sin x) \, dx$$

919 world-054

R6 Boss+ • 340s • fourier / log-integral

$$\int_0^{\pi/2} \log^2(\cos x) \, dx$$

920 world-055

R5 Boss • 240s • substitution / world-universities

$$\int_0^{\pi/2} \cos x \log(\sin x) \, dx$$

921 world-056

R6 Boss+ • 340s • dilogarithm / world-universities

$$\int_0^1 \log x \log(1+x) \, dx$$

922 world-057

R6 Boss+ • 340s • feynman / world-universities

$$\int_0^\infty \frac{\log(1+x^2)}{1+x^2} dx$$

923 world-058

R6 Boss+ • 340s • gaussian / feynman

$$\int_0^\infty e^{-x^2} \cos x \, dx$$

924 world-059

R6 Boss+ • 340s • partial-fractions / world-universities

$$\int_0^1 \frac{dx}{1+x^3}$$

925 world-061

R6 Boss+ • 340s • dilogarithm / world-universities

$$\int_0^1 \frac{\log(1-x)}{x} dx$$

926 world-062

R6 Boss+ • 340s • dilogarithm / world-universities

$$\int_0^1 \frac{\log x}{1-x^2} dx$$

927 world-063

R6 Boss+ • 340s • beta-function / world-universities

$$\int_0^\infty \frac{x^2}{(1+x^2)^3} dx$$

928 world-064

R6 Boss+ • 340s • frullani / world-universities

$$\int_0^1 \frac{x^3-1}{\log x} dx$$

929 world-065

R6 Boss+ • 340s • beta-function / world-universities

$$\int_0^{\pi/2} \sqrt{\tan x} \, dx$$

930 world-069

R6 Boss+ • 340s • partial-fractions / world-universities

$$\int_0^\infty \frac{dx}{(1+x^2)(4+x^2)}$$

931 world-070

R6 Boss+ • 340s • frullani / laplace

$$\int_0^{\infty} e^{-x} \frac{\sin x}{x} dx$$

932 world-072

R6 Boss+ • 340s • weierstrass / world-universities

$$\int_0^{2\pi} \frac{dx}{(2 + \cos x)^2}$$

933 world-073

R6 Boss+ • 340s • fourier / world-universities

$$\int_0^{\pi} \log(2 - \cos x) dx$$

934 world-076

R6 Boss+ • 340s • substitution / world-universities

$$\int_0^{\infty} \frac{\arctan x}{1 + x^2} dx$$

935 world-077

R6 Boss+ • 340s • feynman / world-universities

$$\int_0^1 \frac{\arctan x}{1 + x} dx$$

936 world-079

R5 Boss • 240s • gamma-function / world-universities

$$\int_0^1 (\log x)^3 dx$$

937 world-080

R6 Boss+ • 340s • partial-fractions / world-universities

$$\int_0^{\infty} \frac{dx}{1 + x^2 + x^4}$$

938 world-081

R5 Boss • 240s • hyperbolic / world-universities

$$\int_0^{\infty} \frac{dx}{\cosh x}$$

939 world-082

R5 Boss • 240s • integration-by-parts / inverse-trig

$$\int_0^1 (\arcsin x)^2 dx$$

940 world-083

R6 Boss+ • 340s • eta / fermi

$$\int_0^{\infty} \frac{x}{e^x + 1} dx$$

941 world-084

R6 Boss+ • 340s • dilogarithm / world-universities

$$\int_0^1 \frac{\log(1-x)}{1+x} dx$$

942 world-085

R6 Boss+ • 340s • weierstrass / world-universities

$$\int_0^{\pi/2} \frac{d\theta}{1 + \sin^2 \theta}$$

943 world-088

R6 Boss+ • 340s • dirichlet / world-universities

$$\int_0^{\infty} \frac{\sin x}{x} dx$$

944 world-089

R6 Boss+ • 340s • dirichlet / world-universities

$$\int_0^{\infty} \frac{x - \sin x}{x^3} dx$$

945 world-090

R6 Boss+ • 340s • gaussian / world-universities

$$\int_0^{\infty} e^{-x^2} dx$$

946 world-092

R5 Boss • 240s • integration-by-parts / world-universities

$$\int_0^1 x \log(1-x) dx$$

947 world-094

R6 Boss+ • 340s • beta-function / world-universities

$$\int_0^\infty \frac{dx}{1+x^4}$$

948 world-095

R5 Boss • 240s • integration-by-parts / world-universities

$$\int_0^{\pi/2} \sin x \log(\sin x) dx$$

949 world-098

R6 Boss+ • 340s • beta-function / world-universities

$$\int_0^\infty \frac{x^2}{(1+x^4)} dx$$

950 world-100

R6 Boss+ • 340s • fourier / world-universities

$$\int_0^{\pi/2} \frac{x^2}{\sin^2 x} dx$$

951 putnam-004

R5 Boss • 240s • putnam / symmetry

$$\int_{-1}^1 \frac{dx}{(1+e^x)(1+x^2)}$$

952 putnam-005

R6 Boss+ • 340s • putnam / kings-property

$$\int_0^{\pi/2} \frac{dx}{1+(\tan x)^{\sqrt{2}}}$$

953 putnam-009

R6 Boss+ • 340s • putnam / symmetry

$$\int_0^\infty \frac{dx}{(1+x^2)(1+x^3)}$$

954 putnam-012

R6 Boss+ • 340s • putnam / frullani

$$\int_0^\infty \frac{\arctan(2x) - \arctan x}{x} dx$$

級數 Series

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955 ser-001

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\sum_{n=1}^\infty \frac{1}{n^2}$$

956 ser-002

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\sum_{n=1}^\infty \frac{1}{n}$$

957 ser-003

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\sum_{n=0}^\infty \left(\frac{1}{3}\right)^n$$

958 ser-004

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\sum_{n=1}^\infty \frac{(-1)^{n+1}}{n}$$

959 ser-005

R2 Basic • 45s • rank-2 / beginner-friendly

$$\sum_{n=1}^\infty \frac{n}{n^2+1}$$

960 ser-006

R2 Basic • 50s • rank-2 / beginner-friendly

$$\sum_{n=1}^\infty \frac{n!}{n^n}$$

961 ser-007

R2 Basic • 55s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}$$

962 ser-008

R2 Basic • 45s • rank-2 / beginner-friendly

Radius of convergence of $\sum_{n=0}^{\infty} n!x^n$

963 ser-009

R3 Standard • 45s • rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$

964 ser-010

R3 Standard • 45s • rank-3

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

965 ser-011

R3 Standard • 70s • rank-3

$$\sum_{n=1}^{\infty} \frac{2^n + n}{3^n}$$

966 ser-012

R2 Basic • 60s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

967 ser-013

R2 Basic • 55s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$$

968 ser-014

R3 Standard • 55s • rank-3

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

969 ser-015

R3 Standard • 70s • rank-3

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

970 ser-016

R4 Advanced • 70s • rank-4

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

971 ser-017

R3 Standard • 60s • rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$

972 ser-018

R2 Basic • 45s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

973 ser-019

R2 Basic • 75s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

974 ser-020

R3 Standard • 60s • rank-3

$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

975 td-ser-001

R6 Boss+ • 100s • rank-6 / boss-rank

Coefficient of x^6 in $(1+x)^x$

976 td-ser-002

R6 Boss+ • 110s • rank-6 / boss-rank

Coefficient of x^5 in $e^{2 \arcsin x}$

977 td-ser-003

R6 Boss+ • 105s • rank-6 / boss-rank

$$\sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \cdot \frac{1}{n}$$

978 ser-024

R2 Basic • 60s • integral-test / log

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}$$

979 ser-027

R2 Basic • 50s • taylor / coefficient

Coefficient of x^4 in e^{2x}

980 ser-030

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n5^n}$

981 ser-031

R3 Standard • 45s • p-series / rank-3

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

982 ser-032

R3 Standard • 65s • term-test / rank-3

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^n$$

983 ser-034

R3 Standard • 60s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} n!x^n$

984 ser-035

R3 Standard • 60s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n!}$

985 ser-036

R3 Standard • 70s • comparison / geometric-series

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n + n}$$

986 ser-037

R3 Standard • 70s • integral-test / log

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$$

987 ser-038

R2 Basic • 50s • taylor / coefficient

Coefficient of x^3 in $\log(1+x)$

988 ser-039

R3 Standard • 55s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$

989 gap-ser-root-001

R3 Standard • 75s • root-test / power-series

Radius of convergence of $\sum_{n=1}^{\infty} \frac{3^n}{n^2} x^n$

990 gap-ser-root-002

R3 Standard • 85s • root-test / power-series

Radius of convergence of $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^n x^n$

991 gap-ser-lc-001

R3 Standard • 75s • limit-comparison / harmonic

$\sum_{n=1}^{\infty} \frac{3n^2 + 1}{n^3 + 2}$ converges or diverges?

992 gap-ser-lc-002

R3 Standard • 75s • limit-comparison / p-series

$\sum_{n=1}^{\infty} \frac{n+1}{n^3+1}$ converges or diverges?

993 gap-ser-cond-001

R3 Standard • 75s • alternating-series / absolute-conditional

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ absolute, conditional, or divergent?

994 gap-ser-end-001

R4 Advanced • 85s • endpoint-analysis / power-series

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x = -1$ is absolute, conditional, or divergent?

995 gap-ser-end-002

R4 Advanced • 80s • endpoint-analysis / power-series

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x = 1$ converges or diverges?

996 gap-ser-taylor-001

R3 Standard • 70s • taylor / coefficient

Coefficient of x^5 in the Maclaurin series for $\sin x$

997 gap-ser-taylor-002

R3 Standard • 80s • taylor / full-expansion

Degree 4 Maclaurin polynomial for $\cos x$

998 gap-tech-005

R3 Standard • 45s • technique-recognition / root-test

Technique for $\sum \left(\frac{n}{n+1} \right)^n x^n$

999 mob-tech-007

R2 Basic • 45s • technique-sprint / technique-recognition

$\sum_{n=1}^{\infty} \frac{n!}{3^n}$ Use which technique?

1000 mob-tech-008

R2 Basic • 45s • technique-sprint / technique-recognition

$$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n \quad \text{Use which technique?}$$

1001 mob-tech-019

R2 Basic • 45s • technique-sprint / technique-recognition

Find interval of convergence after radius is known Use which technique?

1002 mob-trap-004

R2 Basic • 45s • trap-drill / rank-2

$$\sum (-1)^n / n \text{ is not absolutely convergent} \quad \text{Main trap?}$$

1003 mob-trap-005

R2 Basic • 45s • trap-drill / rank-2

$$\sum x^n / n \text{ after finding } R = 1 \quad \text{Main trap?}$$

1004 mob-conv-003

R3 Standard • 45s • convergence-test / series-test

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

1005 mob-conv-006

R3 Standard • 45s • convergence-test / series-test

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}$$

1006 mob-conv-010

R3 Standard • 45s • convergence-test / series-test

$$\sum_{n=1}^{\infty} \frac{n+1}{3n+2}$$

1007 rel-basic-019

R1 Warm-up • 45s • geometric-series / beginner-foundation

$$\sum_{n=0}^{\infty} \left(\frac{1}{2} \right)^n$$

1008 rel-basic-022

R1 Warm-up • 45s • power-series / beginner-foundation

$$\text{Radius of convergence of } \sum_{n=0}^{\infty} 3^n x^n$$

1009 rel-basic-024

R1 Warm-up • 45s • geometric-series / beginner-foundation

$$\sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n$$

1010 rel-adv-017

R3 Standard • 75s • taylor / power-series

$$\text{coefficient of } x^6 \text{ in } e^{x^2}$$

1011 rel-adv-019

R3 Standard • 75s • power-series / radius-of-convergence

$$\text{Radius of convergence of } \sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

1012 rel-adv-015

R3 Standard • 55s • root-test / series-test

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n+1} \right)^n$$

1013 rel-adv-018

R3 Standard • 55s • limit-comparison / series-test

$$\sum_{n=1}^{\infty} \frac{3n^2 + 1}{n^3 + n}$$

1014 rel-adv-020

R3 Standard • 55s • endpoint-analysis / power-series

$$\sum_{n=1}^{\infty} \frac{x^n}{n} \text{ at } x = -1$$

1015 rel-hard-ser-001

R5 Boss • 100s • power-series / ratio-test

$$\text{Radius of convergence of } \sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} x^n$$

1016 rel-hard-ser-002

R5 Boss • 100s • power-series / ratio-test

$$\text{Radius of convergence of } \sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} x^n$$

1017 rel-hard-ser-004

R5 Boss • 100s • taylor / coefficient

$$\text{coefficient of } x^{10} \text{ in } \sin(x^2)$$

1018 rel-hard-ser-005

R5 Boss • 100s • telescoping / series-boss

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)}$$

1019 rel-hard-ser-008

R5 Boss • 100s • telescoping / partial-fraction

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

1020 rel-hard-ser-012

R5 Boss • 100s • power-series / ratio-test

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$

1021 rel-hard-ser-010

R5 Boss • 80s • term-test / series-boss

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

1022 rel-hard-ser-011

R5 Boss • 80s • endpoint-analysis / alternating

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \text{ at the endpoint}$$

1023 hc-spec-007

R6 Boss+ • 160s • series / telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)(n+4)}$$

1024 hc-spec-009

R6 Boss+ • 160s • series / power-series

Radius of convergence of $\sum_{n=0}^{\infty} \frac{(3n)!}{(n!)^3} x^n$

1025 hc-spec-010

R6 Boss+ • 160s • series / bessel

coefficient of x^8 in $J_0(x)$ **1026 exam-ser-001**

R5 Boss • 180s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{nx^n}{3^n}$

1027 exam-ser-002

R5 Boss • 180s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$

1028 exam-ser-003

R5 Boss • 180s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$

1029 exam-ser-008

R5 Boss • 180s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)!}$$

1030 exam-ser-009

R5 Boss • 180s • taylor / coefficient

求 e^{2x} 的 x^5 係數**1031 exam-ser-010**

R5 Boss • 180s • taylor / coefficient

求 $\sin(x^2)$ 的 x^6 係數**1032 exam-ser-011**

R5 Boss • 180s • taylor / coefficient

求 $\log(1+x)$ 的 x^4 係數**1033 exam-ser-013**

R5 Boss • 180s • power-series / sum-series

$$\sum_{n=1}^{\infty} \frac{n(n+1)}{2^n}$$

1034 exam-ser-014

R5 Boss • 180s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{4^n x^n}{\sqrt{n+1}}$

1035 exam-ser-017

R5 Boss • 180s • taylor / sum-series

$\sum_{n=0}^{\infty} \frac{2^n}{n!}$

1036 exam-ser-018

R5 Boss • 180s • binomial / coefficient

求 $(1+x)^7$ 的 x^3 係數

1037 exam-ser-019

R5 Boss • 180s • geometric-series / coefficient

求 $\frac{1}{1-2x}$ 的 x^4 係數

1038 exam-ser-020

R5 Boss • 180s • power-series / sum-series

$\sum_{n=1}^{\infty} \frac{n}{3^n}$

1039 uni-ser-001

R6 Boss+ • 210s • sum-series / power-series

$\sum_{n=1}^{\infty} \frac{n}{4^n}$

1040 uni-ser-004

R6 Boss+ • 210s • taylor / coefficient

求 x^8 在 $\sin(x^2)$ 的係數

1041 uni-ser-005

R6 Boss+ • 210s • taylor / coefficient

求 x^6 在 $e^x \cos x$ 的係數

1042 uni-ser-006

R6 Boss+ • 210s • taylor / coefficient

求 x^5 在 $x^2 \log(1+x)$ 的係數

1043 uni-ser-007

R6 Boss+ • 210s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{n^2}{5^n} x^n$

1044 uni-ser-008

R6 Boss+ • 210s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$

1045 uni-ser-009

R6 Boss+ • 210s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$

1046 uni-ser-010

R6 Boss+ • 210s • taylor / alternating-series

$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$

1047 uni-ser-012

R6 Boss+ • 210s • sum-series / power-series

$\sum_{n=0}^{\infty} \frac{n(n-1)}{2^n}$

1048 uni-ser-013

R6 Boss+ • 210s • coefficient / binomial

求 x^4 在 $\frac{1}{(1-x)^3}$ 的係數

1049 uni-ser-014

R6 Boss+ • 210s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{3^n}{n^2} (x-1)^n$ **1050 uni-ser-015**

R6 Boss+ • 210s • sum-series / alternating-series

 $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2^n}$ **1051 uni-ser-016**

R6 Boss+ • 210s • taylor / coefficient

求 x^7 在 $\sin x \cos x$ 的係數**1052 uni-ser-018**

R6 Boss+ • 210s • coefficient / binomial

求 x^6 在 $(1+x)^{10}$ 的係數**1053 uni-ser-019**

R6 Boss+ • 210s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$ **1054 depth-ser-001**

R5 Boss • 230s • sum-series / power-series

 $\sum_{n=1}^{\infty} \frac{n(n+1)}{3^n}$ **1055 depth-ser-002**

R5 Boss • 230s • sum-series / power-series

 $\sum_{n=1}^{\infty} \frac{n^2}{4^n}$ **1056 depth-ser-003**

R5 Boss • 230s • telescoping-series / exam-style

 $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$ **1057 depth-ser-004**

R5 Boss • 230s • taylor / coefficient

coefficient of x^7 in e^{3x} **1058 depth-ser-005**

R5 Boss • 230s • taylor / coefficient

coefficient of x^8 in $\cos(x^2)$ **1059 depth-ser-006**

R5 Boss • 230s • taylor / coefficient

coefficient of x^6 in $\frac{\sin x}{x}$ **1060 depth-ser-007**

R5 Boss • 230s • taylor / sum-series

 $\sum_{n=0}^{\infty} \frac{1}{(2n)!}$ **1061 depth-ser-008**

R5 Boss • 230s • sum-series / power-series

 $\sum_{n=1}^{\infty} \frac{n}{3^{n+1}}$ **1062 depth-ser-009**

R6 Boss+ • 230s • power-series / radius

radius of convergence of $\sum_{n=0}^{\infty} \binom{2n}{n} x^n$ **1063 depth-ser-010**

R6 Boss+ • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} 3^n x^n$

1064 depth-ser-011

R5 Boss • 230s • coefficient / geometric-series

coefficient of x^5 in $\frac{1}{(1-2x)^2}$ **1065 depth-ser-012**

R5 Boss • 230s • sum-series / power-series

$$\sum_{n=1}^{\infty} \frac{n^2}{2^{n+1}}$$

1066 depth-ser-013

R5 Boss • 230s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$$

1067 depth-ser-014

R5 Boss • 230s • telescoping-series / exam-style

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

1068 depth-ser-015

R6 Boss+ • 230s • taylor / coefficient

coefficient of x^3 in $e^x \log(1+x)$ **1069 depth-ser-016**

R6 Boss+ • 230s • sum-series / power-series

$$\sum_{n=1}^{\infty} \frac{n^3}{3^n}$$

1070 depth-ser-017

R5 Boss • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{n^3+1}{5^n} x^n$ **1071 depth-ser-018**

R5 Boss • 230s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}$$

1072 depth-ser-019

R6 Boss+ • 230s • taylor / coefficient

coefficient of x^4 in $(\log(1+x))^2$ **1073 depth-ser-020**

R5 Boss • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n^2} (x+1)^n$ **1074 burst-boss-ser-001**

R5 Boss • 240s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^3}{2^n}$$

1075 burst-boss-ser-002

R5 Boss • 240s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}$$

1076 burst-boss-ser-003

R6 Boss+ • 300s • true-boss / boss-rank

$$H_n = \sum_{k=1}^n \frac{1}{k}. \text{ Find } \sum_{n=1}^{\infty} \frac{H_n}{2^n}$$

1077 burst-boss-ser-004

R6 Boss+ • 300s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

1078 burst-boss-ser-005

R6 Boss+ • 300s • true-boss / boss-rank

Coefficient of x^6 in $(\log(1+x))^2$

1079 burst-boss-ser-006

R6 Boss+ • 300s • true-boss / boss-rank

Coefficient of x^8 in $\frac{x^2 e^{3x}}{1-x}$ **1080 burst-boss2-ser-001**

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{1}{n^4}$$

1081 burst-boss2-ser-002

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4}$$

1082 burst-boss2-ser-003

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$$

1083 burst-boss2-ser-004

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6}$$

1084 burst-boss2-ser-005

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^4}{3^n}$$

1085 burst-boss2-ser-006

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^5}{3^n}$$

1086 burst-boss2-ser-007

R6 Boss+ • 360s • true-boss / boss-rank

$$H_n = \sum_{k=1}^n \frac{1}{k}. \text{ Find } \sum_{n=1}^{\infty} \frac{H_n}{3^n}$$

1087 burst-boss2-ser-008

R6 Boss+ • 360s • true-boss / boss-rank

$$H_n = \sum_{k=1}^n \frac{1}{k}. \text{ Find } \sum_{n=1}^{\infty} \frac{H_n}{4^n}$$

1088 burst-boss3-ser-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^6}{2^n}$$

1089 burst-boss3-ser-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^7}{2^n}$$

1090 burst-boss3-ser-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^5}{4^n}$$

1091 burst-boss3-ser-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^6}{3^n}$$

1092 burst-boss3-ser-005

R6 Boss+ • 420s • true-boss / boss-rank

$$H_n = \sum_{k=1}^n \frac{1}{k}. \text{ Find } \sum_{n=1}^{\infty} \frac{H_n}{n2^n}$$

1093 burst-boss3-ser-006

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{20} in $(\log(1+x))^2$

1094 burst-boss3-ser-007

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{18} in $\frac{e^{2x}}{(1-x)^3}$ **1095 burst-boss3-ser-008**

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{16} in $\frac{e^{2x}}{(1-x)^4}$ **1096 world-006**

R2 Basic • 55s • telescoping / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

1097 world-009

R2 Basic • 55s • power-series / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{n 3^n}$$

1098 world-020

R4 Advanced • 150s • exponential-series / world-universities

$$\sum_{n=1}^{\infty} \frac{n^2}{n!}$$

1099 world-021

R4 Advanced • 150s • exponential-series / world-universities

$$\sum_{n=1}^{\infty} \frac{n^3}{n!}$$

1100 world-022

R4 Advanced • 150s • log-series / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{n(2n+1)}$$

1101 world-023

R4 Advanced • 150s • zeta / telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)^2}$$

1102 world-028

R3 Standard • 90s • power-series / radius

收斂半徑： $\sum_{n=1}^{\infty} \binom{2n}{n} x^n$ **1103 world-034**

R5 Boss • 240s • euler-sum / dilogarithm

$$\sum_{n=1}^{\infty} \frac{H_n}{n 2^n}$$

1104 world-039

R5 Boss • 240s • central-binomial / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{n^2 \binom{2n}{n}}$$

1105 world-042

R5 Boss • 240s • power-series / radius

$$\text{收斂半徑：} \sum_{n=1}^{\infty} \frac{(3n)!}{(n!)^3} x^n$$

1106 world-048

R5 Boss • 240s • binomial-series / world-universities

$$\sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!(2n+1)}$$

1107 world-053

R5 Boss • 240s • telescoping / alternating

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 - 1}$$

1108 world-060

R6 Boss+ • 340s • dirichlet-beta / world-universities

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3}$$

1109 world-075

R5 Boss • 240s • zeta / telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}$$

1110 world-078

R6 Boss+ • 340s • dilogarithm / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \left(\frac{1}{2}\right)^n$$

1111 world-086

R6 Boss+ • 340s • zeta / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^6}$$

1112 world-087

R6 Boss+ • 340s • zeta / world-universities

$$\sum_{n=1}^{\infty} \frac{1}{n^6}$$

1113 world-097

R6 Boss+ • 340s • euler-sum / generating-function

$$\sum_{n=1}^{\infty} \frac{H_n}{2^n}$$

1114 putnam-001

R5 Boss • 240s • putnam / telescoping

$$\sum_{n=1}^{\infty} \arctan \frac{1}{n^2 + n + 1}$$

1115 putnam-002

R5 Boss • 240s • putnam / product

$$\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}$$

1116 putnam-006

R6 Boss+ • 340s • putnam / wallis

$$\prod_{n=1}^{\infty} \left(1 - \frac{1}{(2n)^2}\right)$$

1117 putnam-007

R5 Boss • 240s • putnam / telescoping

$$\sum_{n=1}^{\infty} \frac{n}{n^4 + n^2 + 1}$$

1118 putnam-011

R5 Boss • 240s • putnam / alternating

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)}$$

答案速查

0001. lim-001 1
0002. lim-002 $1/2$
0003. lim-003 4
0004. lim-004 $3/2$
0005. lim-005 1
0006. lim-006 3
0007. lim-007 2
0008. lim-008 $1/2$
0009. lim-009 $-1/6$
0010. lim-010 2
0011. lim-011 0
0012. lim-012 $1/3$
0013. lim-013 $1/2$
0014. lim-014 $-1/2$
0015. lim-015 $1/2$
0016. lim-016 $-1/8$
0017. lim-017 $1/3$
0018. lim-018 $1/3$
0019. lim-019 $-e/2$
0020. lim-020 $-1/3$
0021. td-lim-001
 $\exp(1/2)$
0022. td-lim-002
 $\sqrt{e}/3$
0023. td-lim-004
 $-1/2$
0024. lim-021 $1/3$
0025. lim-022 $3/2$
0026. lim-024 $-1/2$
0027. lim-025 1
0028. lim-026 0
0029. lim-027 dne
0030. lim-028 $1/2$
0031. lim-029 0
0032. lim-030 $9/2$
0033. lim-031 0
0034. lim-032 dne
0035. lim-033 1

0036. lim-034 0
0037. lim-035 $5/2$
0038. lim-036 $1/2$
0039. lim-037 2
0040. lim-039 dne
0041. lim-040 2
0042. gap-tech-001
rationalize
0043. gap-tech-002
taylor
0044. gap-tech-008
pathtest
0045. mob-tech-001
rationalize
0046. mob-tech-002
Taylor
0047. mob-tech-009
pathtest
0048. mob-trap-009
indeterminate
0049. mob-limtrap-003
 $-1/24$
0050. mob-limtrap-004
 $1/3$
0051. mob-limtrap-005
 $1/6$
0052. mob-limtrap-006
 $-1/8$
0053. mob-limtrap-008
0
0054. rel-basic-001
5
0055. rel-basic-003
4
0056. rel-basic-004
2
0057. rel-basic-005
 $1/2$
0058. rel-adv-002
 $1/24$
0059. rel-boss-019
1
0060. rel-hard-lim-001
 $-1/5040$
0061. rel-hard-lim-002
 $17/315$

0062. rel-hard-lim-003
 $1/5$

0063. rel-hard-lim-004
 $1/360$

0064. rel-hard-lim-007
 $1/2$

0065. rel-hard-lim-008
 1

0066. rel-hard-lim-009
 $1/5$

0067. rel-hard-lim-010
 $11*e/24$

0068. rel-hard-lim-006
DNE

0069. exam-lim-001
 $81/40$

0070. exam-lim-002
 $8/3$

0071. exam-lim-003
 1

0072. exam-lim-005
 2

0073. exam-lim-006
 -2

0074. exam-lim-007
 $-9/8$

0075. exam-lim-008
 $1/3$

0076. exam-lim-009
 $1/3$

0077. exam-lim-010
 $-8/3$

0078. exam-lim-011
 1

0079. exam-lim-012
 $\exp(6)$

0080. exam-lim-015
 1

0081. exam-lim-017
 $-1/12$

0082. exam-lim-019
 1

0083. exam-lim-020
 $1/6$

0084. uni-lim-001
 $1/120$

0085. uni-lim-002
 $-1/4$

0086. uni-lim-003
 $3/2$

0087. uni-lim-005
 $1/8$

0088. uni-lim-006
 $1/2$

0089. uni-lim-007
 -1

0090. uni-lim-008
 $-9*\exp(3)/2$

0091. uni-lim-009
 1

0092. uni-lim-010
 $-1/2$

0093. uni-lim-011
 $-1/2$

0094. uni-lim-012
 1

0095. uni-lim-014
 $1/3$

0096. depth-lim-001
 $-1/5040$

0097. depth-lim-002
 $1/5$

0098. depth-lim-003
 $1/2$

0099. depth-lim-004
 $1/6$

0100. depth-lim-005
 $2/3$

0101. depth-lim-006
 $1/2$

0102. depth-lim-008
 $1/2$

0103. depth-lim-009
 $1/3$

0104. depth-lim-010
 $1/2$

0105. depth-lim-011
 $-1/2$

0106. depth-lim-012
 $\exp(1)/2$

0107. burst-boss-lim-001
 $2/15$

0108. burst-boss-lim-002
 $1/120$

0109. burst-boss-lim-003
 $-1/4$

0110. burst-boss-lim-004
 $-1/8$

0111. burst-boss-lim-005
 $1/6$

0112. burst-boss-lim-006
 $-1/7$

0113. burst-boss-lim-008
 $-e/2$

0114. burst-boss2-lim-001
 $\exp(-1/6)$

0115. burst-boss2-lim-002
 $-1/45$

0116. burst-boss2-lim-003
 $7/360$

0117. burst-boss2-lim-004
 $-1/12$

0118. burst-boss2-lim-005
 $1/10$

0119. burst-boss2-lim-006
 $1/21$

0120. burst-boss2-lim-007
 $-1/4$

0121. burst-boss2-lim-008
 $-1/4$

0122. burst-boss2-lim-009
 $(1/\sqrt{2})-1)/2$

0123. burst-boss2-lim-010
 $(2-e)/2$

0124. burst-boss3-lim-001
 $1/\exp(1)$

0125. burst-boss3-lim-002
 $4/\exp(2)$

0126. burst-boss3-lim-003
 $1/2$

0127. burst-boss3-lim-004
 $(e-1)/12$

0128. burst-boss3-lim-005
 $-1/24$

0129. burst-boss3-lim-006
 $-1/24$

0130. burst-boss3-lim-007
 $11/72$

0131. burst-boss3-lim-008
 $-1/42$

0132. burst-boss3-lim-009
 $-1/180$

0133. burst-boss3-lim-010
 $-1/12$

0134. burst-boss3-lim-011
 $1/16$

0135. burst-boss3-lim-012
 $\exp(1/6)$

0136. world-001
 $5/3$

0137. world-002
 $\log(2)$

0138. world-003
 $\exp(3)$

0139. world-007
 $1/6$

0140. world-008
 2

0141. world-011
 4

0142. world-012
 $2/3$

0143. world-013
 $\exp(1/2)$

0144. world-026
 $1/6$

0145. world-030
 $\exp(-1/2)$

0146. world-035
 $4/\exp(1)$

0147. world-040
 $11*\exp(1)/24$

0148. world-046
 $1/2$

0149. world-049
 $11*\exp(1)/24$

0150. world-066
 $\exp(1/3)$

0151. world-067
 $27/4$

0152. world-071
 $\pi/4$

0153. world-074
 $\sqrt{\pi}$

0154. world-091
 $-\exp(1)/2$

0155. world-096
 $\exp(1/6)$

0156. world-099
 $1/30$

0157. putnam-003
 $4/\exp(1)$

0158. putnam-008
 $\exp(2)$

0159. putnam-010
 $\pi/4$

0160. der-001	$5x^4-6x$	0196. der-033	$2y-\sin(x)$
0161. der-002	$\cos(x)-\sin(x)$	0197. der-034	$x-y$
0162. der-003	$\exp(x)*\log(x)+\exp(x)/x$	0198. der-035	$\exp(t)*(4t^3+t^4)$
0163. der-004	$1/(2*\sqrt{x})$	0199. der-036	4
0164. der-005	$2x/(1+x^2)$	0200. der-037	27
0165. der-006	$2x*\sin(x)+x^2*\cos(x)$	0201. der-038	$(x^2-2x-1)/(x-1)^2$
0166. der-007	$1/(1+x^2)$	0202. der-039	$2*\sin(x)*\cos(x)$
0167. der-008	$1/(x+1)^2$	0203. der-040	$2*x*\log(x)*\log(x)/x$
0168. der-009	$x^x*(\log(x)+1)$	0204. der-041	-0.04
0169. der-010	$\exp(-x)*(2x*\cos(x^2)-\sin(x^2))$	0205. der-042	-0.028
0170. der-011	$\cos(x)/\sin(x)$	0206. der-043	0
0171. der-012	$((x^2+1)/x-2x*\log(x))/(x^2+1)^2$	0207. der-044	$y*\exp(x*y)*dx+x*\exp(x*y)*dy$
0172. der-013	$x^{\sin(x)}*(\cos(x)*\log(x)+\sin(x)/x)$	0208. der-045	$-y*dx/(x^2+y^2)+x*dy/(x^2+y^2)$
0173. der-014	$1/(1+x^2)$	0209. der-046	4.92
0174. der-015	$1/\sqrt{1+x^2}$	0210. der-047	$-9/2$
0175. der-016	$1/(1+\cos(x))$	0211. der-048	-9
0176. der-017	$\exp(x^2)*(2x*\cos(3x)-3*\sin(3x))$	0212. der-049	-1
0177. der-018	$1/\sqrt{x^2-1}$	0213. der-050	5
0178. der-019	$10*x*(x^2+1)^4$	0214. der-051	3
0179. der-020	$1/(2*\sqrt{x}*(1+x))$	0215. der-052	144
0180. td-der-001	$2*(1-3*x^2)/(1+x^2)^3$	0216. der-053	-1
0181. td-der-002	$(35*x^3-15*x)/2$	0217. der-054	128
0182. td-der-003	$\log(1+x)+x/(1+x)$	0218. der-055	localminimum
0183. td-der-004	$-3*\sin(x)/(5+3*\cos(x))$	0219. der-056	saddlepoint
0184. der-021	$2*x*y+y*\cos(x*y)$	0220. der-057	0
0185. der-022	$x^2*\exp(x*y)+3*y^2$	0221. der-058	-4
0186. der-023	$\exp(x*y)*(1+x*y)$	0222. der-059	-1
0187. der-024	22/5	0223. der-060	1
0188. der-025	$2x/(x^2+y^2)$	0224. der-061	$\exp(3*x)$
0189. der-026	$2x*\sin(3x)+3x^2*\cos(3x)$	0225. der-062	x^2
0190. der-027	$x/(1+x^2)$	0226. der-063	2
0191. der-028	$\exp(x)/(1+\exp(2*x))$	0227. der-064	$\exp(2*x)$
0192. der-029	$\exp(x)*(x+2)$	0228. der-065	$1/x$
0193. der-030	$-(2*x+y)/(x+2*y)$	0229. der-066	x^4
0194. der-031	$2*x*\exp(x^2+y^2)$	0230. der-067	$4*x$
0195. der-032	$x/(x^2+y^2)$	0231. der-068	$\exp(\sin(x))*(2*x+x^2*\cos(x))$
		0232. der-069	$4*(x^2+y^2)$
		0233. der-070	$-4*y$
		0234. der-071	r
		0235. der-072	$-2*x/y$

0236. der-073 $20*((2*x+y)^2+(x+3*y)^2)$

0237. der-074 -4

0238. der-075 $x^3-3*x*y^2$

0239. der-076 $3*x^2*y-y^3$

0240. der-077 $\sqrt{2}$

0241. der-078 -4

0242. der-079 holomorphic

0243. der-080 notholomorphic

0244. der-081 $1/2$

0245. der-082 0

0246. der-083 $1/2$

0247. der-084 1

0248. der-085 -8

0249. der-086 $4*(x^2+y^2)$

0250. der-087 0

0251. der-088 harmonic

0252. gap-der-param-001
 $3*t/2$

0253. gap-der-param-002
 $-\cos(t)/\sin(t)$

0254. gap-der-param-003
 2

0255. gap-der-param-004
 3

0256. gap-der-polar-001
 1

0257. gap-der-param-005
 5

0258. gap-der-app-001
 $72*\pi$

0259. gap-der-app-002
 $2*\pi$

0260. gap-der-app-003
 $-3/2$

0261. gap-der-app-004
 $-1/4$

0262. gap-der-app-005
 2.01

0263. gap-der-app-006
 $3/2$

0264. gap-der-app-007
 2

0265. gap-der-ode-001
 $2*\exp(-x)$

0266. gap-der-ode-002
 $3*\exp(2*x)$

0267. gap-der-ode-003
 $(\exp(x)-\exp(-x))/2$

0268. gap-der-ode-004
 $\sin(x)$

0269. gap-der-ode-005
 $1/(1+\exp(-x))$

0270. gap-tech-007
parametricdifferentiation

0271. mob-tech-012
Lagrangemultiplier

0272. mob-tech-014
nabla

0273. mob-tech-018
Besselequation

0274. mob-tech-020
logarithmicdifferentiation

0275. mob-trap-001
missingchainfactor

0276. mob-trap-007
treatyconstant

0277. mob-trap-011
zero

0278. mob-lm-001
 25

0279. mob-lm-002
 18

0280. mob-lm-003
 64

0281. mob-lm-004
 16

0282. mob-lm-005
 16

0283. mob-lm-006
 3

0284. mob-lm-007
 25

0285. mob-lm-008
 $1/3$

0286. mob-lm-009
 25

0287. mob-lm-010
 2

0288. mob-bessel-001
 1

0289. mob-bessel-002
 0

0290. mob-bessel-003
 $-1/4$

0291. mob-bessel-004	0315. rel-adv-006
1/2	8/3
0292. mob-bessel-005	0316. rel-adv-007
-J ₁ (x)	128
0293. mob-bessel-006	0317. rel-adv-008
Besselequation	9
0294. mob-bessel-007	0318. rel-boss-007
regularsingular	1/64
0295. mob-bessel-008	0319. rel-hard-mv-005
J ₂ (x)	4
0296. mob-bessel-009	0320. rel-hard-mv-006
-J ₁ (x)	1/27
0297. mob-bessel-010	0321. rel-hard-mv-007
second-orderODE	2
0298. mob-nabla-001	0322. rel-hard-der-001
3	1
0299. mob-nabla-002	0323. rel-hard-der-002
-2	-1
0300. mob-nabla-003	0324. rel-hard-der-003
6	-4
0301. mob-nabla-004	0325. rel-hard-der-004
2*sqrt(2)	1/(2*sqrt(2))
0302. mob-nabla-005	0326. rel-hard-der-005
0	5/3
0303. mob-nabla-006	0327. rel-hard-der-006
0	3/8
0304. mob-nabla-007	0328. rel-hard-der-007
2	5/3
0305. mob-nabla-008	0329. rel-hard-der-008
Laplacian	6
0306. mob-nabla-009	0330. rel-hard-der-009
conservativefield	0
0307. mob-nabla-010	0331. rel-hard-der-010
curl	2
0308. rel-basic-007	0332. hc-der-003
12	exp(exp(2*x)+x^2)*(2*exp(2*x)+2*x)
0309. rel-basic-008	0333. hc-der-004
1	360
0310. rel-basic-009	0334. hc-der-005
2	42
0311. rel-basic-010	0335. hc-der-006
1/4	1680
0312. rel-basic-011	0336. hc-der-007
1/e	-1/11
0313. rel-basic-012	0337. hc-der-008
1	16
0314. rel-adv-005	0338. hc-der-009
20	4096
	0339. hc-der-010
	16

0340. exam-der-003	$(1-\log(x))/x^2$	0363. exam-multi-010	$-2*y$
0341. exam-der-004	$1/(1+x^2)$	0364. uni-der-001	$x^{\cos(x)}*(-\sin(x)*\log(x)+\cos(x)/x)$
0342. exam-der-005	$\exp(-x) * (2*x*\sin(x) - x^2*\sin(x) + x^2*\cos(x))$	0365. uni-der-002	$\sin(x)^x*(\log(\sin(x))+x*\cot(x))$
0343. exam-der-006	$1/\sqrt{1+x^2}$	0366. uni-der-003	11
0344. exam-der-007	$(2+4*x^2)*\exp(x^2)$	0367. uni-der-004	12
0345. exam-der-009	$-1/(1+\sin(x))$	0368. uni-der-005	$-x/(1+x^2)^{(3/2)}$
0346. exam-der-010	$\log(1+x^2)+2*x^2/(1+x^2)$	0369. uni-der-006	$\cos(x)*\exp(\sin(x)^2)$
0347. exam-der-011	$(x^2+1)^2*(6*x*(x-1)-(x^2+1))/(x-1)^2$	0370. uni-der-007	$2*\log(1+4*x^2)-\log(1+x^2)$
0348. exam-der-012	$2/x+1$	0371. uni-der-008	$2/(1+x^2)$
0349. exam-der-013	$11/4$	0372. uni-der-009	$5/3$
0350. exam-der-014	$2*x*\exp(x^4)$	0373. uni-der-010	1
0351. exam-der-015	$2*x*\log(1+x^4)-\log(1+x^2)$	0374. uni-der-011	-1
0352. exam-der-016	$1/2$	0375. uni-der-012	128
0353. exam-der-017	$1/(2*\sqrt{x}*(1+x))$	0376. uni-der-013	2
0354. exam-der-018	$\sec(x)$	0377. uni-der-014	$6/5$
0355. exam-der-019	$1/(1+x^2)^{(3/2)}$	0378. uni-der-015	$\exp(y)+\exp(z)+\exp(x)$
0356. exam-der-020	$\exp(x)/(1+\exp(x))^2$	0379. uni-der-016	-x
0357. exam-multi-002	$6*x*y^2+(1+x*y)*\exp(x*y)$	0380. uni-der-017	$1/(x*\log(x)*\log(\log(x)))$
0358. exam-multi-003	20	0381. uni-der-018	$3*\log(x)^2*\exp(\log(x)^3)/x$
0359. exam-multi-004	27	0382. uni-der-019	32
0360. exam-multi-005	$16/5$	0383. uni-der-020	$(1+x^2)^x*(\log(1+x^2)+2*x^2/(1+x^2))$
0361. exam-multi-008	$4*(x^2+y^2)$	0384. depth-der-001	$x^{(x^2)}*(2*x*\log(x)+x)$
0362. exam-multi-009	$2*x*y+2*y*z+2*z*x$	0385. depth-der-002	$\sin(x)^{\cos(x)}*(-\sin(x)*\log(\sin(x))+\cos(x)*\cot(x))$
		0386. depth-der-004	$2/x$

0387. depth-der-005
 $3x^2\cos(x^6)-2x\cos(x^4)$

0388. depth-der-006
 $1/\sqrt{x^2+4}$

0389. depth-der-007
 -1

0390. depth-der-008
 1

0391. depth-der-009
 $3/4$

0392. depth-der-010
 -3

0393. depth-der-011
 8

0394. depth-der-012
 0

0395. depth-der-013
 $y*z$

0396. depth-der-014
 y^2-x^2

0397. depth-der-015
 $2/(1+x^2)$

0398. depth-der-016
 $(2*\log(x)-\log(x)^2)/x^2$

0399. depth-der-017
 81

0400. depth-der-018
 $(1+1/(2*\sqrt{x}))/ (2*\sqrt{x+\sqrt{x}})$

0401. burst-der-001
 1048576

0402. burst-der-002
 -14348907

0403. burst-der-003
 262144

0404. burst-der-004
 720

0405. burst-der-005
 56320

0406. burst-der-006
 -5040

0407. burst-der-007
 479001600

0408. burst-der-008
 -64

0409. burst-der-009
 24024

0410. burst-der-010
 -1199

0411. burst-boss-der-001
 241087680

0412. burst-boss-der-002
 -21257280

0413. burst-boss-der-003
 -22591

0414. burst-boss-der-004
 16124

0415. burst-boss-der-005
 164833

0416. burst-boss-der-006
 351261243360

0417. burst-boss-der-007
 27525120

0418. burst-boss-der-008
 13172296626

0419. burst-boss-der-009
 598752000

0420. burst-boss-der-010
 1307674368000

0421. burst-boss2-der-001
 -1024

0422. burst-boss2-der-002
 -1024

0423. burst-boss2-der-003
 3041525760

0424. burst-boss2-der-004
 54747463680

0425. burst-boss2-der-005
 -1334792724768

0426. burst-boss2-der-006
 239500800

0427. burst-boss2-der-007
 1796256000

0428. burst-boss2-der-008
 271996268544000

0429. burst-boss3-der-001
 863130293635276800

0430. burst-boss3-der-002
 248156004834017280000

0431. burst-boss3-der-003
 37719712734770626560000

0432. burst-boss3-der-004
 1621934672117760000

0433. burst-boss3-der-005
 $17166620433372086476800000$

0434. burst-boss3-der-006
 553761949463615692800000

0435. burst-boss3-der-007
 -272789137666805760000

0436. burst-boss3-der-008
 18458731648787189760000

0437. world-004
 $1/(1+x^2)$

0438. world-024
 -4

0439. world-025
 $2/15$

0440. world-036
 $61/720$

0441. world-068
 $5/6$

0442. world-093
 $-1/8$

0443. hd-001 5040

0444. hd-002 12

0445. hd-003 120

0446. hd-004 10

0447. hd-005 24

0448. hd-006 -4

0449. hd-007 -720

0450. hd-008 $17/315$

0451. hd-009 $2*\arcsin(x)/\sqrt{1-x^2}$

0452. hd-010 $x^{(x^2)}*(2*x*\log(x)+x)$

0453. hd-011 $1/\sin(x)$

0454. hd-012 $2/3$

0455. app-001
 $1/\pi$

0456. app-002
12

0457. app-003
 $1/2$

0458. app-004
 $2*\sqrt{2}$

0459. app-005
 $3/2$

0460. app-006
 $81/40$

0461. app-007
4

0462. app-008
2

0463. app-009
 $4*\pi/(3*\sqrt{3})$

0464. app-010
 $32*\pi/81$

0465. app-011
12

0466. app-012
 $2/(3*\sqrt{3})$

0467. int-001 $2*x^3$

0468. int-002 $\sin(x)$

0469. int-003 $\log(\abs{x})$

0470. int-004 $\exp(3*x)/3$

0471. int-005 $\exp(x^2)/2$

0472. int-006 $\log(1+x^2)$

0473. int-007 $\sin(x^2)/2$

0474. int-008 1

0475. int-009 $x^2*\log(x)/2-x^2/4$

0476. int-010 $-\cos(x)+\cos(x)^3/3$

0477. int-011 $\operatorname{atan}(x/2)/2$

0478. int-012 $\pi/4$

0479. int-013 $x*(\log(x)^2-2*\log(x)+2)$

0480. int-014 $x^2/2-\log(x^2+1)/2$

0481. int-015 $\exp(x)*(\sin(x)-\cos(x))/2$

0482. int-016 $2*\log(2)-1$

0483. int-017 1

0484. int-018 $-\log(1+\cos(x))$

0485. int-019 $2*\sqrt{x}-2*\log(1+\sqrt{x})$

0486. int-020 $\log(2)/3$

0487. td-int-001
 $\pi*\exp(-3)/2$

0488. td-int-002
 $\pi*\log(9/2)$

0489. td-int-003
 $\log(2/5)$

0490. td-int-004
 $\log(7/3)/2$

0491. td-int-005
 $\pi/\sqrt{5}$

0492. td-int-006
 $\sqrt{\pi}/(12*\sqrt{3})$

0493. int-021 1

0494. int-022 2

0495. int-023 $1/8$

0496. int-024 $1/3$

0497. int-025 $\pi/2$

0498. int-026 9
 0499. int-027 $\log(1+x^2)/2$
 0500. int-028 $x \cdot \log(x) - x$
 0501. int-029 $\operatorname{atan}(x/3)/3$
 0502. int-030 $1/4$
 0503. int-031 $1/4$
 0504. int-032 3π
 0505. int-033 4π
 0506. int-034 $\pi/4$
 0507. int-035 $\pi/2$
 0508. int-036 $1/6$
 0509. int-038 $x \cdot \sin(x) + \cos(x)$
 0510. int-039 $\pi/4$
 0511. int-040 1
 0512. int-041 $\log(1+x^4)/4$
 0513. int-042 $(2/9) \cdot (1+x^3)^{(3/2)}$
 0514. int-043 $\log(1+\sqrt{1+x^2})$
 0515. int-044 $\log(1+\exp(x^2))/2$
 0516. int-045 $\log(x)^{3/3}$
 0517. int-046 $3/8$
 0518. int-047 π
 0519. int-048 $\pi/4$
 0520. int-049 π
 0521. int-050 $\pi/3$
 0522. int-051 $x/(4\sqrt{x^2+4})$
 0523. int-052 $\exp(x) \cdot (x^2-2x+2)$
 0524. int-053 $-x^2 \cos(x) + 2x \sin(x) + 2 \cos(x)$
 0525. int-055 $x \cdot (\log(x)^2 - 2 \log(x) + 2)$
 0526. int-056 $\pi/4 - 1/2$
 0527. int-057 $(\exp(x) - \sin(x) - \cos(x))/2$
 0528. int-058 $(\exp(2x) - 2x - 1)/4$
 0529. int-059 $x^2 - 2x + 2 - 2\exp(-x)$
 0530. int-060 $x \cdot \sin(x)/2$
 0531. int-061 $\pi/4$
 0532. int-062 $1/2$
 0533. int-063 $\pi/4$
 0534. int-064 $\pi^2/4$
 0535. int-065 $\log(7/3)$
 0536. int-066 $\log(4)$

0537. int-067 $\log(4)$
 0538. int-068 $\log(5/2)$
 0539. int-069 $\log(4)$
 0540. int-070 $\log(3)$
 0541. int-071 $\log(11/5)/2$
 0542. int-072 $\log(2)/2$
 0543. int-073 $\log(5/3)/2$
 0544. int-074 $\pi \cdot \log(5/2)/2$
 0545. int-075 $\log(5/2)$
 0546. int-076 $\log(6)$
 0547. gap-int-pf-001
 $(\log(x-1) - \log(x+1))/2$
 0548. gap-int-pf-002
 $\log(x)$
 0549. gap-int-pf-003
 $(\log(x) - \log(x+2))/2$
 0550. gap-int-pf-004
 $\log(x+1) + \log(x+2)$
 0551. gap-int-pf-005
 $-1/(x+1)$
 0552. gap-int-pf-006
 $8 \cdot \log(x-1)/3 + \log(x+2)/3$
 0553. gap-int-pf-007
 $2 \cdot \log(x+2) - \log(x+3)$
 0554. gap-int-polar-001
 π
 0555. gap-int-polar-002
 $\pi/4$
 0556. gap-int-triple-001
 $1/8$
 0557. gap-int-triple-002
 $3/2$
 0558. gap-int-triple-003
 $1/6$
 0559. gap-int-triple-004
 $1/6$
 0560. gap-int-triple-005
 12π
 0561. gap-int-triple-006
 $32\pi/3$
 0562. gap-int-cov-001
 $1/2$
 0563. gap-int-cov-002
 1
 0564. gap-tech-003
 substitution

0565. gap-tech-004
integrationbyparts

0566. gap-tech-006
partialfraction

0567. mob-tech-003
u-substitution

0568. mob-tech-004
integrationbyparts

0569. mob-tech-005
partialfraction

0570. mob-tech-006
trigsubstitution

0571. mob-tech-010
King'sproperty

0572. mob-tech-011
Frullaniintegral

0573. mob-tech-013
gradienttheorem

0574. mob-tech-015
Betafunction

0575. mob-tech-016
Gammafunction

0576. mob-tech-017
Wallisreduction

0577. mob-trap-002
dividebyinnercoefficient

0578. mob-trap-003
arctannotlog

0579. mob-trap-006
requiresapositive

0580. mob-trap-008
multiplybyJacobian

0581. mob-trap-010
secx

0582. mob-trap-012
denominatorGamma(a+b)

0583. mob-beta-001
1

0584. mob-beta-002
1/2

0585. mob-beta-003
1/6

0586. mob-beta-004
1/12

0587. mob-beta-005
pi

0588. mob-beta-006
2

0589. mob-beta-007
pi/2

0590. mob-beta-009
1/30

0591. mob-beta-010
1/4

0592. mob-gamma-001
1

0593. mob-gamma-002
sqrt(pi)

0594. mob-gamma-003
sqrt(pi)/2

0595. mob-gamma-004
3*sqrt(pi)/4

0596. mob-gamma-005
120

0597. mob-gamma-006
2

0598. mob-gamma-007
sqrt(pi)/2

0599. mob-gamma-008
6

0600. mob-gamma-009
15*sqrt(pi)/8

0601. mob-gamma-010
1

0602. mob-wallis-002
2/3

0603. mob-wallis-003
3*pi/16

0604. mob-wallis-004
8/15

0605. mob-wallis-005
pi/4

0606. mob-wallis-006
3*pi/16

0607. mob-wallis-007
pi/16

0608. mob-wallis-008
1

0609. mob-wallis-009
1

0610. mob-wallis-010
1/2

0611. rel-basic-013
1/3

0612. rel-basic-014
2

0613. rel-basic-015
(e^2-1)/2

0614. rel-basic-016	$\log(2)$	0638. rel-hard-int-009	$\log(5/2)$
0615. rel-basic-018	6	0639. rel-hard-int-011	$\pi/4$
0616. rel-adv-011	$1/9$	0640. rel-hard-int-012	$\pi/(2*\sqrt{2})$
0617. rel-adv-014	$5/6$	0641. rel-hard-int-015	-4
0618. rel-adv-009	$\exp(x^2)*(x^2-1)/2$	0642. rel-hard-int-017	$-\pi*\log(2)$
0619. rel-adv-010	$(1+x^2)*\log(1+x^2)/2-(1+x^2)/2$	0643. rel-hard-int-018	$\pi/32$
0620. rel-boss-001	$\pi/2$	0644. rel-hard-cpx-001	$-1/6$
0621. rel-boss-002	$\log(5/2)$	0645. rel-hard-cpx-002	$-1/2$
0622. rel-boss-003	$\log(7/2)$	0646. rel-hard-cpx-003	$1/6$
0623. rel-boss-004	$15*\sqrt{\pi}/8$	0647. rel-hard-cpx-004	$-1/2$
0624. rel-boss-005	$3*\pi/8$	0648. rel-hard-cpx-005	1
0625. rel-boss-008	0	0649. rel-hard-cpx-006	e
0626. rel-boss-009	$1/6$	0650. rel-hard-cpx-007	1
0627. rel-boss-010	$1/18$	0651. rel-hard-cpx-008	1
0628. rel-boss-011	$1/720$	0652. rel-hard-cpx-009	2
0629. rel-boss-012	$128*\pi/5$	0653. rel-hard-cpx-010	$-1/24$
0630. rel-boss-017	2	0654. rel-hard-mv-001	$1/60$
0631. rel-boss-020	$1/2$	0655. rel-hard-mv-002	$\pi/24$
0632. rel-hard-int-001	$15/8$	0656. rel-hard-mv-003	$4*\pi/5$
0633. rel-hard-int-002	$3*\sqrt{\pi}/128$	0657. rel-hard-mv-004	$36*\pi$
0634. rel-hard-int-004	$1/32$	0658. rel-hard-mv-008	$1/2$
0635. rel-hard-int-005	$35*\pi/256$	0659. rel-hard-mv-009	$64*\pi/3$
0636. rel-hard-int-007	$\pi/32$	0660. rel-hard-mv-010	$1/2520$
0637. rel-hard-int-008	$\log(11/3)$	0661. hc-usub-001	$(1+x^2)^{(3/2)}/3-\sqrt{1+x^2}$
		0662. hc-usub-002	$\exp(x^2)*(x^4-2*x^2+2)/2$

0663. hc-usub-003

$$\frac{(1+x^2) * (\log(1+x^2)^2 - 2 * \log(1+x^2) + 2)}{2}$$

0664. hc-usub-004

$$\operatorname{atan}(x^3)/3$$

0665. hc-usub-005

$$\log(1+\sqrt{1+x^4})/2$$

0666. hc-usub-006

$$2 * (1 + \exp(x))^{3/2} / 3$$

0667. hc-usub-007

$$\pi/6$$

0668. hc-usub-008

$$1/16$$

0669. hc-usub-009

$$1/4$$

0670. hc-usub-010

$$\log((1+e)/2)/3$$

0671. hc-ibp-002

$$\frac{x^2 * \sin(3x)}{9 - 2 * \sin(3x)} + \frac{2 * x * \cos(3x)}{27}$$

0672. hc-ibp-003

$$\frac{\exp(x) * ((x^2 - 1) * \sin(x) + (2x - 1 - x^2) * \cos(x))}{2}$$

0673. hc-ibp-005

$$\frac{(x^2 + 1) * \operatorname{atan}(x)^2}{2 - x * \operatorname{atan}(x)} + \log(1+x^2)/2$$

0674. hc-ibp-006

$$\exp(2x) * (2 * \sin(3x) - 3 * \cos(3x)) / 13$$

0675. hc-ibp-007

$$x^4 * \log(x) / 4 - x^4 / 16$$

0676. hc-ibp-008

$$2 * \log(2) / 5 - 47 / 300$$

0677. hc-ibp-009

$$\pi^{2/4 - 2}$$

0678. hc-ibp-010

$$1/6$$

0679. hc-rad-001

$$\sqrt{x^2 - 1} / x$$

0680. hc-rad-002

$$\operatorname{atan}((x+1)/2)/2$$

0681. hc-rad-003

$$-\log(\operatorname{abs}(x)) + \log(\operatorname{abs}(x-1)) + \log(\operatorname{abs}(x+1))$$

0682. hc-rad-004

$$\frac{(\log(\operatorname{abs}(x-1)) - \log(\operatorname{abs}(x+1)))}{4 - \operatorname{atan}(x)/2}$$

0683. hc-rad-005

$$2 * \sqrt{3} - 2 * \pi / 3$$

0684. hc-rad-006

$$1$$

0685. hc-rad-007

$$8/315$$

0686. hc-rad-008

$$\pi/3$$

0687. hc-rad-010

$$\pi/4$$

0688. hc-spec-002

$$\pi/8$$

0689. hc-spec-005

$$8 * \pi^6 / 63$$

0690. hc-spec-006

$$3 * \sqrt{\pi} / (32 * \sqrt{2})$$

0691. exam-int-001

$$-1 / (2 * (1 + x^2))$$

0692. exam-int-002

$$x^2 / 2 - \log(1 + x^2) / 2$$

0693. exam-int-004

$$x^2 * \log(x) / 2 - x^2 / 4$$

0694. exam-int-005

$$\log(x)^2 / 2$$

0695. exam-int-007

$$\tan(x)^2 / 2$$

0696. exam-int-010

$$2 * \log(1 + x^{3/2}) / 3$$

0697. exam-int-011

$$-\sqrt{4 - x^2}$$

0698. exam-int-012

$$\exp(2x) * (2 * \sin(x) - \cos(x)) / 5$$

0699. exam-int-013

$$x * \operatorname{atan}(x) - \log(1 + x^2) / 2$$

0700. exam-int-014

$$x - \operatorname{atan}(x)$$

0701. exam-int-015

$$\log(\log(x))$$

0702. exam-int-016

$$\sin(\log(x))$$

0703. exam-int-017

$$\log(x^2 + 3x + 5)$$

0704. exam-int-018

$$\operatorname{atan}(x^2) / 2$$

0705. exam-int-019

$$2 * \operatorname{atan}(\sqrt{x})$$

0706. exam-int-020

$$2 * \sqrt{1 + x^3} / 3$$

0707. exam-int-021

$$\log(2) / 2$$

0708. exam-int-024

$$\log(2)$$

0709. exam-int-025
 -1

0710. exam-int-026
 $1/2$

0711. exam-int-028
 $\log(2)-1/2$

0712. exam-int-029
 π

0713. exam-int-030
 $1/6$

0714. exam-multi-006
 $1/2$

0715. exam-multi-007
 π

0716. uni-int-003
 $-(\log(x)+1)/x$

0717. uni-int-004
 $-\operatorname{atan}(x)/x+\log(x)-\log(1+x^2)/2$

0718. uni-int-005
 $(1+x^2)^{(3/2)}/3$

0719. uni-int-006
 $\sin(x)^6/6$

0720. uni-int-007
 $\tan(x)+\tan(x)^3/3$

0721. uni-int-008
 $\operatorname{atan}((x+1)/2)/2$

0722. uni-int-009
 $\log(x^2+x+1)$

0723. uni-int-010
 $\exp(x)*(\sin(x)+\cos(x))/2$

0724. uni-int-011
 $x^3*\log(x)/3-x^3/9$

0725. uni-int-012
 $(x^2+1)*\operatorname{atan}(x)/2-x/2$

0726. uni-int-013
 $1/2-\log(2)/2$

0727. uni-int-014
 $\pi/4$

0728. uni-int-015
 $1/4$

0729. uni-int-016
 $2/15$

0730. uni-int-018
 $1/2$

0731. uni-int-020
 $\pi/(3*\sqrt{3})$

0732. uni-int-023
 $6*\pi$

0733. uni-int-025
 $\pi/4$

0734. depth-int-002
 $x^3/3-\log(1+x^3)/3$

0735. depth-int-003
 $-1/\sqrt{1+x^2}$

0736. depth-int-005
 $\exp(x)*(x*(\sin(x)-\cos(x))+\cos(x))/2$

0737. depth-int-006
 $\exp(2*x)*(2*\cos(3*x)+3*\sin(3*x))/13$

0738. depth-int-007
 $2*\sqrt{x}*\operatorname{atan}(\sqrt{x})-\log(1+x)$

0739. depth-int-008
 $2*\sqrt{\log(x)}$

0740. depth-int-009
 $x*(\sin(\log(x))-\cos(\log(x)))/2$

0741. depth-int-010
 $2*\log(1+\sqrt{x})$

0742. depth-int-011
 $\operatorname{atan}(x)/2-x/(2*(1+x^2))$

0743. depth-int-013
 $\log(1+\tan(x))$

0744. depth-int-014
 $x^2*\log(1+x)/2-x^2/4+x/2-\log(1+x)/2$

0745. depth-int-015
 $\log(3)/2-\pi/(3*\sqrt{3})$

0746. depth-int-016
 $\pi/2-1$

0747. depth-int-017
 $\pi^2/12$

0748. depth-int-018
 $\log(2)/2-1/4$

0749. depth-int-019
 $1/2$

0750. depth-int-020
 $\pi/12$

0751. depth-int-021
 $3*\pi/8$

0752. depth-int-022
 $1/3$

0753. depth-int-023
 $3*\pi/2$

0754. depth-int-026
 $1/2$

0755. depth-int-027
 $\pi/2$

0756. depth-int-028
 $-1/9$

0757. burst-int-001
 $8/81$

0758. burst-int-002	$1/2$	0782. burst-int-026	$\pi^2/6$
0759. burst-int-003	$4/125$	0783. burst-int-027	$\pi/(2\sqrt{2})$
0760. burst-int-004	$-1/36$	0784. burst-int-028	0
0761. burst-int-005	$2/125$	0785. burst-int-029	$\pi/\sqrt{2}$
0762. burst-int-006	$-2/27$	0786. burst-int-030	$-\pi\pi/12$
0763. burst-int-007	$1/280$	0787. burst-int-031	$3\pi/16$
0764. burst-int-008	$\pi/8$	0788. burst-int-032	$5\pi/2048$
0765. burst-int-009	$3\sqrt{\pi}/8$	0789. burst-int-033	$231\pi/2048$
0766. burst-int-010	1	0790. burst-int-034	$1024/3003$
0767. burst-int-011	$-\pi\log(2)/2$	0791. burst-int-035	$1/210$
0768. burst-int-012	$\pi\log(2)^2/2+\pi^3/24$	0792. burst-int-036	$\exp(2x)(x^3/2-3x^2/4+3x/4-3/8)$
0769. burst-int-013	$35\pi^2/256$	0793. burst-int-037	$-(x^2)\cos(3x)/3+2x\sin(3x)/9+2\cos(3x)/27$
0770. burst-int-014	$63\pi/512$	0794. burst-int-038	$x(\log(x)^3-3\log(x)^2+6\log(x)-6)$
0771. burst-int-015	$256/693$	0795. burst-int-039	$x^5\log(x)/5-x^5/25$
0772. burst-int-016	$3\pi/512$	0796. burst-int-040	$(x^4-1)\operatorname{atan}(x)/4-x^3/12+x/4$
0773. burst-int-017	$3\pi/256$	0797. burst-int-041	$-\exp(-x)(x^3+3x^2+6x+6)$
0774. burst-int-018	$1/24$	0798. burst-int-042	$x^3\log(1+x)/3-x^3/9+x^2/6-x/3+\log(1+x)/3$
0775. burst-int-019	$7\pi/512$	0799. burst-rec-001	$-1/(n+1)^2$
0776. burst-int-020	$35\pi/128$	0800. burst-rec-002	$2/(n+1)^3$
0777. burst-int-021	$16/35$	0801. burst-rec-003	$2/((n+1)(n+2)(n+3))$
0778. burst-int-022	$35\pi/256$	0802. burst-rec-004	$(n-1)/n$
0779. burst-int-023	$2/63$	0803. burst-rec-005	$(2n-1)/(2n)$
0780. burst-int-024	$5\pi/256$	0804. burst-rec-006	$(2n)/(2n+1)$
0781. burst-int-025	$\pi^4/15$	0805. burst-rec-007	n

0806. burst-rec-008

$$-n$$

0807. burst-rec-009

$$1/(n+1)+1/(n+2)$$

0808. burst-rec-010

$$1/((n+1)*(n+2))$$

0809. burst-boss-int-001

$$\pi^{3/8}$$

0810. burst-boss-int-002

$$-(\pi^2)/(8*\sqrt{2})$$

0811. burst-boss-int-003

$$\pi^2/(2*\sqrt{2})$$

0812. burst-boss-int-004

$$\pi/3$$

0813. burst-boss-int-005

$$\pi/6$$

0814. burst-boss-int-006

$$\pi/3$$

0815. burst-boss-int-007

$$\pi/4$$

0816. burst-boss-int-008

$$\pi/(2*\sqrt{2})$$

0817. burst-boss-int-009

$$2-\pi^{2/6}$$

0818. burst-boss-int-010

$$-(\pi^2)/6$$

0819. burst-boss-int-011

$$24$$

0820. burst-boss-int-012

$$3/128$$

0821. burst-boss-int-013

$$-40/243$$

0822. burst-boss-int-014

$$\pi*\log(2)^2/2-\pi^{3/48}$$

0823. burst-boss-int-015

$$\pi^{3/8}$$

0824. burst-boss-int-016

$$63*\pi^{2/512}$$

0825. burst-boss-int-017

$$35*\pi/65536$$

0826. burst-boss-int-018

$$429*\pi/4096$$

0827. burst-boss-int-019

$$2048/6435$$

0828. burst-boss-int-020

$$5280/15625$$

0829. burst-boss-anti-001

$$\exp(x)*(x^4-4*x^3+12*x^2-24*x+24)$$

0830. burst-boss-anti-002

$$-\exp(-2*x)*(x^4/2+x^3+3*x^2/2+3*x/2+3/4)$$

0831. burst-boss-anti-003

$$-(x^3)*\cos(2*x)/2+3*x^2*\sin(2*x)/4+3*x*\cos(2*x)/4-3*\sin(2*x)/8$$

0832. burst-boss-anti-004

$$x^3*\sin(2*x)/2+3*x^2*\cos(2*x)/4-3*x*\sin(2*x)/4-3*\cos(2*x)/8$$

0833. burst-boss-anti-005

$$x^5*(\log(x)^2/5-2*\log(x)/25+2/125)$$

0834. burst-boss-anti-006

$$x*(\log(x)^4-4*\log(x)^3+12*\log(x)^2-24*\log(x)+24)$$

0835. burst-boss-anti-007

$$x^3*\operatorname{atan}(x)/3-x^2/6+\log(1+x^2)/6$$

0836. burst-boss-anti-008

$$x^5*\operatorname{atan}(x)/5-x^4/20+x^2/10-\log(1+x^2)/10$$

0837. burst-boss-anti-009

$$x^3*\log(1+x^2)/3-2*x^3/9+2*x/3-2*\operatorname{atan}(x)/3$$

0838. burst-boss-anti-010

$$x^4*\log(1+x^2)/4-x^4/8+x^2/4-\log(1+x^2)/4$$

0839. burst-boss-param-001

$$1/(a^2+1)$$

0840. burst-boss-param-002

$$1/a^2$$

0841. burst-boss-param-003

$$\pi/(2*\sqrt{a})$$

0842. burst-boss-param-004

$$\pi*\log((1+\sqrt{1+a}))/2$$

0843. burst-boss-param-005

$$\pi/(2*\sin(\pi*(n+1)/2))$$

0844. burst-boss-param-006

$$-6/(n+1)^4$$

0845. burst-boss2-int-001

$$-(2*\pi^2)/27$$

0846. burst-boss2-int-002

$$10*\pi^3/(81*\sqrt{3})$$

0847. burst-boss2-int-003

$$\pi/3$$

0848. burst-boss2-int-004

$$\pi/\sqrt{3}$$

0849. burst-boss2-int-005

$$\pi*\log(2)/2$$

0850. burst-boss2-int-006

$$\pi/2$$

0851. burst-boss2-int-007

$$\pi/2$$

0852. burst-boss2-int-008

$$-\pi^4/15$$

0853. burst-boss2-int-009

$$-8\pi^6/63$$

0854. burst-boss2-int-010

$$-7\pi^4/120$$

0855. burst-boss2-int-011

$$-31\pi^6/252$$

0856. burst-boss2-int-012

$$\pi^2/16+1/4$$

0857. burst-boss2-int-013

$$\pi/8$$

0858. burst-boss2-int-014

$$\pi^2-4$$

0859. burst-boss2-param-001

$$b/(a^2+b^2)$$

0860. burst-boss2-param-002

$$a/(a^2+b^2)$$

0861. burst-boss2-param-003

$$2a/(a^2+1)^2$$

0862. burst-boss2-param-004

$$(a^2-1)/(a^2+1)^2$$

0863. burst-boss2-param-005

$$\log(a+1)$$

0864. burst-boss2-param-006

$$-120/(a+1)^6$$

0865. burst-boss2-anti-001

$$\exp(2x) * (x^5 / 2 - 5x^4 / 4 + 5x^3 / 2 - 15x^2/4 + 15x/4 - 15/8)$$

0866. burst-boss2-anti-002

$$-(x^4) * \cos(x) + 4x^3 * \sin(x) + 12x^2 * \cos(x) - 24x * \sin(x) - 24 * \cos(x)$$

0867. burst-boss2-anti-003

$$x^4 * \sin(x) + 4x^3 * \cos(x) - 12x^2 * \sin(x) - 24x * \cos(x) + 24 * \sin(x)$$

0868. burst-boss2-anti-004

$$x * (\log(x))^5 - 5 * \log(x)^4 + 20 * \log(x)^3 - 60 * \log(x)^2 + 120 * \log(x) - 120$$

0869. burst-boss2-anti-005

$$x^6 * \operatorname{atan}(x) / 6 - x^5 / 30 + x^3 / 18 - x / 6 + \operatorname{atan}(x) / 6$$

0870. burst-boss2-anti-006

$$x^4/4 - x^2/2 + \log(1+x^2)/2$$

0871. burst-boss3-int-001

$$1/27$$

0872. burst-boss3-int-002

$$-1/54$$

0873. burst-boss3-int-003

$$8\pi^2/3$$

0874. burst-boss3-int-004

$$\pi^2/3$$

0875. burst-boss3-int-005

$$-\pi^2 * \log(2)/2$$

0876. burst-boss3-int-006

$$\pi^3 - 6\pi$$

0877. burst-boss3-int-007

$$\pi^4 - 12\pi^2 + 48$$

0878. burst-boss3-int-008

$$-714/28561$$

0879. burst-boss3-int-009

$$-720/28561$$

0880. burst-boss3-int-010

$$984/3125$$

0881. burst-boss3-int-011

$$-912/3125$$

0882. burst-boss3-int-012

$$31\pi^6/252$$

0883. burst-boss3-int-013

$$7\pi^4/120$$

0884. burst-boss3-param-001

$$18(a^2-3)/(a^2+9)^3$$

0885. burst-boss3-param-002

$$2(a^3-3a)/(a^2+1)^3$$

0886. burst-boss3-param-003

$$24a(a^2-1)/(a^2+1)^4$$

0887. burst-boss3-param-004

$$24/(a+1)^5$$

0888. burst-boss3-param-005

$$720/(a+1)^7$$

0889. burst-boss3-param-006

$$2/(a+1)^3$$

0890. burst-boss3-anti-001

$$-\exp(-3x) * (x^6/3 + 2x^5/3 + 10x^4/9 + 40x^3/27 + 40x^2/27 + 80x/81 + 80/243)$$

0891. burst-boss3-anti-002

$$-x^5 * \cos(2x) / 2 + 5x^4 * \sin(2x) / 4 + 5x^3 * \cos(2x) / 2 - 15x^2 * \sin(2x) / 4 - 15x * \cos(2x) / 4 + 15 * \sin(2x) / 8$$

0892. burst-boss3-anti-003

$$x^5 * \sin(2x) / 2 + 5x^4 * \cos(2x) / 4 - 5x^3 * \sin(2x) / 2 - 15x^2 * \cos(2x) / 4 + 15x * \sin(2x) / 4 + 15 * \cos(2x) / 8$$

0893. burst-boss3-anti-004

$$x^7 * \operatorname{atan}(x) / 7 - x^6/42 + x^4/28 - x^2/14 + \log(1+x^2)/14$$

0894. burst-boss3-anti-005

$$x^7 * \log(1+x) / 7 - x^7/49 + x^6/42 - x^5/35 + x^4/28 - x^3/21 + x^2/14 - x/7 + \log(1+x)/7$$

0895. burst-boss3-anti-006

$$x^6/6 - x^4/4 + x^2/2 - \log(1+x^2)/2$$

0896. world-005
 $8/15$

0897. world-010
 $3\pi/8$

0898. world-014
 $\pi \log(2)/8$

0899. world-015
 $\pi \log(2)/8$

0900. world-016
 $\log(3/2)$

0901. world-017
 $2\pi/(3\sqrt{3})$

0902. world-018
 $-\log(2)$

0903. world-019
 $\log(3)$

0904. world-027
 $1/2$

0905. world-029
 $-\pi/4$

0906. world-031
 $\pi^{2/4}$

0907. world-032
 $\pi^{3/16}$

0908. world-033
 π

0909. world-037
 $\sqrt{\pi} \exp(-1)/2$

0910. world-038
 $3\pi/8$

0911. world-041
 $\log(2)$

0912. world-043
 $-\pi^{2/3}/(6\sqrt{3})$

0913. world-044
 $\pi \log(2)/2$

0914. world-045
 $\pi/\exp(1)$

0915. world-047
 $\pi/3$

0916. world-050
 $-1/16$

0917. world-051
 $5\pi^{2/96}$

0918. world-052
 $\pi \log(2)^{2/2} + \pi^{3/24}$

0919. world-054
 $\pi \log(2)^{2/2} + \pi^{3/24}$

0920. world-055
 -1

0921. world-056
 $2 - 2\log(2) - \pi^{2/12}$

0922. world-057
 $\pi \log(2)$

0923. world-058
 $\sqrt{\pi} \exp(-1/4)/2$

0924. world-059
 $\log(2)/3 + \pi/(3\sqrt{3})$

0925. world-061
 $-\pi^{2/6}$

0926. world-062
 $-\pi^{2/8}$

0927. world-063
 $\pi/16$

0928. world-064
 $\log(4)$

0929. world-065
 $\pi/\sqrt{2}$

0930. world-069
 $\pi/12$

0931. world-070
 $\pi/4$

0932. world-072
 $4\pi/(3\sqrt{3})$

0933. world-073
 $\pi \log((2+\sqrt{3}))/2$

0934. world-076
 $\pi^{2/8}$

0935. world-077
 $\pi \log(2)/8$

0936. world-079
 -6

0937. world-080
 $\pi/(2\sqrt{3})$

0938. world-081
 $\pi/2$

0939. world-082
 $\pi^{2/4-2}$

0940. world-083
 $\pi^{2/12}$

0941. world-084
 $\log(2)^{2/2} - \pi^{2/12}$

0942. world-085
 $\pi/(2\sqrt{2})$

0943. world-088
 $\pi/2$

0944. world-089
 $\pi/4$

0945. world-090
 $\sqrt{\pi}/2$

0946. world-092
 $-3/4$

0947. world-094
 $\pi/(2\sqrt{2})$

0948. world-095
 $\log(2)-1$

0949. world-098
 $\pi/(2\sqrt{2})$

0950. world-100
 $\pi\log(2)$

0951. putnam-004
 $\pi/4$

0952. putnam-005
 $\pi/4$

0953. putnam-009
 $\pi/4$

0954. putnam-012
 $\pi\log(2)/2$

0955. ser-001 收斂

0956. ser-002 發散

0957. ser-003 $3/2$

0958. ser-004 條件收斂

0959. ser-005 發散

0960. ser-006 收斂

0961. ser-007 $\exp(3)-1$

0962. ser-008 0

0963. ser-009 1

0964. ser-010 條件收斂

0965. ser-011 $9/4$

0966. ser-012 1

0967. ser-013 發散

0968. ser-014 2

0969. ser-015 6

0970. ser-016 收斂

0971. ser-017 3

0972. ser-018 絕對收斂

0973. ser-019 $3/4$

0974. ser-020 發散

0975. td-ser-001
 $33/40$

0976. td-ser-002
 $13/12$

0977. td-ser-003
 $2\log(2)$

0978. ser-024 發散

0979. ser-027 $2/3$

0980. ser-030 5

0981. ser-031 發散

0982. ser-032 發散

0983. ser-034 0

0984. ser-035 Infinity

0985. ser-036 收斂

0986. ser-037 收斂

0987. ser-038 $1/3$

0988. ser-039 $1/2$

0989. gap-ser-root-001
 $1/3$

0990. gap-ser-root-002
1

0991. gap-ser-lc-001
divergent

0992. gap-ser-lc-002
convergent

0993. gap-ser-cond-001
conditional

0994. gap-ser-end-001
conditional

0995. gap-ser-end-002
divergent

0996. gap-ser-taylor-001
 $1/120$

0997. gap-ser-taylor-002
 $1-x^2/2+x^4/24$

0998. gap-tech-005
roottest

0999. mob-tech-007
ratiotest

1000. mob-tech-008
roottest

1001. mob-tech-019
endpointanalysis

1002. mob-trap-004
conditional

1003. mob-trap-005
checkendpoints

1004. mob-conv-003
conditional

1005. mob-conv-006
convergent

1006. mob-conv-010
divergent

1007. rel-basic-019
 2

1008. rel-basic-022
 $1/3$

1009. rel-basic-024
 $1/2$

1010. rel-adv-017
 $1/6$

1011. rel-adv-019
 1

1012. rel-adv-015
convergent

1013. rel-adv-018
divergent

1014. rel-adv-020
conditional

1015. rel-hard-ser-001
 $1/4$

1016. rel-hard-ser-002
 4

1017. rel-hard-ser-004
 $1/120$

1018. rel-hard-ser-005
 $1/18$

1019. rel-hard-ser-008
 $1/2$

1020. rel-hard-ser-012
 $1/e$

1021. rel-hard-ser-010
divergent

1022. rel-hard-ser-011
conditional

1023. hc-spec-007
 $1/96$

1024. hc-spec-009
 $1/27$

1025. hc-spec-010
 $1/147456$

1026. exam-ser-001
 3

1027. exam-ser-002
 e

1028. exam-ser-003
 1

1029. exam-ser-008
 $(\exp(1)-\exp(-1))/2$

1030. exam-ser-009
 $4/15$

1031. exam-ser-010
 $-1/6$

1032. exam-ser-011
 $-1/4$

1033. exam-ser-013
 8

1034. exam-ser-014
 $1/4$

1035. exam-ser-017
 $\exp(2)$

1036. exam-ser-018
 35

1037. exam-ser-019
 16

1038. exam-ser-020
 $3/4$

1039. uni-ser-001
 $4/9$

1040. uni-ser-004
 0

1041. uni-ser-005
 0

1042. uni-ser-006
 $1/3$

1043. uni-ser-007
 5

1044. uni-ser-008
 4

1045. uni-ser-009
 $1/e$

1046. uni-ser-010
 $\pi/4$

1047. uni-ser-012
 4

1048. uni-ser-013
 15

1049. uni-ser-014
 $1/3$

1050. uni-ser-015
 $2/9$

1051. uni-ser-016
 $-4/315$

1052. uni-ser-018
 210

1053. uni-ser-019
 1

1054. depth-ser-001	9/4	1078. burst-boss-ser-005	137/180
1055. depth-ser-002	20/27	1079. burst-boss-ser-006	1553/80
1056. depth-ser-003	1/2	1080. burst-boss2-ser-001	$\pi^4/90$
1057. depth-ser-004	243/560	1081. burst-boss2-ser-002	$\pi^4/96$
1058. depth-ser-005	1/24	1082. burst-boss2-ser-003	$7\pi^4/720$
1059. depth-ser-006	-1/5040	1083. burst-boss2-ser-004	$31\pi^6/30240$
1060. depth-ser-007	$(\exp(1)+\exp(-1))/2$	1084. burst-boss2-ser-005	15
1061. depth-ser-008	1/4	1085. burst-boss2-ser-006	273/4
1062. depth-ser-009	1/4	1086. burst-boss2-ser-007	$3\log(3/2)/2$
1063. depth-ser-010	4/3	1087. burst-boss2-ser-008	$4\log(4/3)/3$
1064. depth-ser-011	192	1088. burst-boss3-ser-001	9366
1065. depth-ser-012	3	1089. burst-boss3-ser-002	94586
1066. depth-ser-013	$\cos(1)$	1090. burst-boss3-ser-003	4108/243
1067. depth-ser-014	1/4	1091. burst-boss3-ser-004	1491/4
1068. depth-ser-015	1/3	1092. burst-boss3-ser-005	$\pi^2/12$
1069. depth-ser-016	33/8	1093. burst-boss3-ser-006	275295799/775975200
1070. depth-ser-017	5	1094. burst-boss3-ser-007	111165691613818/97692469875
1071. depth-ser-018	$\sin(1)$	1095. burst-boss3-ser-008	1094053506107/212837625
1072. depth-ser-019	11/12	1096. world-006	1/3
1073. depth-ser-020	1/2	1097. world-009	$\log(3/2)$
1074. burst-boss-ser-001	26	1098. world-020	$2\exp(1)$
1075. burst-boss-ser-002	3/2	1099. world-021	$5\exp(1)$
1076. burst-boss-ser-003	$2\log(2)$	1100. world-022	$2-2\log(2)$
1077. burst-boss-ser-004	$\pi^2/12$	1101. world-023	$\pi^2/3-3$
		1102. world-028	1/4

1103. world-034
 $\pi^{2/12}$

1104. world-039
 $\pi^{2/18}$

1105. world-042
 $1/27$

1106. world-048
 $\pi/2$

1107. world-053
 $1/4$

1108. world-060
 $\pi^{3/32}$

1109. world-075
 $\pi^{2/6-1}$

1110. world-078
 $\pi^{2/12-\log(2)^{2/2}}$

1111. world-086
 $\pi^{6/960}$

1112. world-087
 $\pi^{6/945}$

1113. world-097
 $2*\log(2)$

1114. putnam-001
 $\pi/4$

1115. putnam-002
 $2/3$

1116. putnam-006
 $2/\pi$

1117. putnam-007
 $1/2$

1118. putnam-011
 $2*\log(2)-1$

答案與解法提示

0001 lim-001

極限・R1

Answer. 1

Note. 標準極限： $\sin x$ 和 x 在 0 附近同階，所以答案是 1。

0002 lim-002

極限・R1

Answer. $1/2$

Note. 用 $1 - \cos x$ 約等於 $x^2/2$ 。

0003 lim-003

極限・R1

Answer. 4

Note. 因式分解為 $(x-2)(x+2)$ ，約掉後代入 $x=2$ 。

0004 lim-004

極限・R1

Answer. $3/2$

Note. 最高次項係數比是 $3/2$ 。

0005 lim-005

極限・R2

Answer. 1

Note. e^x 在 0 的線性項是 x 。

0006 lim-006

極限・R2

Answer. 3

Note. $\ln(1+u)$ 約等於 u ，令 $u=3x$ 。

0007 lim-007

極限・R2

Answer. 2

Note. $\tan(4x)$ 約等於 $4x$ ， $\sin(2x)$ 約等於 $2x$ 。

0008 lim-008

極限・R2

Answer. $1/2$

Note. 有理化後是 $x/(\sqrt{x^2+1}+x)$ ，趨近 $1/2$ 。

0009 lim-009

極限・R3

Answer. $-1/6$

Note. $\sin x = x - x^3/6 + \dots$ 。

0010 lim-010

極限・R3

Answer. 2

Note. $e^{2x} = 1 + 2x + 2x^2 + \dots$ 。

0011 lim-011

極限・R3

Answer. 0

Note. 令 $x=1/t$ ，變成 $-(\ln t)/t$ ，趨近 0。

0012 lim-012

極限・R3

Answer. $1/3$

Note. $\ln(1+x) = x - x^2/2 + x^3/3 - \dots$ 。

0013	lim-013	極限・R3
Answer. $1/2$		
Note. $\tan x = x + x^3/3 + \dots$, $\sin x = x - x^3/6 + \dots$ 。		
0014	lim-014	極限・R4
Answer. $-1/2$		
Note. 二階項來自 $\ln(1+x)$ 的 $-x^2/2$ 。		
0015	lim-015	極限・R3
Answer. $1/2$		
Note. $\sec x = 1 + x^2/2 + \dots$ 。		
0016	lim-016	極限・R4
Answer. $-1/8$		
Note. 展開 $\sqrt{1+1/x}$, 下一項是 $-1/(8x^2)$ 。		
0017	lim-017	極限・R3
Answer. $1/3$		
Note. $e^x - e^{-x} = 2x + x^3/3 + \dots$ 。		
0018	lim-018	極限・R3
Answer. $1/3$		
Note. $\sin x$ 與 $x \cos x$ 的三階項相減。		
0019	lim-019	極限・R4
Answer. $-e/2$		
Note. $(1+x)^{1/x} = e(1-x/2+\dots)$ 。		
0020	lim-020	極限・R3
Answer. $-1/3$		
Note. $\arctan x = x - x^3/3 + \dots$ 。		
0021	td-lim-001	極限・R6
Answer. $\exp(1/2)$		
Note. 取 $\log$ 後, $\sum k/n^2$ 趨近 $1/2$, 其餘高階項趨近 0 。		
0022	td-lim-002	極限・R6
Answer. $\sqrt{e}/3$		
Note. 把 $\log$ 展開到 $1/n$ 項: $\log P_n = 1/2 + 1/(3n) + O(1/n^2)$ 。		
0023	td-lim-004	極限・R4
Answer. $-1/2$		
Note. $(1+x)^x = \exp(x \log(1+x)) = 1 + x^2 - x^3/2 + \dots$ 。		
0024	lim-021	極限・R2
Answer. $1/3$		
Note. $\tan x = x + x^3/3 + O(x^5)$, 所以分子約為 $x^3/3$, 極限為 $1/3$ 。		
0025	lim-022	極限・R2
Answer. $3/2$		

Note. 有理化後為 $3x/(\sqrt{x^2+3x}+x)$ ，上下同除以 x 得 $3/(\sqrt{1+3/x}+1)$ ，極限為 $3/2$ 。

0026 lim-024

極限 • R2

Answer. $-1/2$

Note. $\log(1+x)=x-x^2/2+O(x^3)$ ，分子為 $-x^2/2+O(x^3)$ ，極限為 $-1/2$ 。

0027 lim-025

極限 • R4

Answer. 1

Note. 令 $u=xy$ ，則 $u \rightarrow 0$ ，標準極限 $\sin u/u = 1$ 。

0028 lim-026

極限 • R4

Answer. 0

Note. 令 $x=r \cos \theta$, $y=r \sin \theta$ ，原式 $= r \cos^2 \theta \sin \theta$ ，當 $r \rightarrow 0$ 時趨近 0。

0029 lim-027

極限 • R4

Answer. dne

Note. 沿 $y=x$ 得 $1/2$ ；沿 $y=-x$ 得 $-1/2$ 。路徑極限不同，所以 DNE。

0030 lim-028

極限 • R4

Answer. $1/2$

Note. 設 $r^2=x^2+y^2$ 。有理化後為 $1/(\sqrt{1+r^2}+1)$ ，令 $r \rightarrow 0$ 得 $1/2$ 。

0031 lim-029

極限 • R4

Answer. 0

Note. 因 $x^4+y^4 \leq (x^2+y^2)^2$ ，所以原式 $\leq x^2+y^2$ ，極限為 0。

0032 lim-030

極限 • R2

Answer. $9/2$

Note. $1-\cos(3x) \sim (3x)^2/2$ ，所以極限為 $9/2$ 。

0033 lim-031

極限 • R4

Answer. 0

Note. With $x=r \cos t$, $y=r \sin t$, the expression is $r^2 \cos^2 t \sin^2 t$, so the limit is 0.

0034 lim-032

極限 • R4

Answer. dne

Note. Along $y=0$ the value is 1; along $x=0$ the value is -1. The limit does not exist.

0035 lim-033

極限 • R4

Answer. 1

Note. Set $u=x+y$. The expression becomes $(e^u-1)/u$, so the limit is 1.

0036 lim-034

極限 • R4

Answer. 0

Note. Let $r=\sqrt{x^2+y^2}$. Since $|x| \leq r$, the absolute value is $\leq r^3/r^2=r$, hence the limit is 0.

0037 lim-035

極限 • R2

Answer. $5/2$

Note. As $x \rightarrow 0$, $\sin(5x) \sim 5x$ and $\sin(2x) \sim 2x$, so the limit is $5/2$.

0038	lim-036	極限 • R3
Answer. $1/2$ Note. $\log(1+x+x^2)=x+x^2-(x^2/2)+O(x^3)=x+x^2/2+O(x^3)$. The limit is $1/2$.		
0039	lim-037	極限 • R2
Answer. 2 Note. After rationalizing, the expression is $4/(\sqrt{1+4x}+1)$, whose limit is 2.		
0040	lim-039	極限 • R4
Answer. dne Note. Along $y=kx^2$ the expression is $k/(1+k^2)$. Since this depends on k , the limit does not exist.		
0041	lim-040	極限 • R2
Answer. 2 Note. Since $\log(1+2/x) \sim 2/x$, the limit is 2.		
0042	gap-tech-001	極限 • R2
Answer. rationalize Note. The right reflex is to rationalize with the conjugate.		
0043	gap-tech-002	極限 • R2
Answer. taylor Note. Use Taylor expansion of e^{2x} around 0.		
0044	gap-tech-008	極限 • R2
Answer. pathtest Note. Use a path test such as $y=mx^2$.		
0045	mob-tech-001	極限 • R2
Answer. rationalize Note. Use rationalize.		
0046	mob-tech-002	極限 • R2
Answer. Taylor Note. Use Taylor.		
0047	mob-tech-009	極限 • R2
Answer. pathtest Note. Use path test.		
0048	mob-trap-009	極限 • R2
Answer. indeterminate Note. indeterminate		
0049	mob-limtrap-003	極限 • R3
Answer. $-1/24$ Note. The limit is $-1/24$.		

0050	mob-limtrap-004	極限・R3
Answer. $1/3$		
Note. The limit is $1/3$.		
0051	mob-limtrap-005	極限・R3
Answer. $1/6$		
Note. The limit is $1/6$.		
0052	mob-limtrap-006	極限・R3
Answer. $-1/8$		
Note. The limit is $-1/8$.		
0053	mob-limtrap-008	極限・R3
Answer. 0		
Note. The limit is 0.		
0054	rel-basic-001	極限・R1
Answer. 5		
Note. $\sin(5x)/(5x)$ tends to 1.		
0055	rel-basic-003	極限・R1
Answer. 4		
Note. Use $e^u - 1 \sim u$.		
0056	rel-basic-004	極限・R1
Answer. 2		
Note. Use $\log(1+u) \sim u$.		
0057	rel-basic-005	極限・R1
Answer. $1/2$		
Note. Rationalize the numerator.		
0058	rel-adv-002	極限・R3
Answer. $1/24$		
Note. The next Taylor coefficient is $1/24$.		
0059	rel-boss-019	極限・R5
Answer. 1		
Note. Rewrite as $n \log(1+1/n)$.		
0060	rel-hard-lim-001	極限・R5
Answer. $-1/5040$		
Note. Use the x^7 term of $\sin x$.		
0061	rel-hard-lim-002	極限・R5
Answer. $17/315$		
Note. Use the x^7 term of $\tan x$.		
0062	rel-hard-lim-003	極限・R5
Answer. $1/5$		

Note. Use the x^5 term of $\log(1+x)$.

0063 rel-hard-lim-004

極限 • R5

Answer. $1/360$

Note. Use the even expansion of $e^x + e^{-x}$.

0064 rel-hard-lim-007

極限 • R5

Answer. $1/2$

Note. Use $1 - \cos u \sim u^2/2$.

0065 rel-hard-lim-008

極限 • R5

Answer. 1

Note. Use $e^u - 1 \sim u$.

0066 rel-hard-lim-009

極限 • R5

Answer. $1/5$

Note. Use $\arctan x = x - x^3/3 + x^5/5 - \dots$

0067 rel-hard-lim-010

極限 • R5

Answer. $11e/24$

Note. The second correction term is $11e/(24n^2)$.

0068 rel-hard-lim-006

極限 • R5

Answer. DNE

Note. Along $y=0$ the limit is 0; along $y=x$ it is $1/4$.

0069 exam-lim-001

極限 • R5

Answer. $81/40$

Note. Expand $\sin(3x)$ through the x^5 term.

0070 exam-lim-002

極限 • R5

Answer. $8/3$

Note. Use $\log(1+u) = u - u^2/2 + u^3/3 - \dots$ with $u=2x$.

0071 exam-lim-003

極限 • R5

Answer. 1

Note. The constant and linear terms cancel; the x^2 coefficient is 1.

0072 exam-lim-005

極限 • R5

Answer. 2

Note. $1 - \cos u$ is asymptotic to $u^2/2$.

0073 exam-lim-006

極限 • R5

Answer. -2

Note. $\sqrt{1+4x} = 1 + 2x - 2x^2 + \dots$

0074 exam-lim-007

極限 • R5

Answer. $-9/8$

Note. Apply $\sqrt{1+u} = 1 + u/2 - u^2/8 + \dots$ with $u=3/x$.

0075	exam-lim-008	極限 • R5
Answer. $1/3$ Note. Compare leading terms.		
0076	exam-lim-009	極限 • R5
Answer. $1/3$ Note. Use $x - \sin x \sim x^3/6$ and $1 - \cos x \sim x^2/2$.		
0077	exam-lim-010	極限 • R5
Answer. $-8/3$ Note. $\arctan u = u - u^3/3 + \dots$		
0078	exam-lim-011	極限 • R5
Answer. 1 Note. Write $x^x = \exp(x \log x)$, and $x \log x \rightarrow 0$.		
0079	exam-lim-012	極限 • R5
Answer. $\exp(6)$ Note. This is $\exp(3x \log(1+2/x)) \rightarrow \exp(6)$.		
0080	exam-lim-015	極限 • R5
Answer. 1 Note. $e^{\sin x} - 1$ is asymptotic to $\sin x$.		
0081	exam-lim-017	極限 • R5
Answer. $-1/12$ Note. Compare x^4 coefficients: $1/24$ and $1/8$.		
0082	exam-lim-019	極限 • R5
Answer. 1 Note. Rewrite as $x \log(1+1/x)$.		
0083	exam-lim-020	極限 • R5
Answer. $1/6$ Note. $\arcsin x = x + x^3/6 + \dots$		
0084	uni-lim-001	極限 • R6
Answer. $1/120$ Note. Keep the x^5 term of $\sin x$.		
0085	uni-lim-002	極限 • R6
Answer. $-1/4$ Note. Use the fourth term of $\log(1+x)$.		
0086	uni-lim-003	極限 • R6
Answer. $3/2$ Note. Compare the x^2 coefficients.		

0087 uni-lim-005

極限 • R6

Answer. $1/8$

Note. Odd terms remain after subtracting the two radicals.

0088 uni-lim-006

極限 • R6

Answer. $1/2$

Note. Put $y=1/x$ and expand $(1/y+1)\log(1+y)$.

0089 uni-lim-007

極限 • R6

Answer. -1

Note. The cubic term is the first nonzero term.

0090 uni-lim-008

極限 • R6

Answer. $-9*\exp(3)/2$

Note. Expand $x^{-1}\log(1+3x)$ through the linear term.

0091 uni-lim-009

極限 • R6

Answer. 1

Note. $\cosh x$ and $\cos x$ have opposite x^2 coefficients.

0092 uni-lim-010

極限 • R6

Answer. $-1/2$

Note. Compare $\arctan x$ and $\arcsin x$ cubic terms.

0093 uni-lim-011

極限 • R6

Answer. $-1/2$

Note. Only the quadratic term is needed.

0094 uni-lim-012

極限 • R6

Answer. 1

Note. Both numerator and denominator start at $x^2/2$.

0095 uni-lim-014

極限 • R6

Answer. $1/3$

Note. Compare fourth-order coefficients.

0096 depth-lim-001

極限 • R6

Answer. $-1/5040$

Note. Keep the x^7 term of $\sin x$.

0097 depth-lim-002

極限 • R6

Answer. $1/5$

Note. Use the fifth term of $\log(1+x)$.

0098 depth-lim-003

極限 • R5

Answer. $1/2$

Note. The quadratic term comes from $(\sin x)^2/2$.

0099 depth-lim-004

極限 • R6

Answer. $1/6$

Note. Compare fourth-order coefficients.

0100 depth-lim-005

極限 • R5

Answer. $2/3$

Note. Use $\tan x \sim x + x^3/3$ and $1 - \cos x \sim x^2/2$.

0101 depth-lim-006

極限 • R6

Answer. $1/2$

Note. Put $u=1/x$ and compare the u^2 term.

0102 depth-lim-008

極限 • R5

Answer. $1/2$

Note. Compare the cubic coefficients.

0103 depth-lim-009

極限 • R5

Answer. $1/3$

Note. The cubic terms have opposite signs.

0104 depth-lim-010

極限 • R5

Answer. $1/2$

Note. Use $1 - \cos x \sim x^2/2$ and $\sin x \sim x$.

0105 depth-lim-011

極限 • R5

Answer. $-1/2$

Note. Use $\cos x = 1 - x^2/2 + \dots$

0106 depth-lim-012

極限 • R6

Answer. $\exp(1)/2$

Note. Expand $x^{-1} \log(1+x+x^2)$.

0107 burst-boss-lim-001

極限 • R5

Answer. $2/15$

Note. Use $\tan x = x + x^3/3 + 2x^5/15 + \dots$

0108 burst-boss-lim-002

極限 • R5

Answer. $1/120$

Note. Use the fifth-order term of $\sin x$.

0109 burst-boss-lim-003

極限 • R5

Answer. $-1/4$

Note. Use the fourth-order term of $\log(1+x)$.

0110 burst-boss-lim-004

極限 • R6

Answer. $-1/8$

Note. Expand $\sin x$ first, then exponentiate to fourth order.

0111 burst-boss-lim-005

極限 • R6

Answer. $1/6$

Note. Compare both cosine expansions through x^4 .

0112	burst-boss-lim-006	極限 • R6
Answer. $-1/7$		
Note. Use the alternating arctan series.		
0113	burst-boss-lim-008	極限 • R6
Answer. $-e/2$		
Note. Use $x \log(1+1/x)=1-1/(2x)+\dots$		
0114	burst-boss2-lim-001	極限 • R6
Answer. $\exp(-1/6)$		
Note. Take logs and use $\log(\sin x/x)=-x^2/6+\dots$		
0115	burst-boss2-lim-002	極限 • R6
Answer. $-1/45$		
Note. Use $x/\tan x=1-x^2/3-x^4/45+\dots$		
0116	burst-boss2-lim-003	極限 • R6
Answer. $7/360$		
Note. Use the Laurent expansion of $\csc x$.		
0117	burst-boss2-lim-004	極限 • R6
Answer. $-1/12$		
Note. Expand $\log(\cos x)$ through the fourth order.		
0118	burst-boss2-lim-005	極限 • R6
Answer. $1/10$		
Note. Integrate the series for e^{-t^2} term by term.		
0119	burst-boss2-lim-006	極限 • R6
Answer. $1/21$		
Note. Integrate the series for $\log(1+t^2)$ term by term.		
0120	burst-boss2-lim-007	極限 • R6
Answer. $-1/4$		
Note. Use the Euler-Maclaurin correction for $H_{2n}-H_n$.		
0121	burst-boss2-lim-008	極限 • R6
Answer. $-1/4$		
Note. Read the first endpoint correction in the Riemann sum.		
0122	burst-boss2-lim-009	極限 • R6
Answer. $(1/\sqrt{2}-1)/2$		
Note. Apply the first Euler-Maclaurin correction.		
0123	burst-boss2-lim-010	極限 • R6
Answer. $(2-e)/2$		
Note. Expand $n \log(1+x/n)$ to the first correction term.		

0124	burst-boss3-lim-001	極限・R6
Answer. $1/\exp(1)$		
Note. Maximize at $x=n/(n+1)$.		
0125	burst-boss3-lim-002	極限・R6
Answer. $4/\exp(2)$		
Note. Maximize at $x=n/(n+2)$.		
0126	burst-boss3-lim-003	極限・R6
Answer. $1/2$		
Note. Only a boundary layer near $x=1$ contributes.		
0127	burst-boss3-lim-004	極限・R6
Answer. $(e-1)/12$		
Note. Use the second Euler-Maclaurin endpoint correction.		
0128	burst-boss3-lim-005	極限・R6
Answer. $-1/24$		
Note. Use the second Euler-Maclaurin endpoint correction.		
0129	burst-boss3-lim-006	極限・R6
Answer. $-1/24$		
Note. Use $f'(1)-f'(0)$ for the Riemann-sum correction.		
0130	burst-boss3-lim-007	極限・R6
Answer. $11/72$		
Note. Expand the integrand before integrating.		
0131	burst-boss3-lim-008	極限・R6
Answer. $-1/42$		
Note. Integrate the sine series term by term.		
0132	burst-boss3-lim-009	極限・R6
Answer. $-1/180$		
Note. Use the logarithmic sine expansion.		
0133	burst-boss3-lim-010	極限・R6
Answer. $-1/12$		
Note. Expand $\log(\cosh x)$.		
0134	burst-boss3-lim-011	極限・R6
Answer. $1/16$		
Note. Use the binomial series.		
0135	burst-boss3-lim-012	極限・R6
Answer. $\exp(1/6)$		
Note. Take logs and use $x/\sin x=1+x^2/6+\dots$		
0136	world-001	極限・R1
Answer. $5/3$		

Note. Both behave like their argument.

0137 world-002

極限・R2

Answer. $\log(2)$

Note. Derivative of 2^x at 0.

0138 world-003

極限・R2

Answer. $\exp(3)$

Note. Standard e-limit.

0139 world-007

極限・R2

Answer. $1/6$

Note. $\sinh x = x + x^3/6 + \dots$

0140 world-008

極限・R2

Answer. 2

Note. $x^3/3$ over $x^3/6$.

0141 world-011

極限・R3

Answer. 4

Note. Geometric-mean limit: $\sqrt[4]{2 \cdot 8} = 4$.

0142 world-012

極限・R4

Answer. $2/3$

Note. $\cot^2 x = 1/x^2 - 2/3 - \dots$

0143 world-013

極限・R4

Answer. $\exp(1/2)$

Note. $(e^x - 1)/x = 1 + x/2 + \dots$; limit $e^{1/2}$.

0144 world-026

極限・R4

Answer. $1/6$

Note. Next term of e^x .

0145 world-030

極限・R4

Answer. $\exp(-1/2)$

Note. $\log \cos x \sim -x^2/2$ and $\sin^2 x \sim x^2$.

0146 world-035

極限・R5

Answer. $4/\exp(1)$

Note. Stirling gives $4/e$.

0147 world-040

極限・R5

Answer. $11 \cdot \exp(1)/24$

Note. $(1+x)^{1/x} = e(1-x/2 + 11x^2/24 - \dots)$.

0148 world-046

極限・R5

Answer. $1/2$

Note. $x^{1/x} = 1 + (\log x)/x + (\log x)^2/(2x^2) + \dots$

0149	world-049	極限 • R5
Answer. $11 \cdot \exp(1)/24$ Note. Next term of $(1+1/n)^n$.		
0150	world-066	極限 • R6
Answer. $\exp(1/3)$ Note. log of product $\rightarrow 1/3$.		
0151	world-067	極限 • R6
Answer. $27/4$ Note. Central trinomial growth $(27/4)^n$.		
0152	world-071	極限 • R6
Answer. $\pi/4$ Note. Riemann sum of $1/(1+x^2)$ on $[0,1]$.		
0153	world-074	極限 • R6
Answer. $\sqrt{\pi}$ Note. $\text{binom}(2n,n) \sim 4^n / \sqrt{\pi n}$.		
0154	world-091	極限 • R6
Answer. $-\exp(1)/2$ Note. $(1+1/n)^n = e(1 - 1/(2n) + \dots)$.		
0155	world-096	極限 • R6
Answer. $\exp(1/6)$ Note. $\log(\sinh x / x) \sim x^2/6$.		
0156	world-099	極限 • R6
Answer. $1/30$ Note. Famous degree-7 expansion: $1/30$.		
0157	putnam-003	極限 • R5
Answer. $4/\exp(1)$ Note. $\log \rightarrow \int_0^1 \log(1+x) dx = 2\log 2 - 1$.		
0158	putnam-008	極限 • R5
Answer. $\exp(2)$ Note. Integrand $\rightarrow e^2$; average $\rightarrow e^2$.		
0159	putnam-010	極限 • R6
Answer. $\pi/4$ Note. Riemann sum of $1/(1+x^2)$.		
0160	der-001	微分 • R1
Answer. $5x^4 - 6x$ Note. 逐項微分。		

0161	der-002	微分・R1
Answer. $\cos(x) - \sin(x)$ Note. $\sin$ 的導數是 $\cos$ ， $\cos$ 的導數是 $-\sin$ 。		
0162	der-003	微分・R1
Answer. $\exp(x) \cdot \log(x) + \exp(x)/x$ Note. 乘法法則。		
0163	der-004	微分・R1
Answer. $1/(2\sqrt{x})$ Note. $\sqrt{x} = x^{1/2}$ 。		
0164	der-005	微分・R2
Answer. $2x/(1+x^2)$ Note. 鏈鎖律： $u=1+x^2$ 。		
0165	der-006	微分・R2
Answer. $2x \sin(x) + x^2 \cos(x)$ Note. 乘法法則。		
0166	der-007	微分・R2
Answer. $1/(1+x^2)$ Note. 反三角函數基本導數。		
0167	der-008	微分・R2
Answer. $1/(x+1)^2$ Note. 商法則或改寫成 $1-1/(x+1)$ 。		
0168	der-009	微分・R3
Answer. $x^x(\log(x)+1)$ Note. 對數微分： $x^x = e^{x \ln x}$ 。		
0169	der-010	微分・R3
Answer. $\exp(-x)(2x \cos(x^2) - \sin(x^2))$ Note. 乘法法則加鏈鎖律。		
0170	der-011	微分・R3
Answer. $\cos(x)/\sin(x)$ Note. 鏈鎖律，答案也可寫成 $\cot x$ 。		
0171	der-012	微分・R4
Answer. $((x^2+1)/x - 2x \log(x))/(x^2+1)^2$ Note. 商法則。		
0172	der-013	微分・R4
Answer. $x^{\sin(x)}(\cos(x) \log(x) + \sin(x)/x)$ Note. 對數微分： $x^{\sin x} = e^{\sin x \ln x}$ 。		
0173	der-014	微分・R3
Answer. $1/(1+x^2)$		

Note. 令 $u=(1+x)/(1-x)$ ，套 $\arctan$ 鏈鎖律後會大幅化簡。

0174 der-015

微分 · R3

Answer. $1/\sqrt{1+x^2}$

Note. 這是 $\operatorname{asinh} x$ 的導數。

0175 der-016

微分 · R3

Answer. $1/(1+\cos(x))$

Note. 商法則後分子化成 $1+\cos x$ 。

0176 der-017

微分 · R3

Answer. $\exp(x^2)*(2*x*\cos(3*x)-3*\sin(3*x))$

Note. 乘法法則加兩個鏈鎖律。

0177 der-018

微分 · R3

Answer. $1/\sqrt{x^2-1}$

Note. 這是 $\operatorname{arcosh} x$ 的導數形式。

0178 der-019

微分 · R2

Answer. $10*x*(x^2+1)^4$

Note. 鏈鎖律。

0179 der-020

微分 · R3

Answer. $1/(2*\sqrt{x}*(1+x))$

Note. 先對 $\arctan u$ 微分，再乘上 $\sqrt{x}$ 的導數。

0180 td-der-001

微分 · R4

Answer. $2*(1-3*x^2)/(1+x^2)^3$

Note. 先算 $y'=-1/(1+x^2)$ ，再連續微分兩次。

0181 td-der-002

微分 · R2

Answer. $(35*x^3-15*x)/2$

Note. 這是 Legendre P_4 的導數，逐項微分即可。

0182 td-der-003

微分 · R3

Answer. $\log(1+x)+x/(1+x)$

Note. 乘法法則。這也是展開 $(1+x)^x$ 時會出現的核心項。

0183 td-der-004

微分 · R3

Answer. $-3*\sin(x)/(5+3*\cos(x))$

Note. 鏈鎖律，內層導數是 $-3 \sin x$ 。

0184 der-021

微分 · R4

Answer. $2*x*y+y*\cos(x*y)$

Note. 偏 x 微分得 $2xy + \cos(xy)*y$ 。

0185 der-022

微分 · R4

Answer. $x^2*\exp(x*y)+3*y^2$

Note. d/dy of $x e^{\{xy\}} = x*e^{\{xy\}}*x = x^2 e^{\{xy\}}$ ，再加 $3y^2$ 。

0186	der-023	微分・R4
Answer. $\exp(xy) \cdot (1+xy)$ Note. 先得 $f_y = x e^{xy}$ ，再對 x 微分為 $e^{xy} + xy e^{xy} = e^{xy}(1+xy)$ 。		
0187	der-024	微分・R4
Answer. $22/5$ Note. $\text{grad } f = (2x, 2y)$ ，在 $(1, 2)$ 為 $(2, 4)$ 。與 $(3/5, 4/5)$ 內積得 $22/5$ 。		
0188	der-025	微分・R4
Answer. $2x/(x^2+y^2)$ Note. 令 $u = x^2 + y^2$ ， $u_x = 2x$ ，所以 $f_x = 2x/(x^2+y^2)$ 。		
0189	der-026	微分・R2
Answer. $2x \sin(3x) + 3x^2 \cos(3x)$ Note. 乘法律給 $2x \sin(3x) + x^2 \cdot 3 \cos(3x)$ 。		
0190	der-027	微分・R2
Answer. $x/(1+x^2)$ Note. 原式為 $(1/2)\log(1+x^2)$ ，微分得 $x/(1+x^2)$ 。		
0191	der-028	微分・R2
Answer. $\exp(x)/(1+\exp(2x))$ Note. 由鏈鎖律得 $e^x/(1+e^{2x})$ 。		
0192	der-029	微分・R2
Answer. $\exp(x) \cdot (x+2)$ Note. $f' = e^x(x+1)$ ， $f'' = e^x(x+1) + e^x = e^x(x+2)$ 。		
0193	der-030	微分・R3
Answer. $-(2x+y)/(x+2y)$ Note. 微分得 $2x+y+xy'+2yy'=0$ ，所以 $y' = -(2x+y)/(x+2y)$ 。		
0194	der-031	微分・R4
Answer. $2x \cdot \exp(x^2+y^2)$ Note. The derivative is $e^{x^2+y^2}$ times $2x$.		
0195	der-032	微分・R4
Answer. $x/(x^2+y^2)$ Note. $(1/x)/(1+y^2/x^2) = x/(x^2+y^2)$.		
0196	der-033	微分・R4
Answer. $2y - \sin(x)$ Note. $f_x = 2xy + \cos x$, so $f_{xx} = 2y - \sin x$.		
0197	der-034	微分・R4
Answer. $x-y$ Note. The Jacobian matrix is $[[1, 1], [y, x]]$, so $\det = x-y$.		

0198 der-035

微分・R4

Answer. $\exp(t) \cdot (4t^3 + t^4)$

Note. After substitution $z = t^4 e^t$, so $dz/dt = e^t(4t^3 + t^4)$.

0199 der-036

微分・R4

Answer. 4

Note. The Laplacian is $2+2=4$.

0200 der-037

微分・R4

Answer. 27

Note. The Hessian is $\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$, whose determinant is $36-9=27$.

0201 der-038

微分・R2

Answer. $(x^2 - 2x - 1)/(x-1)^2$

Note. Derivative is $[2x(x-1) - (x^2+1)]/(x-1)^2 = (x^2-2x-1)/(x-1)^2$.

0202 der-039

微分・R2

Answer. $2\sin(x)\cos(x)$

Note. By chain rule, the derivative is $2\sin(x)\cos(x)$.

0203 der-040

微分・R3

Answer. $2x^{\log(x)} \log(x)/x$

Note. $x^{\log x} = \exp((\log x)^2)$, so the derivative is $x^{\log x} \cdot 2\log(x)/x$.

0204 der-041

微分・R4

Answer. -0.04

Note. $df = 4(0.01) + 4(-0.02) = -0.04$.

0205 der-042

微分・R4

Answer. -0.028

Note. $df = (3/5)(0.06) + (4/5)(-0.08) = 0.036 - 0.064 = -0.028$.

0206 der-043

微分・R4

Answer. 0

Note. $dz = 0.02 + (-0.04)/2 = 0$.

0207 der-044

微分・R4

Answer. $y \exp(xy) dx + x \exp(xy) dy$

Note. $df = y e^{xy} dx + x e^{xy} dy$.

0208 der-045

微分・R4

Answer. $-y dx/(x^2+y^2) + x dy/(x^2+y^2)$

Note. $df = (-y/(x^2+y^2))dx + (x/(x^2+y^2))dy$.

0209 der-046

微分・R4

Answer. 4.92

Note. The estimate is $5 + 2(0.02) + 4(-0.03) = 4.92$.

0210 der-047

微分・R5

Answer. $-9/2$

Note. The critical point is $(3/2, 1/2)$. The value is $-9/2$.

0211 der-048

微分・R5

Answer. -9

Note. The critical point is $(1, 1)$, and the minimum value is -9 .

0212 der-049

微分・R5

Answer. -1

Note. The minimum occurs at $x=-1, y=2$, so the x -coordinate is -1 .

0213 der-050

微分・R5

Answer. 5

Note. The closest point is $(1, 2)$, so the minimum of x^2+y^2 is 5 .

0214 der-051

微分・R4

Answer. 3

Note. The Hessian is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, so $\det=3$.

0215 der-052

微分・R4

Answer. 144

Note. The Hessian is $\text{diag}(12, 12)$, so the determinant is 144 .

0216 der-053

微分・R4

Answer. -1

Note. At $(0, 0)$, the Hessian is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, whose determinant is -1 .

0217 der-054

微分・R4

Answer. 128

Note. The Hessian is $\begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$, so $\det=144-16=128$.

0218 der-055

微分・R4

Answer. local minimum

Note. The Hessian is $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, positive definite, so it is a local minimum.

0219 der-056

微分・R4

Answer. saddle point

Note. The Hessian determinant is -4 , so the critical point is a saddle point.

0220 der-057

微分・R4

Answer. 0

Note. At $(0, 0)$, the Hessian is $\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$, so $\det=0$.

0221 der-058

微分・R4

Answer. -4

Note. The Hessian at $(1, 0)$ is $\begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$, so the determinant is -4 .

0222 der-059

微分・R4

Answer. -1

Note. $W = \sin x(-\sin x) - \cos x \cos x = -1$.

0223 der-060

微分・R4

Answer. 1

Note. $W = \cos x \cos x - (-\sin x) \sin x = 1$.

0224 der-061

微分・R4

Answer. $\exp(3x)$

Note. $W = e^x(2e^{2x}) - e^x e^{2x} = e^{3x}$.

0225 der-062

微分・R4

Answer. x^2

Note. $W = x(2x) - 1 \cdot x^2 = x^2$.

0226 der-063

微分・R4

Answer. 2

Note. The determinant of $\begin{bmatrix} 1, x, x^2 \\ 0, 1, 2x \\ 0, 0, 2 \end{bmatrix}$ is 2.

0227 der-064

微分・R4

Answer. $\exp(2x)$

Note. $W = e^x(e^x + xe^x) - e^x(xe^x) = e^{2x}$.

0228 der-065

微分・R4

Answer. $1/x$

Note. $W = 1 \cdot (1/x) - 0 \cdot \log x = 1/x$.

0229 der-066

微分・R4

Answer. x^4

Note. $W = x^2(3x^2) - (2x)(x^3) = x^4$.

0230 der-067

微分・R5

Answer. $4x$

Note. After substitution, $F = 2x^2 + 2y^2$, so $dF/dx = 4x$.

0231 der-068

微分・R5

Answer. $\exp(\sin x) \cdot (2x + x^2 \cos x)$

Note. $d/dx[x^2 e^{\sin x}] = e^{\sin x}(2x + x^2 \cos x)$.

0232 der-069

微分・R6

Answer. $4(x^2 + y^2)$

Note. The determinant is $4x^2 + 4y^2 = 4(x^2 + y^2)$.

0233 der-070

微分・R5

Answer. $-4y$

Note. The product is $(u-v)(-2)$. Since $u-v=2y$, the determinant is $-4y$.

0234 der-071

微分・R5

Answer. r

Note. The Jacobian is r .

0235 der-072

微分 • R5

Answer. $-2x/y$

Note. The determinant is $y(-x/y^2) - x(1/y) = -2x/y$.

0236 der-073

微分 • R6

Answer. $20((2x+y)^2 + (x+3y)^2)$

Note. By the chain rule, the determinant is $4(u^2 + v^2)^5 = 20((2x+y)^2 + (x+3y)^2)$.

0237 der-074

微分 • R5

Answer. -4

Note. The determinant is $(-2)^2 = -4$.

0238 der-075

微分 • R6

Answer. $x^3 - 3xy^2$

Note. $(x+iy)^3 = x^3 + 3ix^2y - 3xy^2 - iy^3$, so the real part is $x^3 - 3xy^2$.

0239 der-076

微分 • R6

Answer. $3x^2y - y^3$

Note. The imaginary part is $3x^2y - y^3$.

0240 der-077

微分 • R6

Answer. $\sqrt{2}$

Note. $|1+i| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

0241 der-078

微分 • R6

Answer. -4

Note. $(1+i)^4 = (2i)^2 = -4$, so the real part is -4 .

0242 der-079

微分 • R6

Answer. holomorphic

Note. $u_x = 2x = v_y$ and $u_y = -2y = -v_x$, so the function is holomorphic.

0243 der-080

微分 • R6

Answer. notholomorphic

Note. The CR equations fail except at a point, so it is not holomorphic on any open set.

0244 der-081

微分 • R6

Answer. $1/2$

Note. The residue is $1/(i+i) = 1/(2i)$, whose modulus is $1/2$.

0245 der-082

微分 • R6

Answer. 0

Note. There is no $(z-2)^{-1}$ term, hence the residue is 0 .

0246 der-083

微分 • R6

Answer. $1/2$

Note. $e^z/z^3 = \sum z^{n-3}/n!$, so z^{-1} comes from $n=2$ and has coefficient $1/2$.

0247 der-084

微分 • R6

Answer. 1

Note. By the residue theorem, the normalized integral is 1.

0248 der-085

微分・R6

Answer. -8

Note. $f'(1+i)=4(-2+2i)=-8+8i$, so the real part is -8.

0249 der-086

微分・R6

Answer. $4*(x^2+y^2)$

Note. The real Jacobian determinant is $4(x^2+y^2)$.

0250 der-087

微分・R6

Answer. 0

Note. $u=x^3-3xy^2$ has $u_{xx}=6x$ and $u_{yy}=-6x$, so the Laplacian is 0.

0251 der-088

微分・R6

Answer. harmonic

Note. The Laplacian is $6x-6x=0$, so u is harmonic.

0252 gap-der-param-001

微分・R2

Answer. $3t/2$

Note. $dy/dx=3t^2/(2t)=3t/2$.

0253 gap-der-param-002

微分・R3

Answer. $-\cos(t)/\sin(t)$

Note. $dy/dx=\cos t/(-\sin t)=-\cot t$.

0254 gap-der-param-003

微分・R2

Answer. 2

Note. $dy/dx=2t$ and $d/dx=d/dt$, so $d^2y/dx^2=2$.

0255 gap-der-param-004

微分・R2

Answer. 3

Note. $dy/dx=3t/2$, so at $t=2$ the slope is 3.

0256 gap-der-polar-001

微分・R4

Answer. 1

Note. The numerator and denominator are both -1, so the slope is 1.

0257 gap-der-param-005

微分・R2

Answer. 5

Note. The speed is $\sqrt{3^2+4^2}=5$.

0258 gap-der-app-001

微分・R3

Answer. 72π

Note. $dV/dt=4\pi(9)(2)=72\pi$.

0259 gap-der-app-002

微分・R2

Answer. 2π

Note. $dA/dt=2\pi(5)(0.2)=2\pi$.

0260	gap-der-app-003	微分・R3
Answer. $-3/2$		
Note. $dy/dt=-(x/y)dx/dt=-(6/8)*2=-3/2$.		
0261	gap-der-app-004	微分・R2
Answer. $-1/4$		
Note. The normal slope is $-1/4$.		
0262	gap-der-app-005	微分・R2
Answer. 2.01		
Note. $\sqrt{4.04} \approx 2 + (1/4)(0.04) = 2.01$.		
0263	gap-der-app-006	微分・R2
Answer. $3/2$		
Note. $x_1 = 1 - (-1)/2 = 3/2$.		
0264	gap-der-app-007	微分・R3
Answer. 2		
Note. The curvature is 2.		
0265	gap-der-ode-001	微分・R2
Answer. $2 \cdot \exp(-x)$		
Note. $y = Ce^{\{-x\}}$; the initial condition gives $C=2$.		
0266	gap-der-ode-002	微分・R2
Answer. $3 \cdot \exp(2 \cdot x)$		
Note. The solution is $y = 3e^{\{2x\}}$.		
0267	gap-der-ode-003	微分・R3
Answer. $(\exp(x) - \exp(-x))/2$		
Note. $e^x y = (e^{\{2x\}} - 1)/2$, so $y = (e^x - e^{\{-x\}})/2$.		
0268	gap-der-ode-004	微分・R3
Answer. $\sin(x)$		
Note. The solution is $y = \sin x$.		
0269	gap-der-ode-005	微分・R4
Answer. $1/(1 + \exp(-x))$		
Note. With $C=1$, the solution is $y = 1/(1 + e^{\{-x\}})$.		
0270	gap-tech-007	微分・R2
Answer. parametricdifferentiation		
Note. Use parametric differentiation.		
0271	mob-tech-012	微分・R2
Answer. Lagrangemultiplier		
Note. Use Lagrange multiplier.		

0272	mob-tech-014	微分・R2
Answer. nabla		
Note. Use nabla.		
0273	mob-tech-018	微分・R2
Answer. Besselequation		
Note. Use Bessel equation.		
0274	mob-tech-020	微分・R2
Answer. logarithmicdifferentiation		
Note. Use logarithmic differentiation.		
0275	mob-trap-001	微分・R2
Answer. missingchainfactor		
Note. missing chain factor		
0276	mob-trap-007	微分・R2
Answer. treatyconstant		
Note. treat y constant		
0277	mob-trap-011	微分・R2
Answer. zero		
Note. zero		
0278	mob-lm-001	微分・R4
Answer. 25		
Note. The optimal value is 25.		
0279	mob-lm-002	微分・R4
Answer. 18		
Note. The optimal value is 18.		
0280	mob-lm-003	微分・R4
Answer. 64		
Note. The optimal value is 64.		
0281	mob-lm-004	微分・R4
Answer. 16		
Note. The optimal value is 16.		
0282	mob-lm-005	微分・R4
Answer. 16		
Note. The optimal value is 16.		
0283	mob-lm-006	微分・R4
Answer. 3		
Note. The optimal value is 3.		
0284	mob-lm-007	微分・R4
Answer. 25		

Note. The optimal value is 25.

0285 mob-lm-008

微分・R4

Answer. $1/3$

Note. The optimal value is $1/3$.

0286 mob-lm-009

微分・R4

Answer. 25

Note. The optimal value is 25.

0287 mob-lm-010

微分・R4

Answer. 2

Note. The optimal value is 2.

0288 mob-bessel-001

微分・R6

Answer. 1

Note. Use the first terms of Bessel functions.

0289 mob-bessel-002

微分・R6

Answer. 0

Note. Use the first terms of Bessel functions.

0290 mob-bessel-003

微分・R6

Answer. $-1/4$

Note. Use the first terms of Bessel functions.

0291 mob-bessel-004

微分・R6

Answer. $1/2$

Note. Use the first terms of Bessel functions.

0292 mob-bessel-005

微分・R6

Answer. $-J_1(x)$

Note. $-J_1(x)$

0293 mob-bessel-006

微分・R6

Answer. Besselequation

Note. Bessel equation

0294 mob-bessel-007

微分・R6

Answer. regularsingular

Note. regular singular

0295 mob-bessel-008

微分・R6

Answer. $J_2(x)$

Note. $J_2(x)$

0296 mob-bessel-009

微分・R6

Answer. $-J_1(x)$

Note. $-J_1(x)$

0297	mob-bessel-010	微分・R6
Answer. second-order ODE		
Note. second-order ODE		
0298	mob-nabla-001	微分・R4
Answer. 3		
Note. Use standard nabla identities.		
0299	mob-nabla-002	微分・R4
Answer. -2		
Note. Use standard nabla identities.		
0300	mob-nabla-003	微分・R4
Answer. 6		
Note. Use standard nabla identities.		
0301	mob-nabla-004	微分・R4
Answer. $2\sqrt{2}$		
Note. Use standard nabla identities.		
0302	mob-nabla-005	微分・R4
Answer. 0		
Note. Use standard nabla identities.		
0303	mob-nabla-006	微分・R4
Answer. 0		
Note. Use standard nabla identities.		
0304	mob-nabla-007	微分・R4
Answer. 2		
Note. Use standard nabla identities.		
0305	mob-nabla-008	微分・R4
Answer. Laplacian		
Note. Laplacian		
0306	mob-nabla-009	微分・R4
Answer. conservative field		
Note. conservative field		
0307	mob-nabla-010	微分・R4
Answer. curl		
Note. curl		
0308	rel-basic-007	微分・R1
Answer. 12		
Note. The derivative is $3x^2$.		

0309	rel-basic-008	微分・R1
Answer. 1		
Note. The derivative is $\cos x$.		
0310	rel-basic-009	微分・R1
Answer. 2		
Note. The derivative is $2e^{2x}$.		
0311	rel-basic-010	微分・R1
Answer. $1/4$		
Note. The derivative is $1/(2\sqrt{x})$.		
0312	rel-basic-011	微分・R1
Answer. $1/e$		
Note. The derivative is $1/x$.		
0313	rel-basic-012	微分・R1
Answer. 1		
Note. The derivative is $\log x + 1$.		
0314	rel-adv-005	微分・R4
Answer. 20		
Note. The determinant is $4(x^2 + y^2)$.		
0315	rel-adv-006	微分・R4
Answer. $8/3$		
Note. Dot the gradient $(4, 4, 2)$ with $(2, -1, 2)/3$.		
0316	rel-adv-007	微分・R4
Answer. 128		
Note. The Hessian is $\begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$.		
0317	rel-adv-008	微分・R4
Answer. 9		
Note. The minimum distance squared to the plane is 9.		
0318	rel-boss-007	微分・R6
Answer. $1/64$		
Note. $J_0(x) = 1 - x^2/4 + x^4/64 - \dots$		
0319	rel-hard-mv-005	微分・R5
Answer. 4		
Note. Compute the determinant of $\begin{bmatrix} 1/y & -x/y^2 \\ y & x \end{bmatrix}$.		
0320	rel-hard-mv-006	微分・R5
Answer. $1/27$		
Note. The symmetric maximum occurs at $x=y=z=1/3$.		
0321	rel-hard-mv-007	微分・R5
Answer. 2		

Note. The minimum occurs at $x=y=1$.

0322 rel-hard-der-001

微分 • R5

Answer. 1

Note. The Wronskian is e^{2x} .

0323 rel-hard-der-002

微分 • R5

Answer. -1

Note. Compute $f'g - fg'$.

0324 rel-hard-der-003

微分 • R5

Answer. -4

Note. At $(1,0)$, the Hessian is $\text{diag}(-2,2)$.

0325 rel-hard-der-004

微分 • R5

Answer. $1/(2\sqrt{2})$

Note. Use $|y''|/(1+(y')^2)^{3/2}$.

0326 rel-hard-der-005

微分 • R5

Answer. $5/3$

Note. Use $(dy/dt)/(dx/dt)$.

0327 rel-hard-der-006

微分 • R5

Answer. $3/8$

Note. Differentiate dy/dx with respect to t , then divide by dx/dt .

0328 rel-hard-der-007

微分 • R5

Answer. $5/3$

Note. The gradient is $(1,1,1)$.

0329 rel-hard-der-008

微分 • R5

Answer. 6

Note. The divergence is $y+z+x$.

0330 rel-hard-der-009

微分 • R5

Answer. 0

Note. Away from the origin, this field has zero curl.

0331 rel-hard-der-010

微分 • R5

Answer. 2

Note. $f_{xx} + f_{yy} = 2e^{x+y}$.

0332 hc-der-003

微分 • R6

Answer. $\exp(\exp(2x) + x^2) \cdot (2\exp(2x) + 2x)$

Note. Differentiate the outer exponential and then the exponent.

0333 hc-der-004

微分 • R6

Answer. 360

Note. Only the fourth derivative of x^4 survives at $x=0$ in the Leibniz sum.

0334	hc-der-005	微分・R6
Answer. 42 Note. The x^7 coefficient in $x^2 \sin x$ is $1/120$.		
0335	hc-der-006	微分・R6
Answer. 1680 Note. The x^8 coefficient is $1/4!$, then multiply by $8!$.		
0336	hc-der-007	微分・R6
Answer. $-1/11$ Note. Use $\sin x/x = \sum (-1)^k x^{2k}/(2k+1)!$.		
0337	hc-der-008	微分・R6
Answer. 16 Note. From $\log y = xy$, the series begins $y = 1 + x + 3x^2/2 + 8x^3/3 + \dots$		
0338	hc-der-009	微分・R6
Answer. 4096 Note. The twelfth derivative returns cosine with factor 2^{12} .		
0339	hc-der-010	微分・R6
Answer. 16 Note. Use $\text{Im}((1+i)^9)$, or the Taylor coefficient of $e^x \sin x$.		
0340	exam-der-003	微分・R5
Answer. $(1 - \log(x))/x^2$ Note. Use quotient rule or $x^{-1} \log x$.		
0341	exam-der-004	微分・R5
Answer. $1/(1+x^2)$ Note. The derivative simplifies after combining $1+u^2$.		
0342	exam-der-005	微分・R5
Answer. $\exp(-x) \cdot (2x \sin(x) - x^2 \sin(x) + x^2 \cos(x))$ Note. Apply product rule across x^2 , e^{-x} , and $\sin x$.		
0343	exam-der-006	微分・R5
Answer. $1/\sqrt{1+x^2}$ Note. This is the derivative of $\text{asinh } x$.		
0344	exam-der-007	微分・R5
Answer. $(2+4x^2) \exp(x^2)$ Note. Differentiate $2x e^{x^2}$.		
0345	exam-der-009	微分・R5
Answer. $-1/(1+\sin(x))$ Note. The numerator becomes $-(1+\sin x)$.		

0346 exam-der-010

微分 • R5

Answer. $\log(1+x^2)+2*x^2/(1+x^2)$

Note. Product rule plus chain rule.

0347 exam-der-011

微分 • R5

Answer. $(x^2+1)^2*(6*x*(x-1)-(x^2+1))/(x-1)^2$

Note. Factor $(x^2+1)^2$ after quotient rule.

0348 exam-der-012

微分 • R5

Answer. $2/x+1$

Note. Take $\log y=2 \log x+x$, then differentiate.

0349 exam-der-013

微分 • R5

Answer. $11/4$

Note. $dy/dx=(dy/dt)/(dx/dt)=(3t^2-1)/(2t)$.

0350 exam-der-014

微分 • R5

Answer. $2*x*\exp(x^4)$

Note. Use FTC with upper limit x^2 .

0351 exam-der-015

微分 • R5

Answer. $2*x*\log(1+x^4)-\log(1+x^2)$

Note. Differentiate both variable limits.

0352 exam-der-016

微分 • R5

Answer. $1/2$

Note. $(f^{-1})'(1)=1/f'(0)=1/2$.

0353 exam-der-017

微分 • R5

Answer. $1/(2*\sqrt{x}*(1+x))$

Note. Chain rule with $u=\sqrt{x}$.

0354 exam-der-018

微分 • R5

Answer. $\sec(x)$

Note. This standard derivative equals $\sec x$.

0355 exam-der-019

微分 • R5

Answer. $1/(1+x^2)^{3/2}$

Note. Differentiate $x(1+x^2)^{-1/2}$.

0356 exam-der-020

微分 • R5

Answer. $\exp(x)/(1+\exp(x))^2$

Note. Quotient rule cancels to one e^x term.

0357 exam-multi-002

微分 • R5

Answer. $6*x*y^2+(1+x*y)*\exp(x*y)$

Note. Differentiate first with respect to y , then x .

0358 exam-multi-003

微分 • R5

Answer. 20

Note. The gradient is $(2,4)$, whose squared norm is 20.

0359 exam-multi-004

微分 • R5

Answer. 27

Note. The Hessian is $\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$.

0360 exam-multi-005

微分 • R5

Answer. $16/5$

Note. Dot grad $f(1,2) = (4,1)$ with the unit direction.

0361 exam-multi-008

微分 • R5

Answer. $4*(x^2+y^2)$

Note. Compute the 2 by 2 determinant.

0362 exam-multi-009

微分 • R5

Answer. $2*x*y+2*y*z+2*z*x$

Note. Divergence is the sum of matching partial derivatives.

0363 exam-multi-010

微分 • R5

Answer. $-2*y$

Note. The z-component is $\text{partial}_x F_2 - \text{partial}_y F_1$.

0364 uni-der-001

微分 • R6

Answer. $x^{\cos(x)} * (-\sin(x) * \log(x) + \cos(x)/x)$

Note. Use log differentiation.

0365 uni-der-002

微分 • R6

Answer. $\sin(x)^x * (\log(\sin(x)) + x * \cot(x))$

Note. Differentiate $x \log(\sin x)$.

0366 uni-der-003

微分 • R6

Answer. 11

Note. Use the imaginary part of $(2+i)^3$.

0367 uni-der-004

微分 • R6

Answer. 12

Note. Read the x^4 coefficient of $x^2 e^x$.

0368 uni-der-005

微分 • R6

Answer. $-x/(1+x^2)^{(3/2)}$

Note. First derivative is $1/\sqrt{1+x^2}$.

0369 uni-der-006

微分 • R6

Answer. $\cos(x) * \exp(\sin(x)^2)$

Note. FTC plus chain rule.

0370 uni-der-007

微分 • R6

Answer. $2 * \log(1+4*x^2) - \log(1+x^2)$

Note. Differentiate both moving endpoints.

0371	uni-der-008	微分・R6
Answer. $2/(1+x^2)$ Note. Simplify after applying $1+u^2$.		
0372	uni-der-009	微分・R6
Answer. $5/3$ Note. $dy/dx=(1+1/t^2)/(1-1/t^2)$.		
0373	uni-der-010	微分・R6
Answer. 1 Note. Use $x=r \cos \theta$, $y=r \sin \theta$.		
0374	uni-der-011	微分・R6
Answer. -1 Note. Differentiate implicitly.		
0375	uni-der-012	微分・R6
Answer. 128 Note. The Hessian is $[[12,-4],[-4,12]]$.		
0376	uni-der-013	微分・R6
Answer. 2 Note. The positive critical point is (2,1).		
0377	uni-der-014	微分・R6
Answer. $6/5$ Note. $\text{grad } f(0,2)=(2,0)$.		
0378	uni-der-015	微分・R6
Answer. $\exp(y)+\exp(z)+\exp(x)$ Note. Add the matching partial derivatives.		
0379	uni-der-016	微分・R6
Answer. $-x$ Note. The z-component is $\text{partial}_x F_2 - \text{partial}_y F_1$.		
0380	uni-der-017	微分・R6
Answer. $1/(x*\log(x)*\log(\log(x)))$ Note. Three nested logs give three denominator factors.		
0381	uni-der-018	微分・R6
Answer. $3*\log(x)^2*\exp(\log(x)^3)/x$ Note. Differentiate the exponent.		
0382	uni-der-019	微分・R6
Answer. 32 Note. The fifth derivative returns a positive sine coefficient 2^5 .		

0383 uni-der-020

微分・R6

Answer. $(1+x^2)^x (\log(1+x^2) + 2x^2/(1+x^2))$

Note. Use log differentiation.

0384 depth-der-001

微分・R5

Answer. $x^x (x^2)^x (2x \log(x) + x)$

Note. Differentiate $x^2 \log x$.

0385 depth-der-002

微分・R6

Answer. $\sin(x)^{\cos(x)} (-\sin(x) \log(\sin(x)) + \cos(x) \cot(x))$

Note. Differentiate $\cos x \log(\sin x)$.

0386 depth-der-004

微分・R5

Answer. $2/x$

Note. After two derivatives the expression is $2 \log x + 3$.

0387 depth-der-005

微分・R6

Answer. $3x^2 \cos(x^6) - 2x \cos(x^4)$

Note. Apply FTC to both moving limits.

0388 depth-der-006

微分・R5

Answer. $1/\sqrt{x^2+4}$

Note. This is an asinh-type derivative.

0389 depth-der-007

微分・R5

Answer. -1

Note. Differentiate implicitly and substitute $(3,3)$.

0390 depth-der-008

微分・R5

Answer. 1

Note. $dy/dx = (dy/dt)/(dx/dt) = t+1$.

0391 depth-der-009

微分・R6

Answer. $3/4$

Note. Differentiate dy/dx with respect to t , then divide by dx/dt .

0392 depth-der-010

微分・R6

Answer. -3

Note. Use $-(f''(0))/(f'(0))^3$.

0393 depth-der-011

微分・R5

Answer. 8

Note. The Hessian is $[[2,2],[2,6]]$.

0394 depth-der-012

微分・R6

Answer. 0

Note. Away from the origin the Laplacian is zero.

0395 depth-der-013

微分・R5

Answer. $y \cdot z$

Note. Only the first component contributes yz.

0396 depth-der-014

微分・R5

Answer. $y^2 - x^2$

Note. Compute $\text{partial}_x Q - \text{partial}_y P$.

0397 depth-der-015

微分・R6

Answer. $2/(1+x^2)$

Note. For $x > 0$ this angle is $2 \arctan x$.

0398 depth-der-016

微分・R5

Answer. $(2 \log(x) - \log(x)^2)/x^2$

Note. Differentiate $\log(x)^2 x^{-1}$.

0399 depth-der-017

微分・R5

Answer. 81

Note. The fourth derivative returns $3^4 \cos(3x)$.

0400 depth-der-018

微分・R5

Answer. $(1 + 1/(2\sqrt{x})) / (2\sqrt{x + \sqrt{x}})$

Note. Chain rule across both radicals.

0401 burst-der-001

微分・R6

Answer. 1048576

Note. The value is 2^{20} .

0402 burst-der-002

微分・R6

Answer. -14348907

Note. Use $3^{15} \sin(15\pi/2)$.

0403 burst-der-003

微分・R6

Answer. 262144

Note. Even derivatives of $\cosh(2x)$ return 2^{18} at zero.

0404 burst-der-004

微分・R6

Answer. 720

Note. Read the x^{10} coefficient of $x^3 e^x$.

0405 burst-der-005

微分・R6

Answer. 56320

Note. Read the x^{11} coefficient after shifting the sine series by x^2 .

0406 burst-der-006

微分・R6

Answer. -5040

Note. The eighth derivative is $(-1)^7 7!$.

0407 burst-der-007

微分・R6

Answer. 479001600

Note. The x^{12} coefficient is 1.

0408	burst-der-008	微分・R6
Answer. -64 Note. Use the real part of $(1+i)^{12}$.		
0409	burst-der-009	微分・R6
Answer. 24024 Note. Read the x^{14} coefficient of x^4e^{-x} .		
0410	burst-der-010	微分・R6
Answer. -1199 Note. Use the imaginary part of $(2+i)^9$.		
0411	burst-boss-der-001	微分・R6
Answer. 241087680 Note. Use $[x^n](\log(1+x))^2 = (-1)^n 2H_{n-1}/n$.		
0412	burst-boss-der-002	微分・R6
Answer. -21257280 Note. Use $[x^n](\log(1+x))^2 = (-1)^n 2H_{n-1}/n$.		
0413	burst-boss-der-003	微分・R6
Answer. -22591 Note. Multiply the even series of e^{x^2} and $\cos x$.		
0414	burst-boss-der-004	微分・R6
Answer. 16124 Note. Use the imaginary part of $(1+2i)^{14}$.		
0415	burst-boss-der-005	微分・R6
Answer. 164833 Note. Use the real part of $(1+2i)^{16}$.		
0416	burst-boss-der-006	微分・R6
Answer. 351261243360 Note. Read the x^{18} coefficient after shifting by x^4 .		
0417	burst-boss-der-007	微分・R6
Answer. 27525120 Note. Read the x^{13} coefficient of $\sin(2x)$.		
0418	burst-boss-der-008	微分・R6
Answer. 13172296626 Note. Read the x^{16} coefficient of $\cos(3x)$.		
0419	burst-boss-der-009	微分・R6
Answer. 598752000 Note. The x^{10} coefficient is $C(11,3)$.		

0420	burst-boss-der-010	微分・R6
Answer. 1307674368000		
Note. Use $x^3/(1+x^2)=x^3-x^5+x^7-\dots$		
0421	burst-boss2-der-001	微分・R6
Answer. -1024		
Note. Use the imaginary part of $(1+i)^{21}$.		
0422	burst-boss2-der-002	微分・R6
Answer. -1024		
Note. Use the real part of $(1+i)^{20}$.		
0423	burst-boss2-der-003	微分・R6
Answer. 3041525760		
Note. Read the x^{12} coefficient of e^{2x} , then multiply by $17!$.		
0424	burst-boss2-der-004	微分・R6
Answer. 54747463680		
Note. Read the x^{12} coefficient of $\cos(2x)$, then multiply by $18!$.		
0425	burst-boss2-der-005	微分・R6
Answer. -1334792724768		
Note. Read the x^{15} coefficient of $\sin(3x)$, then multiply by $19!$.		
0426	burst-boss2-der-006	微分・R6
Answer. 239500800		
Note. The x^{10} coefficient is $C(12,2)$.		
0427	burst-boss2-der-007	微分・R6
Answer. 1796256000		
Note. Read the x^8 coefficient of $(1-x)^{-5}$.		
0428	burst-boss2-der-008	微分・R6
Answer. 271996268544000		
Note. Read the x^{12} coefficient of $(1+x)^{-2}$.		
0429	burst-boss3-der-001	微分・R6
Answer. 863130293635276800		
Note. Use $[x^n](\log(1+x))^2=(-1)^n 2H_{n-1}/n$.		
0430	burst-boss3-der-002	微分・R6
Answer. 248156004834017280000		
Note. Read the x^{14} coefficient of $(1-x)^{-7}$.		
0431	burst-boss3-der-003	微分・R6
Answer. 37719712734770626560000		
Note. Read the x^{15} coefficient of $(1-x)^{-6}$.		
0432	burst-boss3-der-004	微分・R6
Answer. 1621934672117760000		

Note. Read the x^{13} coefficient of $(1-x)^{-8}$.

0433 burst-boss3-der-005

微分・R6

Answer. 17166620433372086476800000

Note. Read the x^{24} term of e^{-x^2} .

0434 burst-boss3-der-006

微分・R6

Answer. 553761949463615692800000

Note. Read the x^{24} term of $\cosh(x^2)$.

0435 burst-boss3-der-007

微分・R6

Answer. -272789137666805760000

Note. Read the x^{22} term of $\sin(x^2)$.

0436 burst-boss3-der-008

微分・R6

Answer. 18458731648787189760000

Note. Read the x^{24} term of $\cos(x^2)$.

0437 world-004

微分・R1

Answer. $1/(1+x^2)$

Note. Standard derivative.

0438 world-024

微分・R4

Answer. -4

Note. Real part of $(1+i)^4 = -4$.

0439 world-025

微分・R4

Answer. $2/15$

Note. $\tan x = x + x^3/3 + 2x^5/15 + \dots$

0440 world-036

微分・R5

Answer. $61/720$

Note. Euler number $E_6/6! = 61/720$.

0441 world-068

微分・R6

Answer. $5/6$

Note. Expand $x \log(1+x)$ in the exponent.

0442 world-093

微分・R6

Answer. $-1/8$

Note. $e^{\{\sin x\}} = 1 + x + x^2/2 - x^4/8 - \dots$

0443 hd-001

微分・R5

Answer. 5040

Note. $1/(1-x) = \sum x^n$, so $f^{(7)}(0) = 7!$.

0444 hd-002

微分・R4

Answer. 12

Note. x^4 coefficient is $1/2!$, times $4!$.

0445	hd-003	微分 • R5
Answer. 120 Note. x^6 coefficient is $1/3!$, times $6!$.		
0446	hd-004	微分 • R5
Answer. 10 Note. x^{10} coefficient of $x \sin x$ is $1/9!$, times $10!$.		
0447	hd-005	微分 • R5
Answer. 24 Note. $1/(1+x^2)=1-x^2+x^4-\dots$, x^4 coeff 1.		
0448	hd-006	微分 • R6
Answer. -4 Note. Imaginary part of $(1+i)^5 = -4$.		
0449	hd-007	微分 • R6
Answer. -720 Note. x^7 coefficient is $-1/7$, times $7!$.		
0450	hd-008	微分 • R5
Answer. $17/315$ Note. $\tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + \dots$		
0451	hd-009	微分 • R5
Answer. $2 \arcsin(x)/\sqrt{1-x^2}$ Note. Chain rule on $(\arcsin x)^2$.		
0452	hd-010	微分 • R6
Answer. $x^2(x^2)(2x \log(x) + x)$ Note. Differentiate $x^2 \log x$ in the exponent.		
0453	hd-011	微分 • R5
Answer. $1/\sin(x)$ Note. Derivative of $\log \tan(x/2)$ is $\csc x$.		
0454	hd-012	微分 • R5
Answer. $2/3$ Note. $\tan^2 x = x^2 + 2x^4/3 + \dots$		
0455	app-001	微分 • R4
Answer. $1/\pi$ Note. $dV/dt = 4\pi r^2 dr/dt$.		
0456	app-002	微分 • R5
Answer. 12 Note. Max inscribed rectangle area is $2ab$.		

0457	app-003	微分・R4
Answer. $1/2$		
Note. $R = (1+y'^2)^{3/2}/ y'' = 1/2$.		
0458	app-004	微分・R5
Answer. $2\sqrt{2}$		
Note. $R = (1+1)^{3/2}/1 = 2\sqrt{2}$.		
0459	app-005	微分・R4
Answer. $3/2$		
Note. $x_1 = x_0 - (x_0^2 - 2)/(2x_0)$.		
0460	app-006	微分・R4
Answer. $81/40$		
Note. $2 + 0.1/4 = 2.025$.		
0461	app-007	微分・R4
Answer. 4		
Note. AM-GM or derivative gives 4 at $x=2$.		
0462	app-008	微分・R5
Answer. 2		
Note. $ -10 /\sqrt{9+16}=2$.		
0463	app-009	微分・R5
Answer. $4\pi/(3\sqrt{3})$		
Note. Max cylinder volume in a unit sphere.		
0464	app-010	微分・R6
Answer. $32\pi/81$		
Note. Max cone volume $= 32\pi R^3/81$.		
0465	app-011	微分・R5
Answer. 12		
Note. $x \, dx/dt + y \, dy/dt = 0; y=5$.		
0466	app-012	微分・R6
Answer. $2/(3\sqrt{3})$		
Note. $\kappa = x/(x^2+1)^{3/2}$, max at $x=1/\sqrt{2}$.		
0467	int-001	積分・R1
Answer. $2x^3$		
Note. 次方積分公式。常數 C 可省略。		
0468	int-002	積分・R1
Answer. $\sin(x)$		
Note. $\sin x$ 的導數是 $\cos x$ 。		
0469	int-003	積分・R1
Answer. $\log(\text{abs}(x))$		

Note. 基本積分為 $\ln|x|$ 。

0470 int-004

積分 • R1

Answer. $\exp(3x)/3$

Note. 鏈鎖律反向使用。

0471 int-005

積分 • R2

Answer. $\exp(x^2)/2$

Note. 令 $u=x^2$ ， $du=2x dx$ 。

0472 int-006

積分 • R2

Answer. $\log(1+x^2)$

Note. 分子是分母導數。

0473 int-007

積分 • R2

Answer. $\sin(x^2)/2$

Note. 令 $u=x^2$ 。

0474 int-008

積分 • R2

Answer. 1

Note. 原函數 x^3 ，代入 0 到 1。

0475 int-009

積分 • R3

Answer. $x^2 \log(x)/2 - x^2/4$

Note. 分部積分， $u=\ln x$ ， $dv=x dx$ 。

0476 int-010

積分 • R3

Answer. $-\cos(x) + \cos(x)^3/3$

Note. $\sin^3 x = \sin x(1 - \cos^2 x)$ ，令 $u=\cos x$ 。

0477 int-011

積分 • R3

Answer. $\operatorname{atan}(x/2)/2$

Note. 套 $\arctan$ 形式。

0478 int-012

積分 • R2

Answer. $\pi/4$

Note. 用 $\sin^2 x = (1 - \cos 2x)/2$ 。

0479 int-013

積分 • R3

Answer. $x(\log(x)^2 - 2\log(x) + 2)$

Note. 分部積分兩次。

0480 int-014

積分 • R3

Answer. $x^2/2 - \log(x^2+1)/2$

Note. 先做多項式除法： $x^3/(x^2+1) = x - x/(x^2+1)$ 。

0481 int-015

積分 • R4

Answer. $\exp(x)(\sin(x) - \cos(x))/2$

Note. 分部積分兩次，或猜 $e^x(a \sin x + b \cos x)$ 。

0482	int-016	積分・R3
Answer. $2\log(2)-1$ Note. 原函數是 $(1+x)\ln(1+x)-(1+x)$ 。		
0483	int-017	積分・R3
Answer. 1 Note. 分部積分，邊界項留下 1。		
0484	int-018	積分・R3
Answer. $-\log(1+\cos(x))$ Note. 令 $u=1+\cos x$ ， $du=-\sin x dx$ 。		
0485	int-019	積分・R4
Answer. $2\sqrt{x}-2\log(1+\sqrt{x})$ Note. 令 $u=\sqrt{x}$ ， $dx=2u du$ 。		
0486	int-020	積分・R3
Answer. $\log(2)/3$ Note. 令 $u=1+x^3$ 。		
0487	td-int-001	積分・R6
Answer. $\pi\exp(-3)/2$ Note. 使用 Fourier transform 型公式： $\int_0^\infty \cos(tx)/(1+x^2) dx = \pi e^{- t }/2$ 。		
0488	td-int-002	積分・R6
Answer. $\pi\log(9/2)$ Note. 套公式 $\int_0^\pi \log(a+b\cos x) dx = \pi\log((a+\sqrt{a^2-b^2})/2)$ 。		
0489	td-int-003	積分・R4
Answer. $\log(2/5)$ Note. Frullani 公式： $\int_0^\infty (e^{-\alpha x}-e^{-\beta x})/x dx = \log(\beta/\alpha)$ 。		
0490	td-int-004	積分・R4
Answer. $\log(7/3)/2$ Note. 用 $2\sin A\sin B = \cos(A-B)-\cos(A+B)$ ，再套 Frullani 型結果。		
0491	td-int-005	積分・R6
Answer. $\pi/\sqrt{5}$ Note. 二次型矩陣 $[[3,1],[1,2]]$ 的行列式為 5，面積是 $\pi/\sqrt{\det A}$ 。		
0492	td-int-006	積分・R6
Answer. $\sqrt{\pi}/(12\sqrt{3})$ Note. 高斯積分偶次矩： $\int x^{2m}e^{-ax^2} dx = (2m-1)!! \sqrt{\pi}/(2^m a^{m+1/2})$ 。		
0493	int-021	積分・R4
Answer. 1 Note. 內積分為 $x+1/2$ ，外積分為 $1/2+1/2=1$ 。		

0494 int-022

積分・R4

Answer. 2

Note. D 是直角三角形，兩股長皆為 2，面積為 2。

0495 int-023

積分・R4

Answer. $1/8$

Note. 內積分為 $x^2 y^{2/2} \big|_0^x = x^3/2$ ，外積分得 $1/8$ 。

0496 int-024

積分・R4

Answer. $1/3$

Note. 內積分為 $x(1-x) + (1-x)^2/2 = (1-x^2)/2$ ，外積分得 $1/3$ 。

0497 int-025

積分・R4

Answer. $\pi/2$

Note. 極座標下為 $\int_0^{2\pi} \int_0^1 r^3 dr d\theta = 2\pi \cdot (1/4) = \pi/2$ 。

0498 int-026

積分・R4

Answer. 9

Note. 分離為 $(\int_0^2 x dx)(\int_0^3 y dy) = 2 \cdot (9/2) = 9$ 。

0499 int-027

積分・R2

Answer. $\log(1+x^2)/2$

Note. $u=1+x^2$ ，積分為 $(1/2)\log(1+x^2)$ 。

0500 int-028

積分・R3

Answer. $x \log(x) - x$

Note. 分部積分得 $x \log x - \int 1 dx = x \log x - x$ 。

0501 int-029

積分・R2

Answer. $\text{atan}(x/3)/3$

Note. 由標準公式得 $(1/3)\text{atan}(x/3)$ 。

0502 int-030

積分・R2

Answer. $1/4$

Note. 令 $u=\sin x$ ，積分變成 $\int_0^1 u^3 du = 1/4$ 。

0503 int-031

積分・R4

Answer. $1/4$

Note. Swap order: integral is $\int_0^1 x^2 \cdot x dx = \int_0^1 x^3 dx = 1/4$.

0504 int-032

積分・R4

Answer. 3π

Note. Area = $\pi(4-1)=3\pi$.

0505 int-033

積分・R4

Answer. 4π

Note. Integral of r^2 over disk radius 2 is $\int_0^{2\pi} \int_0^2 r^3 dr d\theta = 8\pi$. Half is 4π .

0506 int-034

積分・R4

Answer. $\pi/4$

Note. The region is $x \geq 0, y \geq 0, x^2 + y^2 \leq 1$, so the area is $\pi/4$.

0507 int-035

積分・R4

Answer. $\pi/2$

Note. The integral is $2\pi \cdot (1/4) = \pi/2$.

0508 int-036

積分・R4

Answer. $1/6$

Note. Integral is $(1/2) \int_0^1 (1-y)^2 dy = 1/6$.

0509 int-038

積分・R3

Answer. $x \sin(x) + \cos(x)$

Note. Integration by parts gives $x \sin x - \int \sin x \, dx = x \sin x + \cos x$.

0510 int-039

積分・R3

Answer. $\pi/4$

Note. The integral is $\arctan(1) - \arctan(0) = \pi/4$.

0511 int-040

積分・R5

Answer. 1

Note. The integral equals $\Gamma(2) = 1! = 1$.

0512 int-041

積分・R3

Answer. $\log(1+x^4)/4$

Note. With $u=1+x^4$, the antiderivative is $(1/4)\log(1+x^4)$.

0513 int-042

積分・R3

Answer. $(2/9) \cdot (1+x^3)^{3/2}$

Note. $u=1+x^3$ gives $(1/3) \int u^{1/2} du = (2/9)(1+x^3)^{3/2}$.

0514 int-043

積分・R4

Answer. $\log(1+\sqrt{1+x^2})$

Note. With $u=\sqrt{1+x^2}$, the integrand becomes $du/(1+u)$, so the answer is $\log(1+\sqrt{1+x^2})$.

0515 int-044

積分・R3

Answer. $\log(1+\exp(x^2))/2$

Note. $u=1+e^{x^2}$ gives $(1/2) \int du/u = (1/2)\log(1+e^{x^2})$.

0516 int-045

積分・R3

Answer. $\log(x)^{3/3}$

Note. $u=\log x$ changes the integral to $\int u^2 du = u^3/3$.

0517 int-046

積分・R4

Answer. $3/8$

Note. The integral is $\int_1^2 u^{-3} du = [-1/(2u^2)]_1^2 = 3/8$.

0518 int-047

積分・R3

Answer. π

Note. The graph is the upper quarter of $x^2+y^2=4$, so the area is π .

0519	int-048	積分・R4
Answer. $\pi/4$ Note. After $x=\sin \theta$, the integral is $\int_0^{\pi/2} \sin^2 \theta \, d\theta = \pi/4$.		
0520	int-049	積分・R4
Answer. π Note. With $x=2\sin \theta$, the integral becomes $4\int_0^{\pi/4} d\theta = \pi$.		
0521	int-050	積分・R4
Answer. $\pi/3$ Note. With $x=\sec \theta$, the integrand becomes $d\theta$. The bounds give $\pi/3$.		
0522	int-051	積分・R4
Answer. $x/(4\sqrt{x^2+4})$ Note. The standard result is $\int dx/(x^2+a^2)^{3/2} = x/(a^2 \sqrt{x^2+a^2})$; here $a^2=4$.		
0523	int-052	積分・R3
Answer. $\exp(x)(x^2-2x+2)$ Note. Two rounds of IBP give $e^x(x^2-2x+2)$.		
0524	int-053	積分・R4
Answer. $-x^2\cos(x)+2x\sin(x)+2\cos(x)$ Note. IBP gives $-x^2 \cos x + 2\int x \cos x \, dx = -x^2 \cos x + 2x \sin x + 2\cos x$.		
0525	int-055	積分・R4
Answer. $x(\log(x)^2-2\log(x)+2)$ Note. IBP gives $x(\log x)^2-2\int \log x \, dx = x(\log x)^2-2(x\log x-x)$.		
0526	int-056	積分・R4
Answer. $\pi/4-1/2$ Note. IBP gives $\pi/8 - (1/2)\int_0^1 x^2/(1+x^2)dx = \pi/8 - 1/2 + \pi/8 = \pi/4-1/2$.		
0527	int-057	積分・R5
Answer. $(\exp(x)-\sin(x)-\cos(x))/2$ Note. $F=e^x \int_0^x e^{-t}\sin t \, dt = (e^x\sin x-\cos x)/2$.		
0528	int-058	積分・R5
Answer. $(\exp(2x)-2x-1)/4$ Note. $e^{2x}\int_0^x t e^{-2t}dt = e^{2x}/4-x/2-1/4$.		
0529	int-059	積分・R5
Answer. $x^2-2x+2-2\exp(-x)$ Note. Solving $F'+F=x^2$ with $F(0)=0$ gives $F=x^2-2x+2-2e^{-x}$.		
0530	int-060	積分・R5
Answer. $x\sin(x)/2$ Note. Expanding and integrating gives $x\sin x/2$.		

0531 int-061

積分 • R5

Answer. $\pi/4$

Note. Let I be the integral. The transformed integral has $\cos^3$ numerator. Adding gives $2I = \int_0^{\pi/2} 1 \, dx = \pi/2$.

0532 int-062

積分 • R5

Answer. $1/2$

Note. I plus its transformed version equals $\int_0^1 1 \, dx = 1$, so $I = 1/2$.

0533 int-063

積分 • R5

Answer. $\pi/4$

Note. Adding the integral to its transformed copy gives $\int_0^{\pi/2} 1 \, dx = \pi/2$, hence $I = \pi/4$.

0534 int-064

積分 • R5

Answer. $\pi^2/4$

Note. The symmetric factor gives $I = (\pi/2) \int_0^{\pi} \sin x / (1 + \cos^2 x) \, dx$. With $u = \cos x$, the remaining integral is $\pi/2$, so $I = \pi^2/4$.

0535 int-065

積分 • R5

Answer. $\log(7/3)$

Note. Frullani gives $\int_0^\infty (e^{-ax} - e^{-bx})/x \, dx = \log(b/a)$. Here $a=3, b=7$.

0536 int-066

積分 • R5

Answer. $\log(4)$

Note. The value is $\log(4/1) = \log 4$.

0537 int-067

積分 • R5

Answer. $\log(4)$

Note. Frullani gives $\log(8/2) = \log 4$.

0538 int-068

積分 • R5

Answer. $\log(5/2)$

Note. The cosine Frullani identity gives $\log(5/2)$.

0539 int-069

積分 • R5

Answer. $\log(4)$

Note. $\int_0^\infty (\cos x - \cos 4x)/x \, dx = \log 4$.

0540 int-070

積分 • R5

Answer. $\log(3)$

Note. The integral equals $\log(9/3) = \log 3$.

0541 int-071

積分 • R5

Answer. $\log(11/5)/2$

Note. The integral is $(1/2) \int_0^\infty (\cos 5x - \cos 11x)/x \, dx = (1/2) \log(11/5)$.

0542 int-072

積分 • R5

Answer. $\log(2)/2$

Note. The value is $(1/2) \log(8/4) = \log(2)/2$.

0543 int-073

積分 • R5

Answer. $\log(5/3)/2$

Note. The integral equals $(1/2)\log(5/3)$.

0544 int-074

積分 • R5

Answer. $\pi \cdot \log(5/2)/2$

Note. Frullani gives $(\pi/2)\log(5/2)$.

0545 int-075

積分 • R5

Answer. $\log(5/2)$

Note. Frullani gives $\log(5/2)$.

0546 int-076

積分 • R5

Answer. $\log(6)$

Note. By Frullani, the value is $\log 6$.

0547 gap-int-pf-001

積分 • R3

Answer. $(\log(x-1) - \log(x+1))/2$

Note. Partial fractions give $1/(x^2-1) = 1/2/(x-1) - 1/2/(x+1)$.

0548 gap-int-pf-002

積分 • R3

Answer. $\log(x)$

Note. The integrand reduces to $1/x$, so the answer is $\log x$.

0549 gap-int-pf-003

積分 • R3

Answer. $(\log(x) - \log(x+2))/2$

Note. $1/(x(x+2)) = (1/2)/x - (1/2)/(x+2)$.

0550 gap-int-pf-004

積分 • R3

Answer. $\log(x+1) + \log(x+2)$

Note. The denominator derivative is $2x+3$, so the integral is $\log(x^2+3x+2)$.

0551 gap-int-pf-005

積分 • R3

Answer. $-1/(x+1)$

Note. The antiderivative of $(x+1)^{-2}$ is $-(x+1)^{-1}$.

0552 gap-int-pf-006

積分 • R4

Answer. $8 \cdot \log(x-1)/3 + \log(x+2)/3$

Note. $A=8/3$ and $B=1/3$, so integrate the two simple fractions.

0553 gap-int-pf-007

積分 • R4

Answer. $2 \cdot \log(x+2) - \log(x+3)$

Note. $A=2$ and $B=-1$, giving $2\log(x+2) - \log(x+3)$.

0554 gap-int-polar-001

積分 • R2

Answer. π

Note. $\text{Area} = (1/2) \cdot 4 \cdot (\pi/2) = \pi$.

0555 gap-int-polar-002

積分 • R3

Answer. $\pi/4$

Note. The polar area is $(1/2)(\pi/2) = \pi/4$.

0556	gap-int-triple-001	積分 • R6
Answer. $1/8$		
Note. The value is $(1/2)^3=1/8$.		
0557	gap-int-triple-002	積分 • R6
Answer. $3/2$		
Note. Each coordinate contributes $1/2$, so the total is $3/2$.		
0558	gap-int-triple-003	積分 • R6
Answer. $1/6$		
Note. The standard simplex in $\mathbb{R}^3$ has volume $1/6$.		
0559	gap-int-triple-004	積分 • R6
Answer. $1/6$		
Note. The integral becomes $\int_0^1 x^2/2 \, dx=1/6$.		
0560	gap-int-triple-005	積分 • R6
Answer. 12π		
Note. The volume is $\pi(2^2)(3)=12\pi$.		
0561	gap-int-triple-006	積分 • R6
Answer. $32\pi/3$		
Note. The volume is $(4/3)\pi(2^3)=32\pi/3$.		
0562	gap-int-cov-001	積分 • R6
Answer. $1/2$		
Note. The inverse Jacobian determinant is $-1/2$, so the absolute value is $1/2$.		
0563	gap-int-cov-002	積分 • R6
Answer. 1		
Note. The area is $2*(1/2)=1$.		
0564	gap-tech-003	積分 • R2
Answer. substitution		
Note. Use $u=x^2$.		
0565	gap-tech-004	積分 • R3
Answer. integrationbyparts		
Note. Use integration by parts with $u=\log x$.		
0566	gap-tech-006	積分 • R3
Answer. partialfraction		
Note. Use partial fractions after factoring x^2-1 .		
0567	mob-tech-003	積分 • R2
Answer. u-substitution		
Note. Use u-substitution.		

0568	mob-tech-004	積分・R2
Answer. integrationbyparts		
Note. Use integration by parts.		
0569	mob-tech-005	積分・R2
Answer. partialfraction		
Note. Use partial fraction.		
0570	mob-tech-006	積分・R2
Answer. trigsbstitution		
Note. Use trig substitution.		
0571	mob-tech-010	積分・R2
Answer. King' sproperty		
Note. Use King's property.		
0572	mob-tech-011	積分・R2
Answer. Frullaniintegral		
Note. Use Frullani integral.		
0573	mob-tech-013	積分・R2
Answer. gradienttheorem		
Note. Use gradient theorem.		
0574	mob-tech-015	積分・R2
Answer. Betafunction		
Note. Use Beta function.		
0575	mob-tech-016	積分・R2
Answer. Gammafunction		
Note. Use Gamma function.		
0576	mob-tech-017	積分・R2
Answer. Wallisreduction		
Note. Use Wallis reduction.		
0577	mob-trap-002	積分・R2
Answer. dividebyinnercoefficient		
Note. divide by inner coefficient		
0578	mob-trap-003	積分・R2
Answer. arctannotlog		
Note. arctan not log		
0579	mob-trap-006	積分・R2
Answer. requiresapositive		
Note. requires a positive		
0580	mob-trap-008	積分・R2
Answer. multiplybyJacobian		

Note. multiply by Jacobian

0581 mob-trap-010

積分・R2

Answer. $\sec x$

Note. $\sec x$

0582 mob-trap-012

積分・R2

Answer. $\frac{1}{\Gamma(a+b)}$

Note. $\frac{1}{\Gamma(a+b)}$

0583 mob-beta-001

積分・R5

Answer. 1

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0584 mob-beta-002

積分・R5

Answer. $\frac{1}{2}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0585 mob-beta-003

積分・R5

Answer. $\frac{1}{6}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0586 mob-beta-004

積分・R5

Answer. $\frac{1}{12}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0587 mob-beta-005

積分・R5

Answer. π

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0588 mob-beta-006

積分・R5

Answer. 2

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0589 mob-beta-007

積分・R5

Answer. $\frac{\pi}{2}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0590 mob-beta-009

積分・R5

Answer. $\frac{1}{30}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0591 mob-beta-010

積分・R5

Answer. $\frac{1}{4}$

Note. Use $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

0592 mob-gamma-001

積分・R5

Answer. 1

Note. Use the gamma recurrence and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

0593	mob-gamma-002	積分 • R5
Answer. $\sqrt{\pi}$ Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0594	mob-gamma-003	積分 • R5
Answer. $\sqrt{\pi}/2$ Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0595	mob-gamma-004	積分 • R5
Answer. $3\sqrt{\pi}/4$ Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0596	mob-gamma-005	積分 • R5
Answer. 120 Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0597	mob-gamma-006	積分 • R5
Answer. 2 Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0598	mob-gamma-007	積分 • R5
Answer. $\sqrt{\pi}/2$ Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0599	mob-gamma-008	積分 • R5
Answer. 6 Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0600	mob-gamma-009	積分 • R5
Answer. $15\sqrt{\pi}/8$ Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0601	mob-gamma-010	積分 • R5
Answer. 1 Note. Use the gamma recurrence and $\Gamma(1/2)=\sqrt{\pi}$.		
0602	mob-wallis-002	積分 • R5
Answer. $2/3$ Note. Evaluate by Wallis reduction or beta symmetry.		
0603	mob-wallis-003	積分 • R5
Answer. $3\pi/16$ Note. Evaluate by Wallis reduction or beta symmetry.		
0604	mob-wallis-004	積分 • R5
Answer. $8/15$ Note. Evaluate by Wallis reduction or beta symmetry.		

0605	mob-wallis-005	積分 • R5
Answer. $\pi/4$		
Note. Evaluate by Wallis reduction or beta symmetry.		
0606	mob-wallis-006	積分 • R5
Answer. $3\pi/16$		
Note. Evaluate by Wallis reduction or beta symmetry.		
0607	mob-wallis-007	積分 • R5
Answer. $\pi/16$		
Note. Evaluate by Wallis reduction or beta symmetry.		
0608	mob-wallis-008	積分 • R5
Answer. 1		
Note. Evaluate by Wallis reduction or beta symmetry.		
0609	mob-wallis-009	積分 • R5
Answer. 1		
Note. Evaluate by Wallis reduction or beta symmetry.		
0610	mob-wallis-010	積分 • R5
Answer. $1/2$		
Note. Evaluate by Wallis reduction or beta symmetry.		
0611	rel-basic-013	積分 • R1
Answer. $1/3$		
Note. Use the power rule.		
0612	rel-basic-014	積分 • R1
Answer. 2		
Note. The antiderivative is $-\cos x$.		
0613	rel-basic-015	積分 • R1
Answer. $(e^2 - 1)/2$		
Note. Divide by the inner coefficient 2.		
0614	rel-basic-016	積分 • R1
Answer. $\log(2)$		
Note. The antiderivative is $\log(1+x)$.		
0615	rel-basic-018	積分 • R1
Answer. 6		
Note. The antiderivative is $3x^2/2$.		
0616	rel-adv-011	積分 • R5
Answer. $1/9$		
Note. Use $\Gamma(2)/3^2$.		
0617	rel-adv-014	積分 • R4
Answer. $5/6$		

Note. Expand and integrate term by term.

0618 rel-adv-009

積分・R3

Answer. $\exp(x^2) \cdot (x^2 - 1)/2$

Note. Let $u = x^2$. Then $x^3 dx = (u/2) du$.

0619 rel-adv-010

積分・R3

Answer. $(1+x^2) \cdot \log(1+x^2)/2 - (1+x^2)/2$

Note. Let $u = 1+x^2$ and integrate $\log u$.

0620 rel-boss-001

積分・R5

Answer. $\pi/2$

Note. Dirichlet integral gives $\pi/2$ for positive frequency.

0621 rel-boss-002

積分・R5

Answer. $\log(5/2)$

Note. Frullani gives $\log(5/2)$.

0622 rel-boss-003

積分・R5

Answer. $\log(7/2)$

Note. Use the cosine Frullani identity.

0623 rel-boss-004

積分・R5

Answer. $15 \cdot \sqrt{\pi}/8$

Note. This is $\Gamma(7/2)$.

0624 rel-boss-005

積分・R5

Answer. $3\pi/8$

Note. This is $B(1/2, 5/2)$.

0625 rel-boss-008

積分・R6

Answer. 0

Note. There is no z^{-1} term.

0626 rel-boss-009

積分・R6

Answer. $1/6$

Note. The residue is the z^3 coefficient of e^z .

0627 rel-boss-010

積分・R6

Answer. $1/18$

Note. Use $u = x+y$ and $v = x/(x+y)$.

0628 rel-boss-011

積分・R6

Answer. $1/720$

Note. Use the simplex beta integral.

0629 rel-boss-012

積分・R6

Answer. $128\pi/5$

Note. Use spherical coordinates: integrate $4\pi r^4$.

0630	rel-boss-017	積分・R5
Answer. 2		
Note. Differentiate the beta/gamma integral or use $\Gamma(3)$.		
0631	rel-boss-020	積分・R6
Answer. $1/2$		
Note. The $z^{2/2}$ term in $1-\cos z$ creates the residue.		
0632	rel-hard-int-001	積分・R5
Answer. $15/8$		
Note. Use $\Gamma(6)/2^6$.		
0633	rel-hard-int-002	積分・R5
Answer. $3\sqrt{\pi}/128$		
Note. This is $\Gamma(5/2)/4^{5/2}$.		
0634	rel-hard-int-004	積分・R5
Answer. $1/32$		
Note. Use $\int_0^1 x^a (\log x)^2 dx = 2/(a+1)^3$.		
0635	rel-hard-int-005	積分・R5
Answer. $35\pi/256$		
Note. Use Wallis reduction.		
0636	rel-hard-int-007	積分・R5
Answer. $\pi/32$		
Note. Convert to one half of a beta integral.		
0637	rel-hard-int-008	積分・R5
Answer. $\log(11/3)$		
Note. Frullani gives $\log(11/3)$.		
0638	rel-hard-int-009	積分・R5
Answer. $\log(5/2)$		
Note. Use the cosine Frullani identity.		
0639	rel-hard-int-011	積分・R5
Answer. $\pi/4$		
Note. Let $u=x^2$.		
0640	rel-hard-int-012	積分・R5
Answer. $\pi/(2\sqrt{2})$		
Note. The standard quartic integral equals $\pi/(2\sqrt{2})$.		
0641	rel-hard-int-015	積分・R5
Answer. -4		
Note. Use $\int_0^1 x^{a-1} \log x dx = -1/a^2$.		

0642	rel-hard-int-017	積分・R5
Answer. $-\pi \log(2)$		
Note. Both log sine and log cosine integrals equal $-\pi \log 2 / 2$.		
0643	rel-hard-int-018	積分・R5
Answer. $\pi/32$		
Note. Use $\int_0^\infty dx/(x^2+a^2)^2 = \pi/(4a^3)$.		
0644	rel-hard-cpx-001	積分・R6
Answer. $-1/6$		
Note. Use the z^3 coefficient of $\sin z$.		
0645	rel-hard-cpx-002	積分・R6
Answer. $-1/2$		
Note. Use the z^2 coefficient of $\cos z$.		
0646	rel-hard-cpx-003	積分・R6
Answer. $1/6$		
Note. Use the z^3 coefficient of $e^z - 1 - z$.		
0647	rel-hard-cpx-004	積分・R6
Answer. $-1/2$		
Note. Use the z^2 coefficient of $\log(1+z)$.		
0648	rel-hard-cpx-005	積分・R6
Answer. 1		
Note. Expand $1/(1-z)$.		
0649	rel-hard-cpx-006	積分・R6
Answer. e		
Note. For a second-order pole, take the derivative of e^z .		
0650	rel-hard-cpx-007	積分・R6
Answer. 1		
Note. Use $\csc^2 z = 1/z^2 + \dots$		
0651	rel-hard-cpx-008	積分・R6
Answer. 1		
Note. $\cot z$ has residue 1 at 0.		
0652	rel-hard-cpx-009	積分・R6
Answer. 2		
Note. Use the z^2 coefficient of $e^{2z} - 1$.		
0653	rel-hard-cpx-010	積分・R6
Answer. $-1/24$		
Note. Use the z^4 coefficient of $1 - \cos z - z^2/2$.		
0654	rel-hard-mv-001	積分・R5
Answer. $1/60$		

Note. Use the two-dimensional simplex formula.

0655 rel-hard-mv-002

積分・R5

Answer. $\pi/24$

Note. Use polar coordinates and integrate $\cos^2 \theta \sin^2 \theta$.

0656 rel-hard-mv-003

積分・R6

Answer. $4\pi/5$

Note. Use spherical coordinates.

0657 rel-hard-mv-004

積分・R6

Answer. 36π

Note. Volume of a ball of radius 3.

0658 rel-hard-mv-008

積分・R6

Answer. $1/2$

Note. Use $u=x+y$, $v=x-y$, with Jacobian $1/2$.

0659 rel-hard-mv-009

積分・R5

Answer. $64\pi/3$

Note. Use polar coordinates and integrate r^5 .

0660 rel-hard-mv-010

積分・R6

Answer. $1/2520$

Note. Use the three-dimensional simplex beta formula.

0661 hc-usub-001

積分・R6

Answer. $(1+x^2)^{3/2}/3 - \sqrt{1+x^2}$

Note. Let $u=1+x^2$, so $x^3 dx = (u-1)du/2$.

0662 hc-usub-002

積分・R6

Answer. $\exp(x^2) \cdot (x^4 - 2x^2 + 2)/2$

Note. Let $u=x^2$, then integrate $u^2 e^u$ by parts.

0663 hc-usub-003

積分・R6

Answer. $(1+x^2) \cdot (\log(1+x^2)^2 - 2\log(1+x^2) + 2)/2$

Note. Let $u=1+x^2$, then use the antiderivative of $(\log u)^2$.

0664 hc-usub-004

積分・R6

Answer. $\operatorname{atan}(x^3)/3$

Note. Let $u=x^3$; the integral becomes one third of $\int du/(1+u^2)$.

0665 hc-usub-005

積分・R6

Answer. $\log(1+\sqrt{1+x^4})/2$

Note. Let $u=\sqrt{1+x^4}$; the remaining integral is $du/(2(1+u))$.

0666 hc-usub-006

積分・R6

Answer. $2 \cdot (1+\exp(x))^{3/2}/3$

Note. Let $u=1+e^x$.

0667	hc-usub-007	積分 • R6
Answer. $\pi/6$ Note. Use $u=x^3$ to reduce the integral to one third of $\arcsin u$ from 0 to 1.		
0668	hc-usub-008	積分 • R6
Answer. $1/16$ Note. Let $u=1+x^2$; then $x^3 dx=(u-1)du/2$.		
0669	hc-usub-009	積分 • R6
Answer. $1/4$ Note. Let $u=1+\sin^2 x$; the bounds are 1 and 2.		
0670	hc-usub-010	積分 • R6
Answer. $\log((1+e)/2)/3$ Note. Let $u=e^{x^3}$; then $du=3x^2e^{x^3}dx$.		
0671	hc-ibp-002	積分 • R6
Answer. $x^2\sin(3x)/3+2x\cos(3x)/9-2\sin(3x)/27$ Note. Use integration by parts twice and keep the factor 3 through each step.		
0672	hc-ibp-003	積分 • R6
Answer. $\exp(x)((x^2-1)\sin(x)+(2x-1-x^2)\cos(x))/2$ Note. Use the complex shortcut or cyclic integration by parts.		
0673	hc-ibp-005	積分 • R6
Answer. $(x^2+1)\operatorname{atan}(x)^2/2-x\operatorname{atan}(x)+\log(1+x^2)/2$ Note. After the first IBP, split $x^2/(1+x^2)=1-1/(1+x^2)$.		
0674	hc-ibp-006	積分 • R6
Answer. $\exp(2x)(2\sin(3x)-3\cos(3x))/13$ Note. Two IBP rounds return the original integral; solve algebraically.		
0675	hc-ibp-007	積分 • R6
Answer. $x^4\log(x)/4-x^4/16$ Note. Use $u=\log x$ and $dv=x^3 dx$.		
0676	hc-ibp-008	積分 • R6
Answer. $2\log(2)/5-47/300$ Note. IBP leaves $\int_0^1 x^5/(1+x) dx$; divide the polynomial before integrating.		
0677	hc-ibp-009	積分 • R6
Answer. $\pi^2/4-2$ Note. Two IBP rounds give $x^2 \sin x+2x \cos x-2\sin x$.		
0678	hc-ibp-010	積分 • R6
Answer. $1/6$ Note. IBP reduces the problem to $\int_0^1 x^4/(1+x^2) dx$.		

0679	hc-rad-001	積分 • R6
Answer. $\sqrt{x^2-1}/x$ Note. Differentiate $\sqrt{x^2-1}/x$, or use $x=\sec \theta$.		
0680	hc-rad-002	積分 • R6
Answer. $\operatorname{atan}((x+1)/2)/2$ Note. Complete the square: $x^2+2x+5=(x+1)^2+4$.		
0681	hc-rad-003	積分 • R6
Answer. $-\log(\operatorname{abs}(x))+\log(\operatorname{abs}(x-1))+\log(\operatorname{abs}(x+1))$ Note. Decompose into $-1/x+1/(x-1)+1/(x+1)$.		
0682	hc-rad-004	積分 • R6
Answer. $(\log(\operatorname{abs}(x-1))-\log(\operatorname{abs}(x+1)))/4-\operatorname{atan}(x)/2$ Note. Split x^4-1 into $(x-1)(x+1)(x^2+1)$.		
0683	hc-rad-005	積分 • R6
Answer. $2\sqrt{3}-2\pi/3$ Note. Use the antiderivative $\sqrt{x^2-4}-2\arccos(2/x)$.		
0684	hc-rad-006	積分 • R6
Answer. 1 Note. Rationalize to $(1-\sin x)/\cos^2 x$, whose antiderivative is $\tan x-\sec x$.		
0685	hc-rad-007	積分 • R6
Answer. $8/315$ Note. Use the beta integral or peel off one sine factor.		
0686	hc-rad-008	積分 • R6
Answer. $\pi/3$ Note. Use $\int_0^\infty dx/(1+x^a)=\pi/a\csc(\pi/a)$.		
0687	hc-rad-010	積分 • R6
Answer. $\pi/4$ Note. Let $u=x^2$ to reduce the integral to half of $\arcsin u$.		
0688	hc-spec-002	積分 • R6
Answer. $\pi/8$ Note. This is $B(3/2,3/2)$.		
0689	hc-spec-005	積分 • R6
Answer. $8\pi^6/63$ Note. Use $\Gamma(6)\zeta(6)$.		
0690	hc-spec-006	積分 • R6
Answer. $3\sqrt{\pi}/(32\sqrt{2})$ Note. Use $\int_0^\infty x^m e^{-ax^2} dx=\Gamma((m+1)/2)/(2a^{(m+1)/2})$.		
0691	exam-int-001	積分 • R5
Answer. $-1/(2(1+x^2))$		

Note. Let $u=1+x^2$.

0692 exam-int-002

積分 • R5

Answer. $x^2/2 - \log(1+x^2)/2$

Note. Divide x^3 by $1+x^2$ as $x - x/(1+x^2)$.

0693 exam-int-004

積分 • R5

Answer. $x^2 \log(x)/2 - x^2/4$

Note. Let $u=\log x$ and $dv=x \, dx$.

0694 exam-int-005

積分 • R5

Answer. $\log(x)^2/2$

Note. Let $u=\log x$.

0695 exam-int-007

積分 • R5

Answer. $\tan(x)^2/2$

Note. Let $u=\tan x$.

0696 exam-int-010

積分 • R5

Answer. $2 \log(1+x^{3/2})/3$

Note. Let $u=1+x^{3/2}$.

0697 exam-int-011

積分 • R5

Answer. $-\sqrt{4-x^2}$

Note. Let $u=4-x^2$.

0698 exam-int-012

積分 • R5

Answer. $\exp(2x) \cdot (2 \sin(x) - \cos(x))/5$

Note. Use the standard exponential-trig formula.

0699 exam-int-013

積分 • R5

Answer. $x \cdot \arctan(x) - \log(1+x^2)/2$

Note. Let $u=\arctan x$ and $dv=dx$.

0700 exam-int-014

積分 • R5

Answer. $x - \arctan(x)$

Note. Rewrite as $1 - 1/(x^2+1)$.

0701 exam-int-015

積分 • R5

Answer. $\log(\log(x))$

Note. Let $u=\log x$.

0702 exam-int-016

積分 • R5

Answer. $\sin(\log(x))$

Note. Let $u=\log x$.

0703 exam-int-017

積分 • R5

Answer. $\log(x^2+3x+5)$

Note. The numerator is the derivative of the denominator.

0704	exam-int-018	積分 • R5
Answer. $\text{atan}(x^2)/2$ Note. Let $u=x^2$.		
0705	exam-int-019	積分 • R5
Answer. $2*\text{atan}(\text{sqrt}(x))$ Note. Let $u=\text{sqrt } x$.		
0706	exam-int-020	積分 • R5
Answer. $2*\text{sqrt}(1+x^3)/3$ Note. Let $u=1+x^3$.		
0707	exam-int-021	積分 • R5
Answer. $\log(2)/2$ Note. Let $u=1+x^2$.		
0708	exam-int-024	積分 • R5
Answer. $\log(2)$ Note. Let $u=1+\cos x$.		
0709	exam-int-025	積分 • R5
Answer. -1 Note. Integrate $x \log x - x$ and take the limit at 0.		
0710	exam-int-026	積分 • R5
Answer. $1/2$ Note. The exponential tail gives $1/2$.		
0711	exam-int-028	積分 • R5
Answer. $\log(2)-1/2$ Note. Let $u=1+x^2$ and integrate $\log u$.		
0712	exam-int-029	積分 • R5
Answer. π Note. Integration by parts leaves π .		
0713	exam-int-030	積分 • R5
Answer. $1/6$ Note. Use $u=1-x$ or the beta integral.		
0714	exam-multi-006	積分 • R5
Answer. $1/2$ Note. The inner integral gives $3x^2/2$.		
0715	exam-multi-007	積分 • R5
Answer. π Note. Use one-half integral of r^2 from $-\pi/2$ to $\pi/2$.		

0716 uni-int-003

積分・R6

Answer. $-(\log(x)+1)/x$ **Note.** Integrate by parts with $dv=x^{-2}dx$.**0717 uni-int-004**

積分・R6

Answer. $-\operatorname{atan}(x)/x+\log(x)-\log(1+x^2)/2$ **Note.** After parts, decompose $1/(x(1+x^2))$.**0718 uni-int-005**

積分・R6

Answer. $(1+x^2)^{3/2}/3$ **Note.** Let $u=1+x^2$.**0719 uni-int-006**

積分・R6

Answer. $\sin(x)^6/6$ **Note.** Let $u=\sin x$.**0720 uni-int-007**

積分・R6

Answer. $\tan(x)+\tan(x)^3/3$ **Note.** Write $\sec^4 x=(1+\tan^2 x)\sec^2 x$.**0721 uni-int-008**

積分・R6

Answer. $\operatorname{atan}((x+1)/2)/2$ **Note.** Complete the square.**0722 uni-int-009**

積分・R6

Answer. $\log(x^2+x+1)$ **Note.** The numerator is the derivative of the denominator.**0723 uni-int-010**

積分・R6

Answer. $\exp(x)*(\sin(x)+\cos(x))/2$ **Note.** Use the standard exponential-trig integral.**0724 uni-int-011**

積分・R6

Answer. $x^3*\log(x)/3-x^3/9$ **Note.** Let $u=\log x$.**0725 uni-int-012**

積分・R6

Answer. $(x^2+1)*\operatorname{atan}(x)/2-x/2$ **Note.** Integrate by parts and rewrite $x^2/(1+x^2)$.**0726 uni-int-013**

積分・R6

Answer. $1/2-\log(2)/2$ **Note.** Rewrite as $x-x/(1+x^2)$.**0727 uni-int-014**

積分・R6

Answer. $\pi/4$ **Note.** Use $x=\tan \theta$ or the standard Wallis value.**0728 uni-int-015**

積分・R6

Answer. $1/4$

Note. Integrate by parts; the remaining rational integral cancels log terms.

0729 uni-int-016

積分 • R6

Answer. $2/15$

Note. Use the beta integral.

0730 uni-int-018

積分 • R6

Answer. $1/2$

Note. Let $u=x^2$.

0731 uni-int-020

積分 • R6

Answer. $\pi/(3\sqrt{3})$

Note. Complete the square.

0732 uni-int-023

積分 • R6

Answer. 6π

Note. This is an ellipse with semiaxes 2 and 3.

0733 uni-int-025

積分 • R6

Answer. $\pi/4$

Note. Use the arctangent antiderivative.

0734 depth-int-002

積分 • R6

Answer. $x^3/3 - \log(1+x^3)/3$

Note. Divide as $x^2-x^2/(1+x^3)$.

0735 depth-int-003

積分 • R5

Answer. $-1/\sqrt{1+x^2}$

Note. Let $u=1+x^2$.

0736 depth-int-005

積分 • R6

Answer. $\exp(x) \cdot (x(\sin(x) - \cos(x)) + \cos(x))/2$

Note. Use parts with the standard integral of $e^x \sin x$.

0737 depth-int-006

積分 • R5

Answer. $\exp(2x) \cdot (2\cos(3x) + 3\sin(3x))/13$

Note. Use the exponential-trig formula.

0738 depth-int-007

積分 • R6

Answer. $2\sqrt{x} \cdot \operatorname{atan}(\sqrt{x}) - \log(1+x)$

Note. Let $u=\sqrt{x}$, then integrate $\operatorname{atan} u$.

0739 depth-int-008

積分 • R5

Answer. $2\sqrt{\log(x)}$

Note. Let $u=\log x$.

0740 depth-int-009

積分 • R6

Answer. $x(\sin(\log(x)) - \cos(\log(x)))/2$

Note. Set $x=e^u$.

0741	depth-int-010	積分・R5
Answer. $2\log(1+\sqrt{x})$ Note. Let $u=\sqrt{x}$.		
0742	depth-int-011	積分・R6
Answer. $\arctan(x)/2-x/(2(1+x^2))$ Note. Rewrite as $1/(1+x^2)-1/(1+x^2)^2$.		
0743	depth-int-013	積分・R5
Answer. $\log(1+\tan(x))$ Note. Let $u=1+\tan x$.		
0744	depth-int-014	積分・R6
Answer. $x^2\log(1+x)/2-x^2/4+x/2-\log(1+x)/2$ Note. After parts, divide x^2 by $x+1$.		
0745	depth-int-015	積分・R6
Answer. $\log(3)/2-\pi/(3\sqrt{3})$ Note. Split x as half the derivative minus a constant.		
0746	depth-int-016	積分・R5
Answer. $\pi/2-1$ Note. Integrate by parts.		
0747	depth-int-017	積分・R6
Answer. $\pi^2/12$ Note. Expand $\log(1+x)$ and integrate termwise.		
0748	depth-int-018	積分・R6
Answer. $\log(2)/2-1/4$ Note. Let $u=1+x^2$.		
0749	depth-int-019	積分・R5
Answer. $1/2$ Note. Let $u=1+x^2$.		
0750	depth-int-020	積分・R6
Answer. $\pi/12$ Note. Use partial fractions in x^2 .		
0751	depth-int-021	積分・R5
Answer. $3\pi/8$ Note. Use power reduction or Wallis.		
0752	depth-int-022	積分・R5
Answer. $1/3$ Note. Integrate x first, then y .		

0753 depth-int-023	積分 • R6
Answer. $3\pi/2$	
Note. Use one-half integral of r^2 from 0 to 2π .	
0754 depth-int-026	積分 • R5
Answer. $1/2$	
Note. Let $u=x^2$.	
0755 depth-int-027	積分 • R5
Answer. $\pi/2$	
Note. This is $\arcsin x$ from 0 to 1.	
0756 depth-int-028	積分 • R5
Answer. $-1/9$	
Note. Use the parameter integral or parts.	
0757 burst-int-001	積分 • R6
Answer. $8/81$	
Note. Repeated IBP gives $4!/3^5$.	
0758 burst-int-002	積分 • R6
Answer. $1/2$	
Note. Use the imaginary part of $2!/(1-i)^3$.	
0759 burst-int-003	積分 • R6
Answer. $4/125$	
Note. Use the real part of $2!/(2-i)^3$.	
0760 burst-int-004	積分 • R5
Answer. $-1/36$	
Note. Differentiate the beta integral or integrate by parts.	
0761 burst-int-005	積分 • R6
Answer. $2/125$	
Note. Use $2!/(5^3)$.	
0762 burst-int-006	積分 • R6
Answer. $-2/27$	
Note. Use $-3!/(3^4)$.	
0763 burst-int-007	積分 • R5
Answer. $1/280$	
Note. This is $B(4,5)$.	
0764 burst-int-008	積分 • R6
Answer. $\pi/8$	
Note. This is $B(3/2,3/2)$.	
0765 burst-int-009	積分 • R6
Answer. $3\sqrt{\pi}/8$	

Note. Set $u=x^2$ and get one half of $\Gamma(5/2)$.

0766 burst-int-010

積分・R5

Answer. 1

Note. Set $u=x^2$ and get one half of $\Gamma(3)$.

0767 burst-int-011

積分・R6

Answer. $-\pi \cdot \log(2)/2$

Note. Use Wallis' product or pair with $\log(\cos x)$.

0768 burst-int-012

積分・R6

Answer. $\pi \cdot \log(2)^{2/2} \pi^{3/24}$

Note. Differentiate the beta integral twice.

0769 burst-int-013

積分・R6

Answer. $35\pi^{2/256}$

Note. Symmetry gives $\pi/2$ times the integral of $\sin^8$ on $[0, \pi]$.

0770 burst-int-014

積分・R5

Answer. $63\pi/512$

Note. Apply the even Wallis reduction.

0771 burst-int-015

積分・R5

Answer. $256/693$

Note. Apply the odd Wallis reduction.

0772 burst-int-016

積分・R6

Answer. $3\pi/512$

Note. Use one half of $B(7/2, 5/2)$.

0773 burst-int-017

積分・R5

Answer. $3\pi/256$

Note. Use one half of $B(5/2, 5/2)$.

0774 burst-int-018

積分・R5

Answer. $1/24$

Note. Use one half of $B(3, 2)$.

0775 burst-int-019

積分・R6

Answer. $7\pi/512$

Note. Use one half of $B(9/2, 3/2)$.

0776 burst-int-020

積分・R5

Answer. $35\pi/128$

Note. Double the half-interval Wallis value.

0777 burst-int-021

積分・R5

Answer. $16/35$

Note. Apply the odd Wallis reduction.

0778	burst-int-022	積分 • R5
Answer. $35\pi/256$		
Note. Apply the even Wallis reduction.		
0779	burst-int-023	積分 • R6
Answer. $2/63$		
Note. Use one half of $B(2,7/2)$.		
0780	burst-int-024	積分 • R6
Answer. $5\pi/256$		
Note. Use one half of $B(3/2,7/2)$.		
0781	burst-int-025	積分 • R6
Answer. $\pi^4/15$		
Note. Expand $1/(e^x-1)$ and use $\Gamma(4)$ times $\zeta(4)$.		
0782	burst-int-026	積分 • R6
Answer. $\pi^2/6$		
Note. Expand the Planck kernel and sum $1/n^2$.		
0783	burst-int-027	積分 • R6
Answer. $\pi/(2\sqrt{2})$		
Note. Use the beta integral or factor x^4+1 .		
0784	burst-int-028	積分 • R6
Answer. 0		
Note. Substitute $x=1/u$; the integral equals its negative.		
0785	burst-int-029	積分 • R6
Answer. $\pi/\sqrt{2}$		
Note. Use the standard $x^{a-1}/(1+x^b)$ beta form.		
0786	burst-int-030	積分 • R6
Answer. $-\pi^2/12$		
Note. Expand $1/(1+x)$ and integrate termwise.		
0787	burst-int-031	積分 • R5
Answer. $3\pi/16$		
Note. Set $u=2x$ and use the full-period symmetry.		
0788	burst-int-032	積分 • R6
Answer. $5\pi/2048$		
Note. Use one half of $B(7/2,7/2)$.		
0789	burst-int-033	積分 • R6
Answer. $231\pi/2048$		
Note. Apply the even Wallis reduction through twelve powers.		

0790	burst-int-034	積分・R6
Answer. 1024/3003		
Note. Apply the odd Wallis reduction through thirteen powers.		
0791	burst-int-035	積分・R6
Answer. $1/210$		
Note. Use one half of $B(5,3)$.		
0792	burst-int-036	積分・R6
Answer. $\exp(2x) \cdot (x^3/2 - 3x^2/4 + 3x/4 - 3/8)$		
Note. Repeated IBP lowers the power of x each time.		
0793	burst-int-037	積分・R6
Answer. $-(x^2) \cdot \cos(3x)/3 + 2x \cdot \sin(3x)/9 + 2 \cdot \cos(3x)/27$		
Note. Integrate by parts twice.		
0794	burst-int-038	積分・R6
Answer. $x \cdot (\log(x))^3 - 3 \log(x)^2 + 6 \log(x) - 6$		
Note. Repeated IBP produces the descending log polynomial.		
0795	burst-int-039	積分・R5
Answer. $x^5 \log(x)/5 - x^5/25$		
Note. One IBP after integrating x^4 .		
0796	burst-int-040	積分・R6
Answer. $(x^4 - 1) \cdot \operatorname{atan}(x)/4 - x^3/12 + x/4$		
Note. After parts, divide x^4 by $1+x^2$.		
0797	burst-int-041	積分・R6
Answer. $-\exp(-x) \cdot (x^3 + 3x^2 + 6x + 6)$		
Note. Repeated IBP gives the factorial polynomial.		
0798	burst-int-042	積分・R6
Answer. $x^3 \log(1+x)/3 - x^3/9 + x^2/6 - x/3 + \log(1+x)/3$		
Note. After parts, divide x^3 by $1+x$.		
0799	burst-rec-001	積分・R6
Answer. $-1/(n+1)^2$		
Note. Differentiate $1/(n+1)$ with respect to n .		
0800	burst-rec-002	積分・R6
Answer. $2/(n+1)^3$		
Note. Differentiate $1/(n+1)$ twice.		
0801	burst-rec-003	積分・R6
Answer. $2/((n+1) \cdot (n+2) \cdot (n+3))$		
Note. This is $B(n+1,3)$.		
0802	burst-rec-004	積分・R6
Answer. $(n-1)/n$		

Note. Integrate by parts to get the Wallis recurrence.

0803 burst-rec-005

積分 • R6

Answer. $(2*n-1)/(2*n)$

Note. Apply the Wallis recurrence with exponent $2n$.

0804 burst-rec-006

積分 • R6

Answer. $(2*n)/(2*n+1)$

Note. Apply the Wallis recurrence with exponent $2n+1$.

0805 burst-rec-007

積分 • R6

Answer. n

Note. $\Gamma(n+1)=n \Gamma(n)$.

0806 burst-rec-008

積分 • R6

Answer. $-n$

Note. IBP gives $J_n=e^{-n}J_{n-1}$.

0807 burst-rec-009

積分 • R5

Answer. $1/(n+1)+1/(n+2)$

Note. Split into two power integrals.

0808 burst-rec-010

積分 • R5

Answer. $1/((n+1)*(n+2))$

Note. Subtract $1/(n+2)$ from $1/(n+1)$.

0809 burst-boss-int-001

積分 • R6

Answer. $\pi^3/8$

Note. Differentiate $\pi/2*\csc(\pi*a/2)$ twice at $a=1$.

0810 burst-boss-int-002

積分 • R6

Answer. $-(\pi^2)/(8*\sqrt{2})$

Note. Differentiate $\pi/4*\csc(\pi*a/4)$ at $a=1$.

0811 burst-boss-int-003

積分 • R6

Answer. $\pi^2/(2*\sqrt{2})$

Note. Use the derivative of $\pi/2*\csc(\pi*a/2)$ at $a=3/2$.

0812 burst-boss-int-004

積分 • R6

Answer. $\pi/3$

Note. Use the standard beta integral for $x^{a-1}/(1+x^b)$.

0813 burst-boss-int-005

積分 • R5

Answer. $\pi/6$

Note. Use $\pi/6*\csc(\pi/2)$.

0814 burst-boss-int-006

積分 • R5

Answer. $\pi/3$

Note. Use $\pi/6*\csc(\pi/6)$.

0815	burst-boss-int-007	積分 • R5
Answer. $\pi/4$ Note. Use $u=x^2$, then $\arctan$.		
0816	burst-boss-int-008	積分 • R6
Answer. $\pi/(2\sqrt{2})$ Note. Use $\pi/4\csc(3\pi/4)$.		
0817	burst-boss-int-009	積分 • R6
Answer. $2-\pi^2/6$ Note. Expand $\log(1-x)$ and integrate $x^n \log x$ termwise.		
0818	burst-boss-int-010	積分 • R6
Answer. $-(\pi^2)/6$ Note. Expand $1/(1-x)$ and sum $-1/n^2$.		
0819	burst-boss-int-011	積分 • R5
Answer. 24 Note. Use $(-1)^4 4!$ from the gamma integral.		
0820	burst-boss-int-012	積分 • R6
Answer. $3/128$ Note. Use $4!/4^5$.		
0821	burst-boss-int-013	積分 • R6
Answer. $-40/243$ Note. Use $-5!/3^6$.		
0822	burst-boss-int-014	積分 • R6
Answer. $\pi \cdot \log(2)^{2/2} - \pi^3/48$ Note. Differentiate the beta integral with respect to both exponents.		
0823	burst-boss-int-015	積分 • R6
Answer. $\pi^3/8$ Note. Subtract the log-sine and log-cosine square identities.		
0824	burst-boss-int-016	積分 • R6
Answer. $63\pi^2/512$ Note. Pair x with $\pi-x$, then use the Wallis value for $\sin^{10}$.		
0825	burst-boss-int-017	積分 • R6
Answer. $35\pi/65536$ Note. Write it as $2^{-8} \sin^8(2x)$ and use Wallis.		
0826	burst-boss-int-018	積分 • R6
Answer. $429\pi/4096$ Note. Use the even Wallis reduction through fourteen powers.		

0827	burst-boss-int-019	積分 • R6
Answer. 2048/6435		
Note. Use the odd Wallis product through fifteen powers.		
0828	burst-boss-int-020	積分 • R6
Answer. 5280/15625		
Note. Use the imaginary part of $5!/(2-i)^6$.		
0829	burst-boss-anti-001	積分 • R6
Answer. $\exp(x) \cdot (x^4 - 4x^3 + 12x^2 - 24x + 24)$		
Note. Repeated IBP produces the alternating factorial polynomial.		
0830	burst-boss-anti-002	積分 • R6
Answer. $-\exp(-2x) \cdot (x^{4/2} + x^3 + 3x^{2/2} + 3x/2 + 3/4)$		
Note. Choose P so that $d(-e^{-2x}P)/dx = x^4 e^{-2x}$.		
0831	burst-boss-anti-003	積分 • R6
Answer. $-(x^3) \cdot \cos(2x)/2 + 3x^2 \sin(2x)/4 + 3x \cos(2x)/4 - 3 \sin(2x)/8$		
Note. Integrate by parts until the power disappears.		
0832	burst-boss-anti-004	積分 • R6
Answer. $x^3 \sin(2x)/2 + 3x^2 \cos(2x)/4 - 3x \sin(2x)/4 - 3 \cos(2x)/8$		
Note. Integrate by parts repeatedly and collect the trig terms.		
0833	burst-boss-anti-005	積分 • R6
Answer. $x^5 \cdot (\log(x)^2/5 - 2 \log(x)/25 + 2/125)$		
Note. IBP lowers the power of log x twice.		
0834	burst-boss-anti-006	積分 • R6
Answer. $x \cdot (\log(x)^4 - 4 \log(x)^3 + 12 \log(x)^2 - 24 \log(x) + 24)$		
Note. The descending log polynomial follows from repeated IBP.		
0835	burst-boss-anti-007	積分 • R5
Answer. $x^3 \cdot \operatorname{atan}(x)/3 - x^2/6 + \log(1+x^2)/6$		
Note. After IBP, divide x^3 by $1+x^2$.		
0836	burst-boss-anti-008	積分 • R6
Answer. $x^5 \cdot \operatorname{atan}(x)/5 - x^4/20 + x^2/10 - \log(1+x^2)/10$		
Note. After IBP, divide x^5 by $1+x^2$.		
0837	burst-boss-anti-009	積分 • R6
Answer. $x^3 \log(1+x^2)/3 - 2x^3/9 + 2x/3 - 2 \operatorname{atan}(x)/3$		
Note. After IBP, reduce $x^4/(1+x^2)$.		
0838	burst-boss-anti-010	積分 • R6
Answer. $x^4 \log(1+x^2)/4 - x^4/8 + x^2/4 - \log(1+x^2)/4$		
Note. After IBP, reduce $x^5/(1+x^2)$.		
0839	burst-boss-param-001	積分 • R6
Answer. $1/(a^2+1)$		

Note. Use the imaginary part of $1/(a-i)$.

0840 burst-boss-param-002

積分 • R5

Answer. $1/a^2$

Note. Differentiate the Laplace integral of e^{-ax} .

0841 burst-boss-param-003

積分 • R5

Answer. $\pi/(2\sqrt{a})$

Note. Set $u=\sqrt{a}x$.

0842 burst-boss-param-004

積分 • R6

Answer. $\pi \cdot \log((1+\sqrt{1+a})/2)$

Note. Differentiate in a , integrate the rational trig form, then use $I(0)=0$.

0843 burst-boss-param-005

積分 • R6

Answer. $\pi/(2\sin(\pi(n+1)/2))$

Note. Use the beta form for $x^{a-1}/(1+x^2)$ with $a=n+1$.

0844 burst-boss-param-006

積分 • R6

Answer. $-6/(n+1)^4$

Note. Differentiate $1/(n+1)$ three times.

0845 burst-boss2-int-001

積分 • R6

Answer. $-(2\pi^2)/27$

Note. Differentiate the beta integral $\pi/3 \csc(\pi a/3)$ at $a=1$.

0846 burst-boss2-int-002

積分 • R6

Answer. $10\pi^3/(81\sqrt{3})$

Note. Differentiate $\pi/3 \csc(\pi a/3)$ twice at $a=1$.

0847 burst-boss2-int-003

積分 • R6

Answer. $\pi/3$

Note. Use the beta form with $a=3/2$ and $b=3$.

0848 burst-boss2-int-004

積分 • R6

Answer. $\pi/\sqrt{3}$

Note. Use the beta form with $a=4/3$ and $b=2$.

0849 burst-boss2-int-005

積分 • R6

Answer. $\pi \cdot \log(2)/2$

Note. Set $x=\tan t$ and integrate $t \cot t$ by parts.

0850 burst-boss2-int-006

積分 • R6

Answer. $\pi/2$

Note. Use the Fourier/Dirichlet kernel identity.

0851 burst-boss2-int-007

積分 • R6

Answer. $\pi/2$

Note. Integrate by parts and use the Dirichlet integral.

0852 burst-boss2-int-008	積分・R6
Answer. $-\pi^4/15$ Note. Expand $1/(1-x)$ and sum $\zeta(4)$.	
0853 burst-boss2-int-009	積分・R6
Answer. $-8\pi^6/63$ Note. Expand $1/(1-x)$ and sum $\zeta(6)$.	
0854 burst-boss2-int-010	積分・R6
Answer. $-7\pi^4/120$ Note. Expand $1/(1+x)$ and use $\eta(4)$.	
0855 burst-boss2-int-011	積分・R6
Answer. $-31\pi^6/252$ Note. Expand $1/(1+x)$ and use $\eta(6)$.	
0856 burst-boss2-int-012	積分・R6
Answer. $\pi^2/16+1/4$ Note. Write $\sin^2 x=(1-\cos 2x)/2$.	
0857 burst-boss2-int-013	積分・R6
Answer. $\pi/8$ Note. Use $\sin x \cos x=\sin 2x/2$.	
0858 burst-boss2-int-014	積分・R6
Answer. π^2-4 Note. Integrate by parts twice.	
0859 burst-boss2-param-001	積分・R6
Answer. $b/(a^2+b^2)$ Note. Use the imaginary part of $1/(a-bi)$.	
0860 burst-boss2-param-002	積分・R6
Answer. $a/(a^2+b^2)$ Note. Use the real part of $1/(a-bi)$.	
0861 burst-boss2-param-003	積分・R6
Answer. $2a/(a^2+1)^2$ Note. Differentiate the Laplace sine transform with respect to a .	
0862 burst-boss2-param-004	積分・R6
Answer. $(a^2-1)/(a^2+1)^2$ Note. Differentiate the Laplace cosine transform with respect to a .	
0863 burst-boss2-param-005	積分・R6
Answer. $\log(a+1)$ Note. Differentiate with respect to a , then use $F(0)=0$.	

0864 burst-boss2-param-006

積分 • R6

Answer. $-120/(a+1)^6$ **Note.** Differentiate $1/(a+1)$ five times.**0865 burst-boss2-anti-001**

積分 • R6

Answer. $\exp(2x) * (x^{5/2} - 5x^{4/4} + 5x^{3/2} - 15x^{2/4} + 15x/4 - 15/8)$ **Note.** Repeated IBP produces the factorial polynomial.**0866 burst-boss2-anti-002**

積分 • R6

Answer. $-(x^4) * \cos(x) + 4x^3 * \sin(x) + 12x^2 * \cos(x) - 24x * \sin(x) - 24 * \cos(x)$ **Note.** Integrate by parts until the polynomial disappears.**0867 burst-boss2-anti-003**

積分 • R6

Answer. $x^4 * \sin(x) + 4x^3 * \cos(x) - 12x^2 * \sin(x) - 24x * \cos(x) + 24 * \sin(x)$ **Note.** Integrate by parts until the polynomial disappears.**0868 burst-boss2-anti-004**

積分 • R6

Answer. $x * (\log(x)^5 - 5 * \log(x)^4 + 20 * \log(x)^3 - 60 * \log(x)^2 + 120 * \log(x) - 120)$ **Note.** Repeated IBP lowers the log power.**0869 burst-boss2-anti-005**

積分 • R6

Answer. $x^6 * \operatorname{atan}(x) / 6 - x^5 / 30 + x^3 / 18 - x / 6 + \operatorname{atan}(x) / 6$ **Note.** After parts, divide x^6 by $1+x^2$.**0870 burst-boss2-anti-006**

積分 • R6

Answer. $x^4 / 4 - x^2 / 2 + \log(1+x^2) / 2$ **Note.** Divide x^5 by $1+x^2$ before integrating.**0871 burst-boss3-int-001**

積分 • R6

Answer. $1/27$ **Note.** Separate the two Gamma-type integrals.**0872 burst-boss3-int-002**

積分 • R6

Answer. $-1/54$ **Note.** Separate variables and use derivatives of $1/(a+1)$.**0873 burst-boss3-int-003**

積分 • R6

Answer. $8 - \pi^2/3$ **Note.** Expand the square and use the log-log beta derivative.**0874 burst-boss3-int-004**

積分 • R6

Answer. $\pi^2/3$ **Note.** The cross term cancels the elementary log-square pieces.**0875 burst-boss3-int-005**

積分 • R6

Answer. $-\pi^2 * \log(2) / 2$ **Note.** Pair x with $\pi-x$ and use the Wallis log-sine integral.**0876 burst-boss3-int-006**

積分 • R6

Answer. $\pi^3 - 6\pi$

Note. Integrate by parts until only the standard sine moment remains.

0877 burst-boss3-int-007

積分 • R6

Answer. $\pi^4 - 12\pi^2 + 48$

Note. Repeated IBP reduces the polynomial power.

0878 burst-boss3-int-008

積分 • R6

Answer. $-714/28561$

Note. Take the real part of $3!/(2-3i)^4$.

0879 burst-boss3-int-009

積分 • R6

Answer. $-720/28561$

Note. Take the imaginary part of $3!/(2-3i)^4$.

0880 burst-boss3-int-010

積分 • R6

Answer. $984/3125$

Note. Take the real part of $4!/(1-2i)^5$.

0881 burst-boss3-int-011

積分 • R6

Answer. $-912/3125$

Note. Take the imaginary part of $4!/(1-2i)^5$.

0882 burst-boss3-int-012

積分 • R6

Answer. $31\pi^6/252$

Note. Use the eta factor $(1-2^{-5}) \Gamma(6) \zeta(6)$.

0883 burst-boss3-int-013

積分 • R6

Answer. $7\pi^4/120$

Note. Use the eta factor $(1-2^{-3}) \Gamma(4) \zeta(4)$.

0884 burst-boss3-param-001

積分 • R6

Answer. $18(a^2-3)/(a^2+9)^3$

Note. Differentiate $3/(a^2+9)$ twice.

0885 burst-boss3-param-002

積分 • R6

Answer. $2(a^3-3a)/(a^2+1)^3$

Note. Take the real part of $2!/(a-i)^3$.

0886 burst-boss3-param-003

積分 • R6

Answer. $24a(a^2-1)/(a^2+1)^4$

Note. Take the imaginary part of $3!/(a-i)^4$.

0887 burst-boss3-param-004

積分 • R6

Answer. $24/(a+1)^5$

Note. Differentiate $1/(a+1)$ four times.

0888 burst-boss3-param-005

積分 • R6

Answer. $720/(a+1)^7$

Note. Differentiate $1/(a+1)$ six times.

0889 burst-boss3-param-006

積分・R6

Answer. $2/(a+1)^3$ **Note.** The integral is $\log(a+1)$.**0890 burst-boss3-anti-001**

積分・R6

Answer. $-\exp(-3x) \cdot (x^6/3 + 2x^5/3 + 10x^4/9 + 40x^3/27 + 40x^2/27 + 80x/81 + 80/243)$ **Note.** Repeated IBP produces the descending factorial polynomial.**0891 burst-boss3-anti-002**

積分・R6

Answer. $-x^5 \cos(2x)/2 + 5x^4 \sin(2x)/4 + 5x^3 \cos(2x)/2 - 15x^2 \sin(2x)/4 - 15x \cos(2x)/4 + 15 \sin(2x)/8$ **Note.** Repeated IBP alternates sine and cosine.**0892 burst-boss3-anti-003**

積分・R6

Answer. $x^5 \sin(2x)/2 + 5x^4 \cos(2x)/4 - 5x^3 \sin(2x)/2 - 15x^2 \cos(2x)/4 + 15x \sin(2x)/4 + 15 \cos(2x)/8$ **Note.** Repeated IBP alternates sine and cosine.**0893 burst-boss3-anti-004**

積分・R6

Answer. $x^7 \operatorname{atan}(x)/7 - x^6/42 + x^4/28 - x^2/14 + \log(1+x^2)/14$ **Note.** After parts, divide x^7 by $1+x^2$.**0894 burst-boss3-anti-005**

積分・R6

Answer. $x^7 \log(1+x)/7 - x^7/49 + x^6/42 - x^5/35 + x^4/28 - x^3/21 + x^2/14 - x/7 + \log(1+x)/7$ **Note.** After parts, divide x^7 by $1+x$.**0895 burst-boss3-anti-006**

積分・R6

Answer. $x^6/6 - x^4/4 + x^2/2 - \log(1+x^2)/2$ **Note.** Divide x^7 by $1+x^2$ before integrating.**0896 world-005**

積分・R2

Answer. $8/15$ **Note.** Wallis odd reduction.**0897 world-010**

積分・R2

Answer. $3\pi/8$ **Note.** Wallis even reduction.**0898 world-014**

積分・R4

Answer. $\pi \log(2)/8$ **Note.** $x \rightarrow \pi/4 - x$ symmetry.**0899 world-015**

積分・R4

Answer. $\pi \log(2)/8$ **Note.** Substitute $x = \tan \theta$; the Putnam log integral.**0900 world-016**

積分・R4

Answer. $\log(3/2)$ **Note.** Frullani for cosines.

0901	world-017	積分・R4
Answer. $2\pi/(3\sqrt{3})$		
Note. $(\pi/3)/\sin(\pi/3)$.		
0902	world-018	積分・R4
Answer. $-\log(2)$		
Note. By parts gives $-\log 2$.		
0903	world-019	積分・R4
Answer. $\log(3)$		
Note. $\int_0^1 (x^a-1)/\log x \, dx = \log(a+1)$.		
0904	world-027	積分・R4
Answer. $1/2$		
Note. Imaginary part of $1/(1-i)^2$.		
0905	world-029	積分・R4
Answer. $-\pi/4$		
Note. Differentiate a Beta integral in a parameter.		
0906	world-031	積分・R5
Answer. $\pi^2/4$		
Note. Expand $1/\sinh$ and sum $1/(2k+1)^2$.		
0907	world-032	積分・R5
Answer. $\pi^3/16$		
Note. Term-by-term gives $\pi^3/16$.		
0908	world-033	積分・R5
Answer. π		
Note. $(\pi/2)(3-1)=\pi$.		
0909	world-037	積分・R5
Answer. $\sqrt{\pi} \exp(-1)/2$		
Note. Gaussian shift gives $(\sqrt{\pi}/2)e^{-1}$.		
0910	world-038	積分・R5
Answer. $3\pi/8$		
Note. Use $\sin^3 = (3 \sin x - \sin 3x)/4$.		
0911	world-041	積分・R5
Answer. $\log(2)$		
Note. $\log(a+1)$ with $a=1$.		
0912	world-043	積分・R5
Answer. $-\pi^2/(6\sqrt{3})$		
Note. Keyhole contour gives $-\pi^2/(6\sqrt{3})$.		
0913	world-044	積分・R5
Answer. $\pi \log(2)/2$		

Note. Equals $(\pi/2) \log 2$.

0914 world-045

積分 • R5

Answer. $\pi/\exp(1)$

Note. Residue gives π/e .

0915 world-047

積分 • R5

Answer. $\pi/3$

Note. Standard value $\pi/3$.

0916 world-050

積分 • R5

Answer. $-1/16$

Note. $x^4/4 \log x - x^4/16$.

0917 world-051

積分 • R6

Answer. $5\pi^2/96$

Note. Ahmed's integral.

0918 world-052

積分 • R6

Answer. $\pi \cdot \log(2)^2/2 + \pi^3/24$

Note. $(\pi/2) \log^2 2 + \pi^3/24$.

0919 world-054

積分 • R6

Answer. $\pi \cdot \log(2)^2/2 + \pi^3/24$

Note. $(\pi/2) \log^2 2 + \pi^3/24$.

0920 world-055

積分 • R5

Answer. -1

Note. $u = \sin x$ gives $\int_0^1 \log u \, du = -1$.

0921 world-056

積分 • R6

Answer. $2 - 2 \log(2) - \pi^2/12$

Note. Equals $2 - 2 \log 2 - \pi^2/12$.

0922 world-057

積分 • R6

Answer. $\pi \cdot \log(2)$

Note. Feynman differentiation.

0923 world-058

積分 • R6

Answer. $\sqrt{\pi} \cdot \exp(-1/4)/2$

Note. $(\sqrt{\pi}/2) e^{-1/4}$.

0924 world-059

積分 • R6

Answer. $\log(2)/3 + \pi/(3\sqrt{3})$

Note. Factor $1+x^3$.

0925 world-061

積分 • R6

Answer. $-\pi^2/6$

Note. $-\text{Li}_2(1)$.

0926	world-062	積分・R6
Answer. $-\pi^2/8$ Note. $-\sum 1/(2n+1)^2$.		
0927	world-063	積分・R6
Answer. $\pi/16$ Note. Beta value $\pi/16$.		
0928	world-064	積分・R6
Answer. $\log(4)$ Note. $\log(a+1)$, $a=3$.		
0929	world-065	積分・R6
Answer. $\pi/\sqrt{2}$ Note. Beta gives $\pi/\sqrt{2}$.		
0930	world-069	積分・R6
Answer. $\pi/12$ Note. $\pi/(2ab(a+b))$.		
0931	world-070	積分・R6
Answer. $\pi/4$ Note. $\arctan(1/s)$ at $s=1$.		
0932	world-072	積分・R6
Answer. $4\pi/(3\sqrt{3})$ Note. Differentiate $\int 1/(a+\cos x)$ in a .		
0933	world-073	積分・R6
Answer. $\pi \cdot \log((2+\sqrt{3})/2)$ Note. $\int_0^\pi \log(a-\cos x) dx = \pi \log((a+\sqrt{a^2-1})/2)$.		
0934	world-076	積分・R6
Answer. $\pi^2/8$ Note. $u=\arctan x$ gives $(\pi/2)^2/2$.		
0935	world-077	積分・R6
Answer. $\pi \cdot \log(2)/8$ Note. Feynman parameter trick.		
0936	world-079	積分・R5
Answer. -6 Note. $\int_0^1 (\log x)^n dx = (-1)^n n!$.		
0937	world-080	積分・R6
Answer. $\pi/(2\sqrt{3})$ Note. $1+x^2+x^4=(x^2+x+1)(x^2-x+1)$.		

0938	world-081	積分・R5
Answer. $\pi/2$ Note. $\int \operatorname{sech} x \, dx = 2 \arctan(\tanh(x/2))$.		
0939	world-082	積分・R5
Answer. $\pi^{2/4}-2$ Note. Equals $\pi^{2/4}-2$.		
0940	world-083	積分・R6
Answer. $\pi^{2/12}$ Note. $\Gamma(2)\eta(2)=\pi^{2/12}$.		
0941	world-084	積分・R6
Answer. $\log(2)^{2/2}-\pi^{2/12}$ Note. Equals $(\log^2 2)/2 - \pi^{2/12}$.		
0942	world-085	積分・R6
Answer. $\pi/(2*\sqrt{2})$ Note. Standard value $\pi/(2\sqrt{2})$.		
0943	world-088	積分・R6
Answer. $\pi/2$ Note. Dirichlet integral.		
0944	world-089	積分・R6
Answer. $\pi/4$ Note. Standard value $\pi/4$.		
0945	world-090	積分・R6
Answer. $\sqrt{\pi}/2$ Note. Gaussian integral.		
0946	world-092	積分・R5
Answer. $-3/4$ Note. By parts gives $-3/4$.		
0947	world-094	積分・R6
Answer. $\pi/(2*\sqrt{2})$ Note. $\pi/(2\sqrt{2})$.		
0948	world-095	積分・R5
Answer. $\log(2)-1$ Note. Equals $\log 2 - 1$.		
0949	world-098	積分・R6
Answer. $\pi/(2*\sqrt{2})$ Note. Same Beta family.		
0950	world-100	積分・R6
Answer. $\pi*\log(2)$		

Note. $\int_0^{\pi/2} x^2 \csc^2 x \, dx = \pi \log 2$.

0951 putnam-004

積分 • R5

Answer. $\pi/4$

Note. Add the $x \rightarrow -x$ reflection.

0952 putnam-005

積分 • R6

Answer. $\pi/4$

Note. King's property, independent of the exponent.

0953 putnam-009

積分 • R6

Answer. $\pi/4$

Note. $x \rightarrow 1/x$ symmetry gives $\pi/4$.

0954 putnam-012

積分 • R6

Answer. $\pi \cdot \log(2)/2$

Note. Frullani for $\arctan$: $(\pi/2)\ln(2/1)$.

0955 ser-001

級數 • R1

Answer. 收斂

Note. p-series, $p=2>1$, 所以收斂。

0956 ser-002

級數 • R1

Answer. 發散

Note. 調和級數發散。

0957 ser-003

級數 • R1

Answer. $3/2$

Note. 等比級數，和為 $1/(1-1/3)=3/2$ 。

0958 ser-004

級數 • R1

Answer. 條件收斂

Note. 交錯調和級數條件收斂。

0959 ser-005

級數 • R2

Answer. 發散

Note. 項行為像 $1/n$ ，和調和級數比較會發散。

0960 ser-006

級數 • R2

Answer. 收斂

Note. 比值測試極限趨近 $e^{-1}<1$ 。

0961 ser-007

級數 • R2

Answer. $\exp(3)-1$

Note. e^3 的展開式少了 $n=0$ 項。

0962 ser-008

級數 • R2

Answer. 0

Note. $n!$ 成長太快，半徑為 0。

0963	ser-009	級數・R3
Answer. 1 Note. 與 $-\ln(1-x)$ 的冪級數相同，半徑 1。		
0964	ser-010	級數・R3
Answer. 條件收斂 Note. 交錯級數收斂，但絕對值級數 $p=1/2$ 發散。		
0965	ser-011	級數・R3
Answer. $9/4$ Note. 拆成 $\sum (2/3)^n$ 加 $\sum n(1/3)^n$ ，分別是 2 和 $3/4$ 。		
0966	ser-012	級數・R2
Answer. 1 Note. 裂項： $1/(n(n+1))=1/n-1/(n+1)$ 。		
0967	ser-013	級數・R2
Answer. 發散 Note. 通項不趨近 0，所以級數發散。		
0968	ser-014	級數・R3
Answer. 2 Note. 公式 $\sum n r^n = r/(1-r)^2$ ，取 $r=1/2$ 。		
0969	ser-015	級數・R3
Answer. 6 Note. 公式 $\sum n^2 r^n = r(1+r)/(1-r)^3$ ，取 $r=1/2$ 。		
0970	ser-016	級數・R4
Answer. 收斂 Note. 這是中央二項係數倒數量級，指數衰減所以收斂。		
0971	ser-017	級數・R3
Answer. 3 Note. n^2 不影響收斂半徑，核心比例是 $ x /3$ 。		
0972	ser-018	級數・R2
Answer. 絕對收斂 Note. 取絕對值後是 p -series， $p=2>1$ 。		
0973	ser-019	級數・R2
Answer. $3/4$ Note. 裂項成 $1/2(1/n-1/(n+2))$ 。		
0974	ser-020	級數・R3
Answer. 發散 Note. 用積分測試， $\int (\ln x)/x \, dx$ 發散。		

0975 td-ser-001

級數・R6

Answer. $33/40$

Note. 令 $(1+x)^x = \exp(x \log(1+x))$ ，展開到 x^6 。

0976 td-ser-002

級數・R6

Answer. $13/12$

Note. $\arcsin x = x + x^3/6 + 3x^5/40 + \dots$ ，再取 $\exp$ 。

0977 td-ser-003

級數・R6

Answer. $2 \cdot \log(2)$

Note. 用 $\int_0^{\pi/2} \sin^{2n} x \, dx$ 表示雙階乘比，再交換求和。

0978 ser-024

級數・R2

Answer. 發散

Note. 由積分判別， $\int_2^{\infty} dx/(x \log x) = \infty$ ，所以級數發散。

0979 ser-027

級數・R2

Answer. $2/3$

Note. 係數為 $2^4/4! = 16/24 = 2/3$ 。

0980 ser-030

級數・R2

Answer. 5

Note. 根值判別得到 $|x-2|/5 < 1$ ，因此收斂半徑為 5。

0981 ser-031

級數・R3

Answer. 發散

Note. Since $p=1/2 < 1$, the p-series diverges.

0982 ser-032

級數・R3

Answer. 發散

Note. The terms tend to e^{-1} , not 0, so the series diverges.

0983 ser-034

級數・R3

Answer. 0

Note. Only $x=0$ converges, so the radius is 0.

0984 ser-035

級數・R3

Answer. Infinity

Note. The series converges for all x , so the radius is Infinity.

0985 ser-036

級數・R3

Answer. 收斂

Note. $0 < 2^n/(3^n + n) < (2/3)^n$, so the series converges by comparison.

0986 ser-037

級數・R3

Answer. 收斂

Note. The integral test shows convergence because $p=2 > 1$.

0987 ser-038

級數・R2

Answer. $1/3$

Note. The coefficient of x^3 in $\log(1+x)$ is $1/3$.

0988 ser-039

級數 • R3

Answer. $1/2$

Note. The radius condition is $|2x| < 1$, so $R=1/2$.

0989 gap-ser-root-001

級數 • R3

Answer. $1/3$

Note. The radius is $1/3$.

0990 gap-ser-root-002

級數 • R3

Answer. 1

Note. The root test gives $|x| < 1$, so $R=1$.

0991 gap-ser-lc-001

級數 • R3

Answer. divergent

Note. Limit comparison with $1/n$ gives a nonzero constant, so the series diverges.

0992 gap-ser-lc-002

級數 • R3

Answer. convergent

Note. Limit comparison with $1/n^2$ shows convergence.

0993 gap-ser-cond-001

級數 • R3

Answer. conditional

Note. It converges by the alternating test but not absolutely.

0994 gap-ser-end-001

級數 • R4

Answer. conditional

Note. The endpoint $x=-1$ is conditionally convergent.

0995 gap-ser-end-002

級數 • R4

Answer. divergent

Note. At $x=1$, the series is the divergent harmonic series.

0996 gap-ser-taylor-001

級數 • R3

Answer. $1/120$

Note. The coefficient is $1/5!=1/120$.

0997 gap-ser-taylor-002

級數 • R3

Answer. $1-x^2/2+x^4/24$

Note. The polynomial is $1-x^2/2+x^4/24$.

0998 gap-tech-005

級數 • R3

Answer. roottest

Note. Use the root test.

0999 mob-tech-007

級數 • R2

Answer. ratiotest

Note. Use ratio test.

1000	mob-tech-008	級數 • R2
Answer. roottest		
Note. Use root test.		
1001	mob-tech-019	級數 • R2
Answer. endpointanalysis		
Note. Use endpoint analysis.		
1002	mob-trap-004	級數 • R2
Answer. conditional		
Note. conditional		
1003	mob-trap-005	級數 • R2
Answer. checkendpoints		
Note. check endpoints		
1004	mob-conv-003	級數 • R3
Answer. conditional		
Note. conditional.		
1005	mob-conv-006	級數 • R3
Answer. convergent		
Note. convergent.		
1006	mob-conv-010	級數 • R3
Answer. divergent		
Note. divergent.		
1007	rel-basic-019	級數 • R1
Answer. 2		
Note. Use $1/(1-r)$.		
1008	rel-basic-022	級數 • R1
Answer. $1/3$		
Note. This is geometric with ratio $3x$.		
1009	rel-basic-024	級數 • R1
Answer. $1/2$		
Note. Use $r/(1-r)$.		
1010	rel-adv-017	級數 • R3
Answer. $1/6$		
Note. Use $e^{x^2} = \sum x^{2k}/k!$.		
1011	rel-adv-019	級數 • R3
Answer. 1		
Note. Polynomial factors do not change the radius from 1.		

1012 rel-adv-015	級數 • R3
Answer. convergent	
Note. The root limit is $1/3 < 1$.	
1013 rel-adv-018	級數 • R3
Answer. divergent	
Note. The terms behave like $3/n$.	
1014 rel-adv-020	級數 • R3
Answer. conditional	
Note. At $x=-1$ this is the alternating harmonic series.	
1015 rel-hard-ser-001	級數 • R5
Answer. $1/4$	
Note. The central binomial coefficient grows like 4^n .	
1016 rel-hard-ser-002	級數 • R5
Answer. 4	
Note. This is the reciprocal central binomial scale.	
1017 rel-hard-ser-004	級數 • R5
Answer. $1/120$	
Note. Use $\sin u = u - u^3/6 + u^5/120 - \dots$ with $u = x^2$.	
1018 rel-hard-ser-005	級數 • R5
Answer. $1/18$	
Note. Write it as a telescoping difference of triple products.	
1019 rel-hard-ser-008	級數 • R5
Answer. $1/2$	
Note. Use $1/(4n^2-1) = 1/2(1/(2n-1) - 1/(2n+1))$.	
1020 rel-hard-ser-012	級數 • R5
Answer. $1/e$	
Note. The n th root of $n^n/n!$ tends to e .	
1021 rel-hard-ser-010	級數 • R5
Answer. divergent	
Note. The terms do not tend to 0.	
1022 rel-hard-ser-011	級數 • R5
Answer. conditional	
Note. Alternating test gives convergence; $p=1/2$ gives no absolute convergence.	
1023 hc-spec-007	級數 • R6
Answer. $1/96$	
Note. Write it as a telescoping difference of four-factor denominators.	
1024 hc-spec-009	級數 • R6
Answer. $1/27$	

Note. The coefficient ratio tends to 27.

1025 hc-spec-010

級數 • R6

Answer. $1/147456$

Note. $J_0(x) = \sum (-1)^k (x^2/4)^k / (k!)^2$; use $k=4$.

1026 exam-ser-001

級數 • R5

Answer. 3

Note. The dominant ratio is $|x|/3$.

1027 exam-ser-002

級數 • R5

Answer. e

Note. Use $(n!/n^n)^{1/n} \rightarrow e^{-1}$.

1028 exam-ser-003

級數 • R5

Answer. 1

Note. The coefficient $1/n$ has n th root 1.

1029 exam-ser-008

級數 • R5

Answer. $(\exp(1) - \exp(-1))/2$

Note. This is $\sinh 1$.

1030 exam-ser-009

級數 • R5

Answer. $4/15$

Note. The coefficient is $2^5/5!$.

1031 exam-ser-010

級數 • R5

Answer. $-1/6$

Note. Use $\sin z = z - z^3/6 + \dots$ with $z = x^2$.

1032 exam-ser-011

級數 • R5

Answer. $-1/4$

Note. The x^n coefficient is $(-1)^{n+1}/n$.

1033 exam-ser-013

級數 • R5

Answer. 8

Note. This is $\sum n^2/2^n$ plus $\sum n/2^n$.

1034 exam-ser-014

級數 • R5

Answer. $1/4$

Note. The n th root behaves like 4.

1035 exam-ser-017

級數 • R5

Answer. $\exp(2)$

Note. This is e^2 .

1036 exam-ser-018

級數 • R5

Answer. 35

Note. Choose 3 factors of x from 7.

1037	exam-ser-019	級數 • R5
Answer. 16 Note. The coefficient is 2^4 .		
1038	exam-ser-020	級數 • R5
Answer. $3/4$ Note. Use $r/(1-r)^2$ with $r=1/3$.		
1039	uni-ser-001	級數 • R6
Answer. $4/9$ Note. Use $\sum n r^n = r/(1-r)^2$.		
1040	uni-ser-004	級數 • R6
Answer. 0 Note. $\sin(x^2)$ only has powers x^{4k+2} .		
1041	uni-ser-005	級數 • R6
Answer. 0 Note. Use $\operatorname{Re}((1+i)^6/6!)$.		
1042	uni-ser-006	級數 • R6
Answer. $1/3$ Note. This is the x^3 coefficient of $\log(1+x)$.		
1043	uni-ser-007	級數 • R6
Answer. 5 Note. The n th root of $n^2/5^n$ is $1/5$.		
1044	uni-ser-008	級數 • R6
Answer. 4 Note. Use the central binomial coefficient growth.		
1045	uni-ser-009	級數 • R6
Answer. $1/e$ Note. Stirling gives n th root asymptotic to e .		
1046	uni-ser-010	級數 • R6
Answer. $\pi/4$ Note. This is $\arctan(1)$.		
1047	uni-ser-012	級數 • R6
Answer. 4 Note. Use the second derivative of the geometric series.		
1048	uni-ser-013	級數 • R6
Answer. 15 Note. The coefficient is $C(6,2)$.		

1049	uni-ser-014	級數・R6
Answer. $1/3$		
Note. The n th root of $3^n/n^2$ is 3.		
1050	uni-ser-015	級數・R6
Answer. $2/9$		
Note. Use $r/(1-r)^2$ with $r=-1/2$ and adjust the sign.		
1051	uni-ser-016	級數・R6
Answer. $-4/315$		
Note. Use $\sin x \cos x = (1/2)\sin(2x)$.		
1052	uni-ser-018	級數・R6
Answer. 210		
Note. This is $C(10,6)$.		
1053	uni-ser-019	級數・R6
Answer. 1		
Note. The n th root of $1/\sqrt{n}$ tends to 1.		
1054	depth-ser-001	級數・R5
Answer. $9/4$		
Note. Use the sums for $n r^n$ and $n^2 r^n$.		
1055	depth-ser-002	級數・R5
Answer. $20/27$		
Note. Use $r(1+r)/(1-r)^3$.		
1056	depth-ser-003	級數・R5
Answer. $1/2$		
Note. Decompose as one-half of consecutive odd reciprocals.		
1057	depth-ser-004	級數・R5
Answer. $243/560$		
Note. The coefficient is $3^7/7!$.		
1058	depth-ser-005	級數・R5
Answer. $1/24$		
Note. Use $\cos z$ with $z=x^2$.		
1059	depth-ser-006	級數・R5
Answer. $-1/5040$		
Note. Divide the sine series by x .		
1060	depth-ser-007	級數・R5
Answer. $(\exp(1)+\exp(-1))/2$		
Note. This is $\cosh(1)$.		
1061	depth-ser-008	級數・R5
Answer. $1/4$		

Note. It is one third of $\sum n/3^n$.

1062 depth-ser-009

級數 • R6

Answer. $1/4$

Note. Central binomial coefficients grow like 4^n .

1063 depth-ser-010

級數 • R6

Answer. $4/3$

Note. Use the reciprocal central binomial growth.

1064 depth-ser-011

級數 • R5

Answer. 192

Note. The coefficient is $(n+1)2^n$ with $n=5$.

1065 depth-ser-012

級數 • R5

Answer. 3

Note. This is half of $\sum n^2/2^n$.

1066 depth-ser-013

級數 • R5

Answer. $\cos(1)$

Note. This is the cosine series at 1.

1067 depth-ser-014

級數 • R5

Answer. $1/4$

Note. Use a telescoping split.

1068 depth-ser-015

級數 • R6

Answer. $1/3$

Note. Convolve the two Taylor series.

1069 depth-ser-016

級數 • R6

Answer. $33/8$

Note. Use the cubic power-series sum.

1070 depth-ser-017

級數 • R5

Answer. 5

Note. Polynomial factors do not affect the radius.

1071 depth-ser-018

級數 • R5

Answer. $\sin(1)$

Note. This is the sine series at 1.

1072 depth-ser-019

級數 • R6

Answer. $11/12$

Note. Square the log series through x^4 .

1073 depth-ser-020

級數 • R5

Answer. $1/2$

Note. The n th root of $2^n/n^2$ is 2.

1074	burst-boss-ser-001	級數・R5
Answer. 26		
Note. Differentiate the geometric series three times.		
1075	burst-boss-ser-002	級數・R5
Answer. $3/2$		
Note. Use $r(1+r)/(1-r)^3$ with $r=1/3$.		
1076	burst-boss-ser-003	級數・R6
Answer. $2\log(2)$		
Note. Use $\sum H_n x^n = -\log(1-x)/(1-x)$.		
1077	burst-boss-ser-004	級數・R6
Answer. $\pi^2/12$		
Note. Use the alternating zeta value at 2.		
1078	burst-boss-ser-005	級數・R6
Answer. $137/180$		
Note. The coefficient is $2H_5/6$.		
1079	burst-boss-ser-006	級數・R6
Answer. $1553/80$		
Note. Collect the x^6 coefficient of $e^{3x}/(1-x)$.		
1080	burst-boss2-ser-001	級數・R6
Answer. $\pi^4/90$		
Note. Use the Basel-type value $\zeta(4)$.		
1081	burst-boss2-ser-002	級數・R6
Answer. $\pi^4/96$		
Note. Remove the even fourth-power terms from $\zeta(4)$.		
1082	burst-boss2-ser-003	級數・R6
Answer. $7\pi^4/720$		
Note. Use $\eta(4) = (1-2^{-3})\zeta(4)$.		
1083	burst-boss2-ser-004	級數・R6
Answer. $31\pi^6/30240$		
Note. Use $\eta(6) = (1-2^{-5})\zeta(6)$.		
1084	burst-boss2-ser-005	級數・R6
Answer. 15		
Note. Apply the fourth derivative of the geometric series at $r=1/3$.		
1085	burst-boss2-ser-006	級數・R6
Answer. $273/4$		
Note. Apply the fifth derivative of the geometric series at $r=1/3$.		

1086	burst-boss2-ser-007	級數・R6
Answer. $3 \cdot \log(3/2)/2$		
Note. Use $\sum H_n x^n = -\log(1-x)/(1-x)$.		
1087	burst-boss2-ser-008	級數・R6
Answer. $4 \cdot \log(4/3)/3$		
Note. Use $\sum H_n x^n = -\log(1-x)/(1-x)$.		
1088	burst-boss3-ser-001	級數・R6
Answer. 9366		
Note. Differentiate the geometric series six times.		
1089	burst-boss3-ser-002	級數・R6
Answer. 94586		
Note. Differentiate the geometric series seven times.		
1090	burst-boss3-ser-003	級數・R6
Answer. 4108/243		
Note. Use the Eulerian polynomial for $\sum n^5 r^n$ at $r=1/4$.		
1091	burst-boss3-ser-004	級數・R6
Answer. 1491/4		
Note. Use the Eulerian polynomial for $\sum n^6 r^n$ at $r=1/3$.		
1092	burst-boss3-ser-005	級數・R6
Answer. $\pi^2/12$		
Note. Integrate the generating function for H_n .		
1093	burst-boss3-ser-006	級數・R6
Answer. 275295799/775975200		
Note. Use $[x^n](\log(1+x))^2 = (-1)^n 2H_{n-1}/n$.		
1094	burst-boss3-ser-007	級數・R6
Answer. 111165691613818/97692469875		
Note. Convolve e^{2x} with the binomial series.		
1095	burst-boss3-ser-008	級數・R6
Answer. 1094053506107/212837625		
Note. Convolve e^{2x} with the binomial series.		
1096	world-006	級數・R2
Answer. 1/3		
Note. Telescoping by 1/3.		
1097	world-009	級數・R2
Answer. $\log(3/2)$		
Note. $-\log(1-1/3)$.		
1098	world-020	級數・R4
Answer. $2 \cdot \exp(1)$		

Note. $\sum n^2/n! = 2e$.

1099 world-021

級數 • R4

Answer. $5 \cdot \exp(1)$

Note. $\sum n^3/n! = 5e$.

1100 world-022

級數 • R4

Answer. $2 - 2 \cdot \log(2)$

Note. Equals $2 - 2 \log 2$.

1101 world-023

級數 • R4

Answer. $\pi^{2/3-3}$

Note. Partial fractions to $\zeta(2)$ terms.

1102 world-028

級數 • R3

Answer. $1/4$

Note. $\text{binom}(2n, n) \sim 4^n$, so $R = 1/4$.

1103 world-034

級數 • R5

Answer. $\pi^{2/12}$

Note. Euler sum equal to $\pi^{2/12}$.

1104 world-039

級數 • R5

Answer. $\pi^{2/18}$

Note. Classic central-binomial sum.

1105 world-042

級數 • R5

Answer. $1/27$

Note. Ratio $\rightarrow 27$, so $R = 1/27$.

1106 world-048

級數 • R5

Answer. $\pi/2$

Note. This is $\arcsin(1) = \pi/2$.

1107 world-053

級數 • R5

Answer. $1/4$

Note. Telescoping with alternating signs.

1108 world-060

級數 • R6

Answer. $\pi^3/32$

Note. $\beta(3) = \pi^3/32$.

1109 world-075

級數 • R5

Answer. $\pi^{2/6-1}$

Note. $1/(n^2(n+1)) = 1/n^2 - 1/n + 1/(n+1)$.

1110 world-078

級數 • R6

Answer. $\pi^{2/12} - \log(2)^{2/2}$

Note. $\text{Li}_2(1/2)$.

1111 world-086 級數 • R6
Answer. $\pi^6/960$
Note. $(63/64)\zeta(6)=\pi^6/960$.

1112 world-087 級數 • R6
Answer. $\pi^6/945$
Note. $\zeta(6)=\pi^6/945$.

1113 world-097 級數 • R6
Answer. $2\log(2)$
Note. $\sum H_n x^n = -\log(1-x)/(1-x), x=1/2$.

1114 putnam-001 級數 • R5
Answer. $\pi/4$
Note. Telescopes to $\pi/2 - \arctan(1) = \pi/4$.

1115 putnam-002 級數 • R5
Answer. $2/3$
Note. Telescoping product equals $2/3$.

1116 putnam-006 級數 • R6
Answer. $2/\pi$
Note. Wallis product gives $2/\pi$.

1117 putnam-007 級數 • R5
Answer. $1/2$
Note. $n^4+n^2+1=(n^2+n+1)(n^2-n+1)$.

1118 putnam-011 級數 • R5
Answer. $2\log(2)-1$
Note. $(-1)^{n+1}(1/n - 1/(n+1))$.
